

(12) **United States Patent**
Kim et al.

(10) **Patent No.:** **US 12,405,044 B2**
(45) **Date of Patent:** **Sep. 2, 2025**

(54) **ICE MAKER AND REFRIGERATOR**

(71) Applicant: **LG ELECTRONICS INC.**, Seoul (KR)

(72) Inventors: **Yonghyun Kim**, Seoul (KR); **Jinil Hong**, Seoul (KR); **Hyunji Park**, Seoul (KR)

(73) Assignee: **LG Electronics Inc.**, Seoul (KR)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 0 days.

(21) Appl. No.: **18/586,795**

(22) Filed: **Feb. 26, 2024**

(65) **Prior Publication Data**

US 2024/0200844 A1 Jun. 20, 2024

Related U.S. Application Data

(63) Continuation of application No. 17/950,626, filed on Sep. 22, 2022, now Pat. No. 11,913,700, which is a (Continued)

(30) **Foreign Application Priority Data**

Nov. 16, 2018 (KR) 10-2018-0142079
Jul. 6, 2019 (KR) 10-2019-0081737
Sep. 6, 2019 (KR) 10-2019-0110813

(51) **Int. Cl.**
F25C 1/10 (2006.01)
F25C 1/243 (2018.01)
(Continued)

(52) **U.S. Cl.**
CPC **F25C 1/10** (2013.01); **F25C 1/243** (2013.01); **F25C 5/22** (2018.01); **F25D 11/02** (2013.01);
(Continued)

(58) **Field of Classification Search**

CPC F25C 1/04; F25C 1/10; F25C 1/18; F25C 1/243; F25C 2305/022; F25C 2305/0221;
(Continued)

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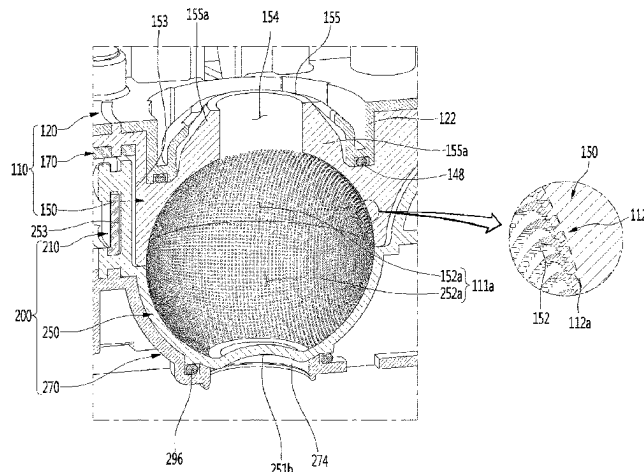
Primary Examiner — Filip Zec

(74) *Attorney, Agent, or Firm* — Fish & Richardson P.C.

(57) **ABSTRACT**

An ice maker includes an upper tray and a lower tray made of an elastic material. Upper chambers in the upper tray and lower chambers in the lower tray contact each other to define spherical ice chambers, respectively. An ejector-receiving opening is opened to each of the upper chambers. An opening-defining wall extends upward along a circumference of each ejector-receiving opening. An upper ejector passes through the ejector-defining wall and vertically moves to remove ice from the ice chamber, and a driver pivots the lower tray to open and close the upper tray and the lower tray. The upper tray includes a horizontal extension forming a top face of the upper tray, and the upper chamber is positioned below the horizontal extension such that the opening-defining wall is formed at a vertical level equal to or lower than a vertical level of the opening-defining wall.

20 Claims, 67 Drawing Sheets



Related U.S. Application Data

continuation of application No. 16/685,793, filed on
Nov. 15, 2019, now Pat. No. 11,480,375.

(51) **Int. Cl.**

F25C 5/20 (2018.01)
F25D 11/02 (2006.01)
F25D 23/02 (2006.01)
F25D 23/12 (2006.01)

(52) **U.S. Cl.**

CPC **F25D 23/028** (2013.01); **F25D 23/126**
(2013.01); **F25C 2305/022** (2013.01); **F25C**
2305/0221 (2021.08); **F25C 2400/06**
(2013.01); **F25C 2400/10** (2013.01)

(58) **Field of Classification Search**

CPC F25C 2400/06; F25C 2400/10; F25C
2500/02; F25C 2600/04; F25C 5/08;
F25C 5/22

See application file for complete search history.

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FIG. 1

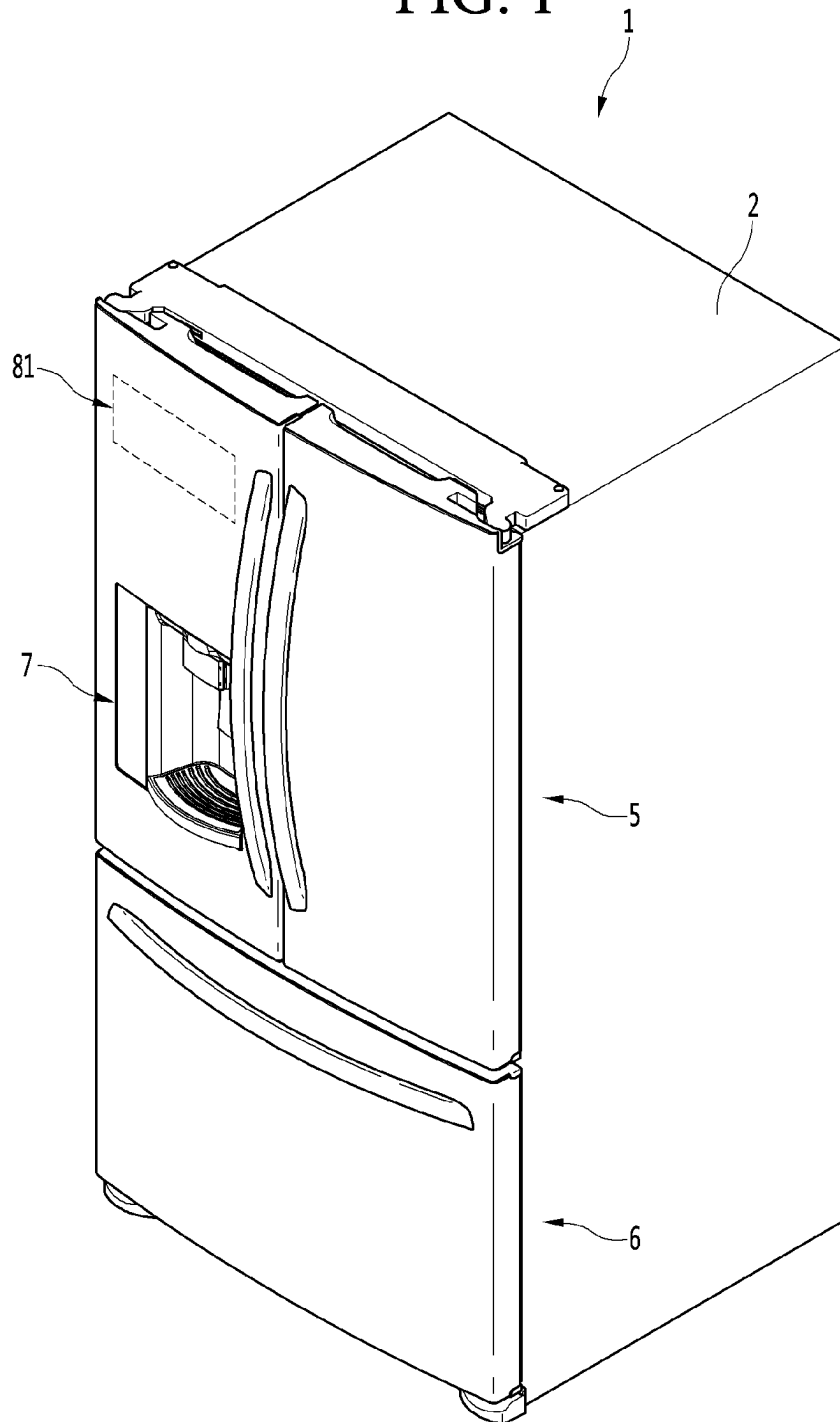


FIG. 2

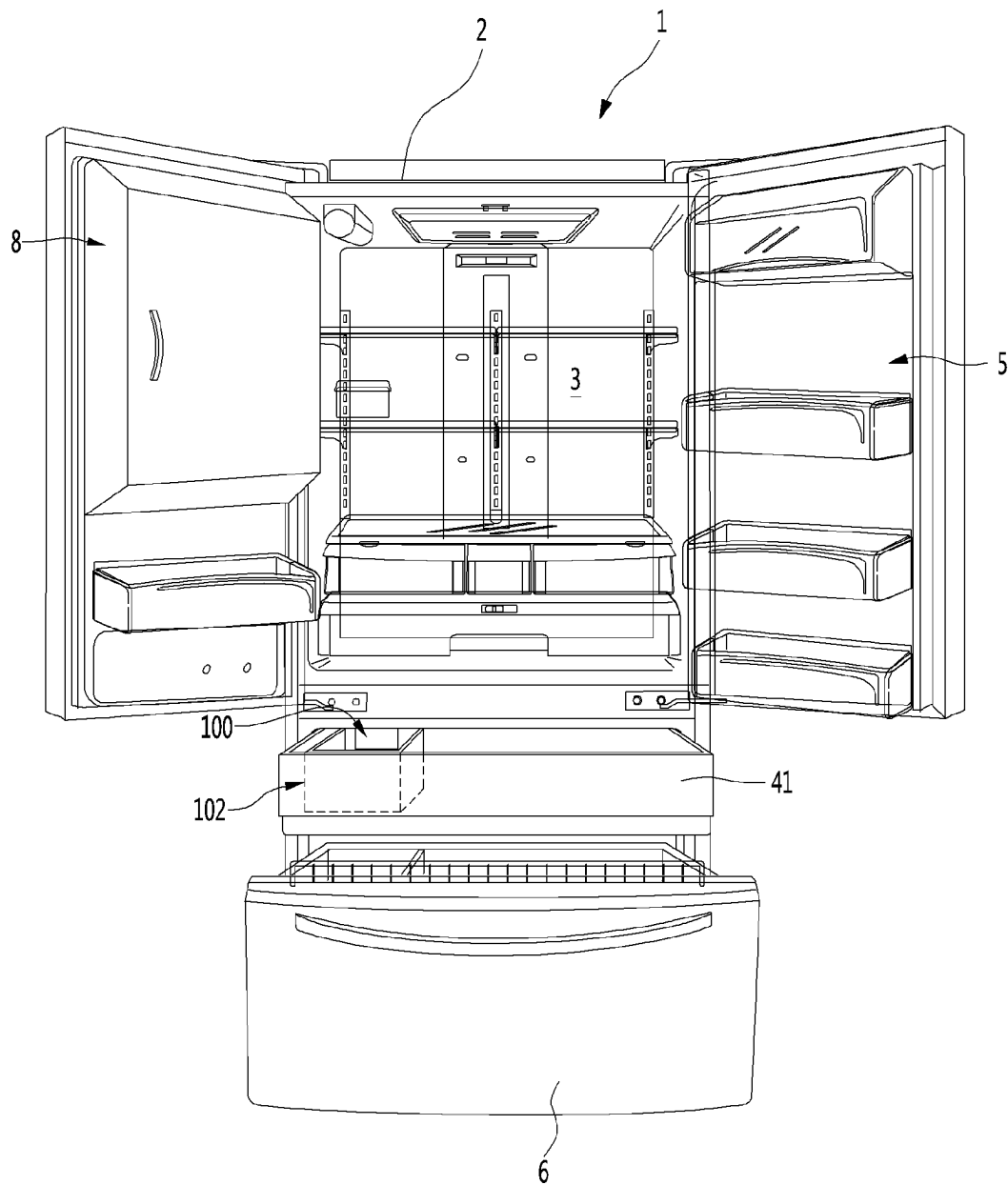


FIG. 3

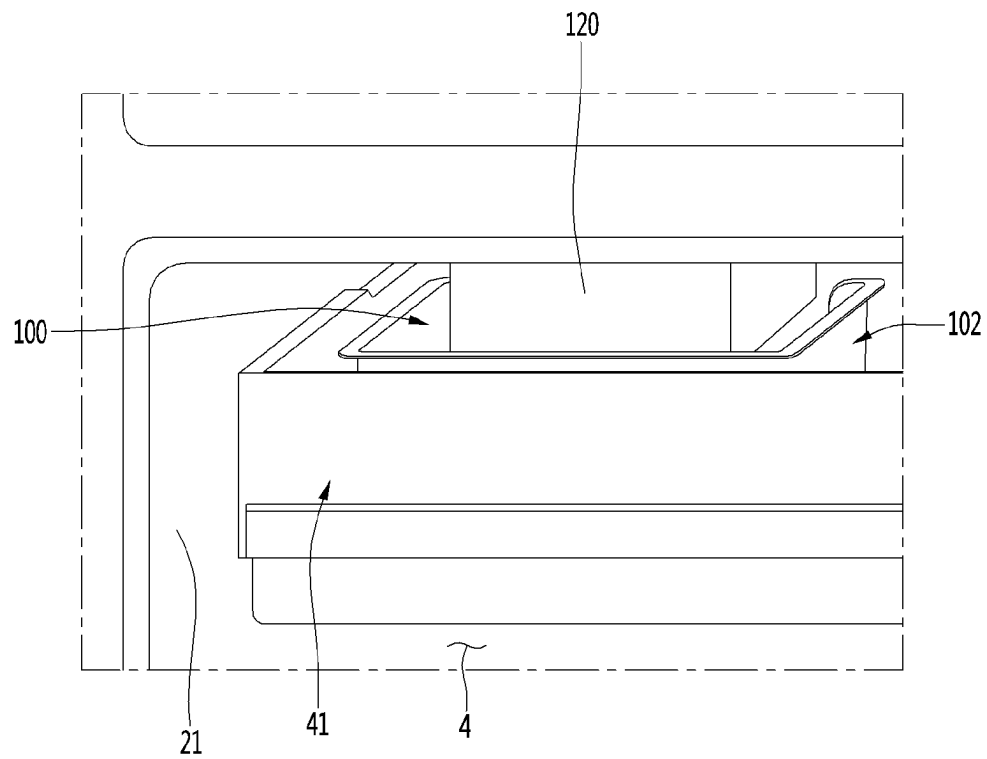


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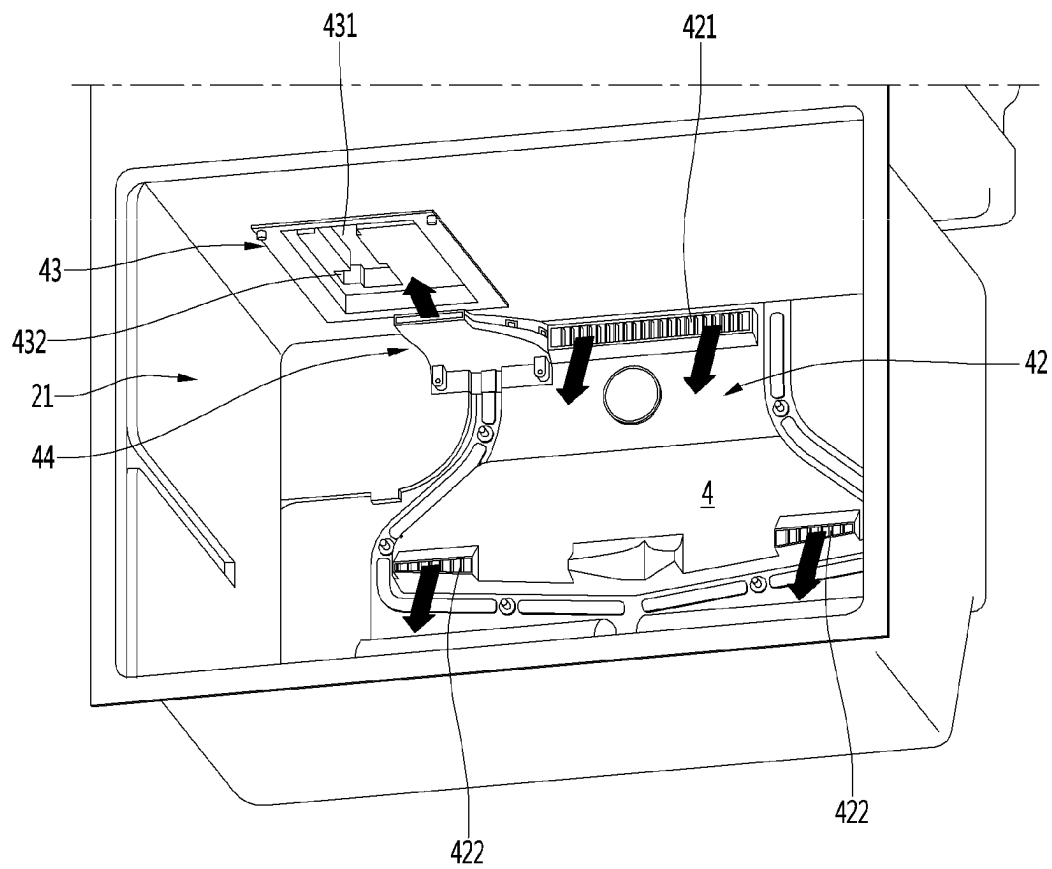


FIG. 5

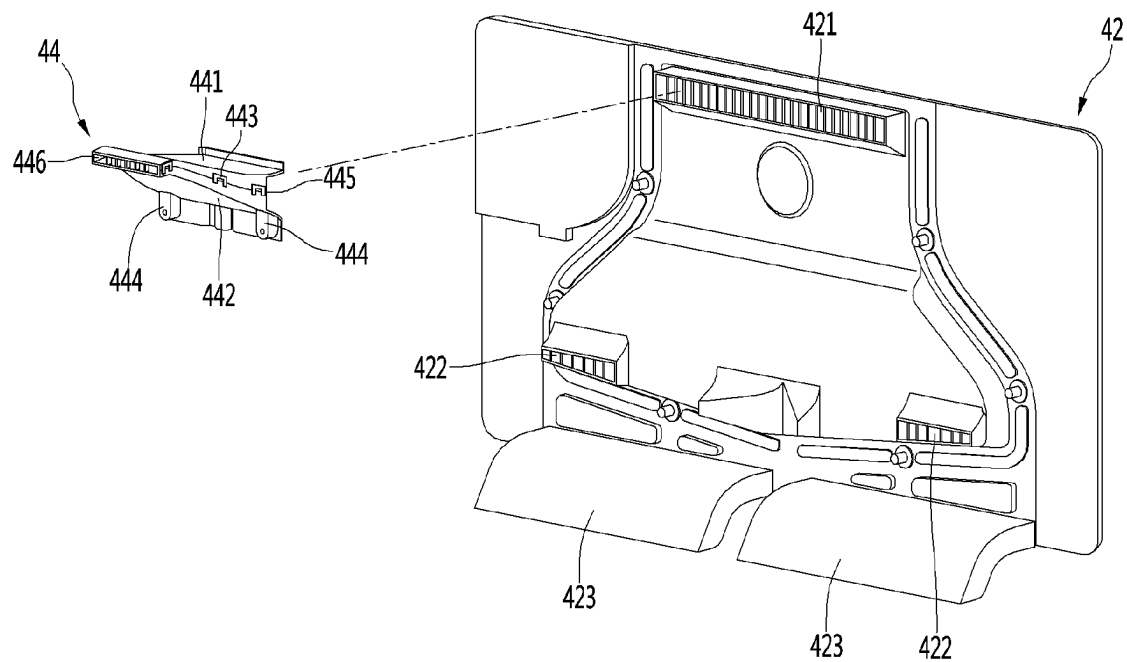


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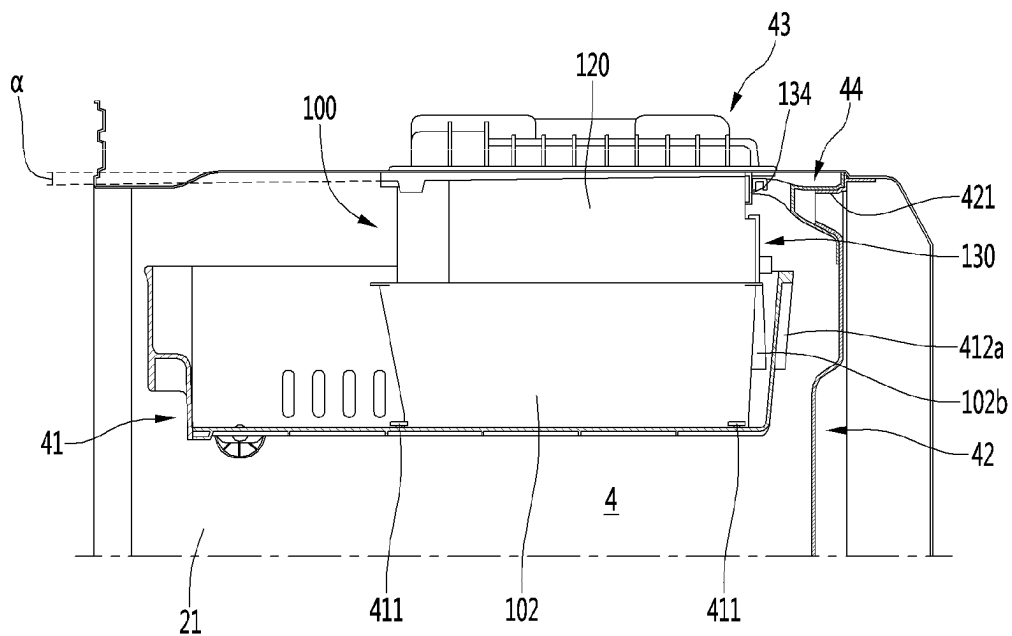


FIG. 7

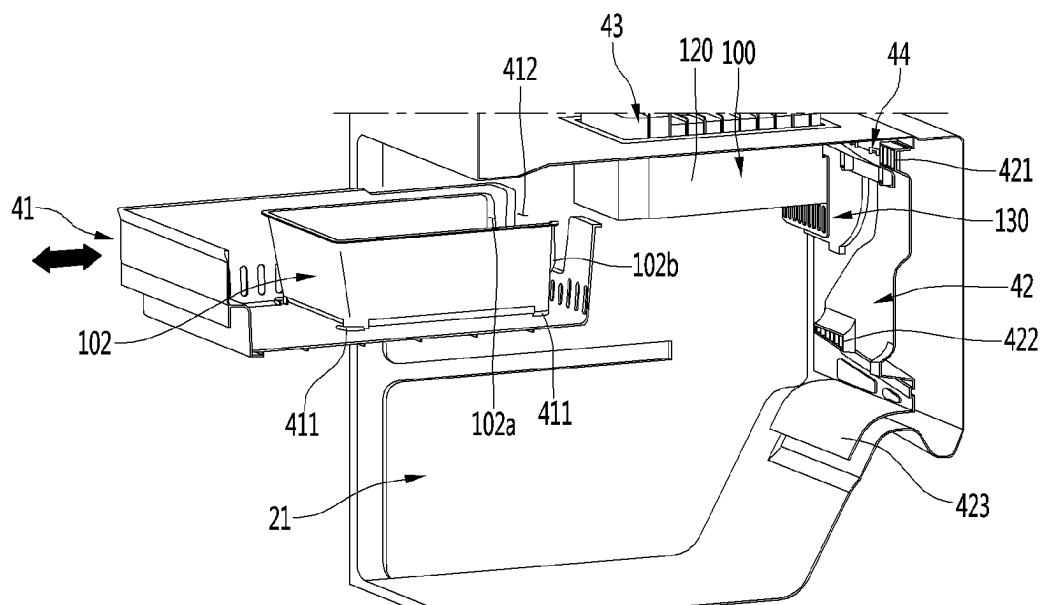


FIG. 8

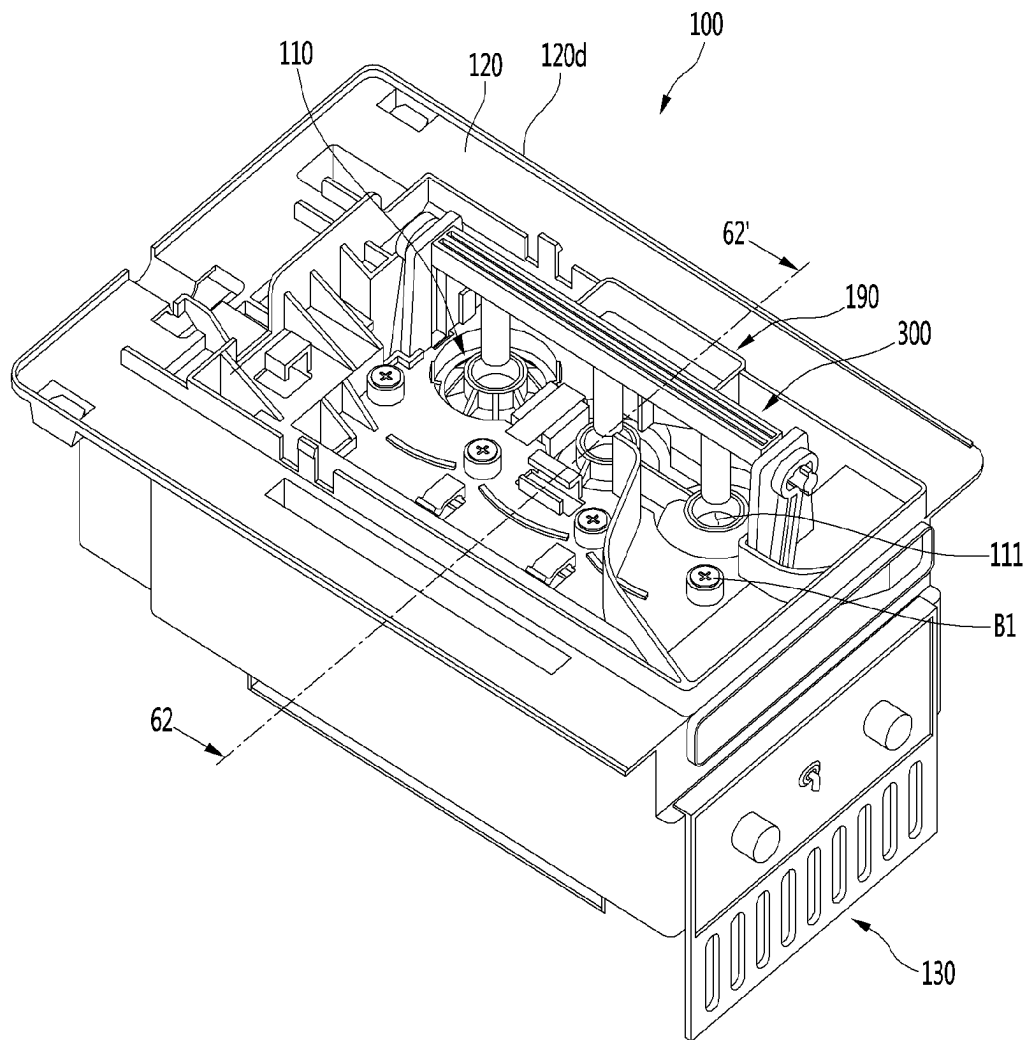


FIG. 9

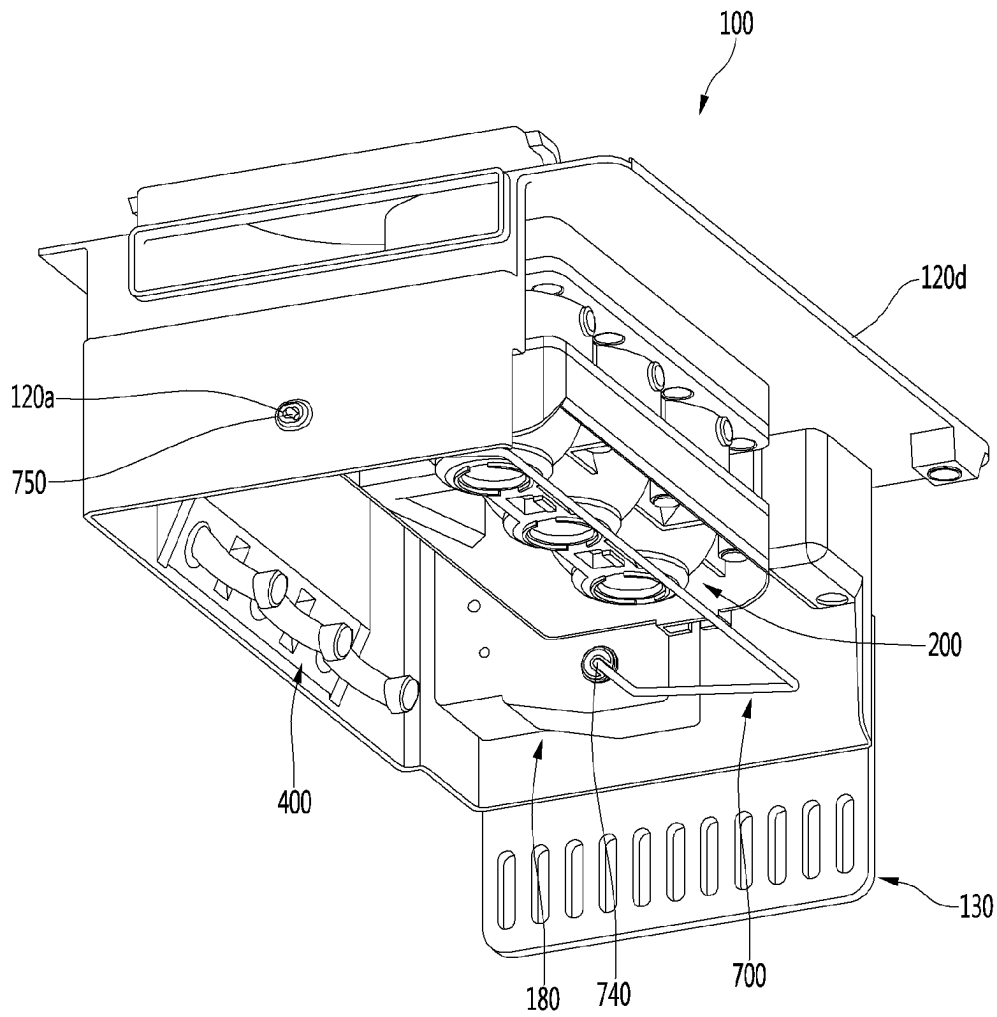


FIG. 10

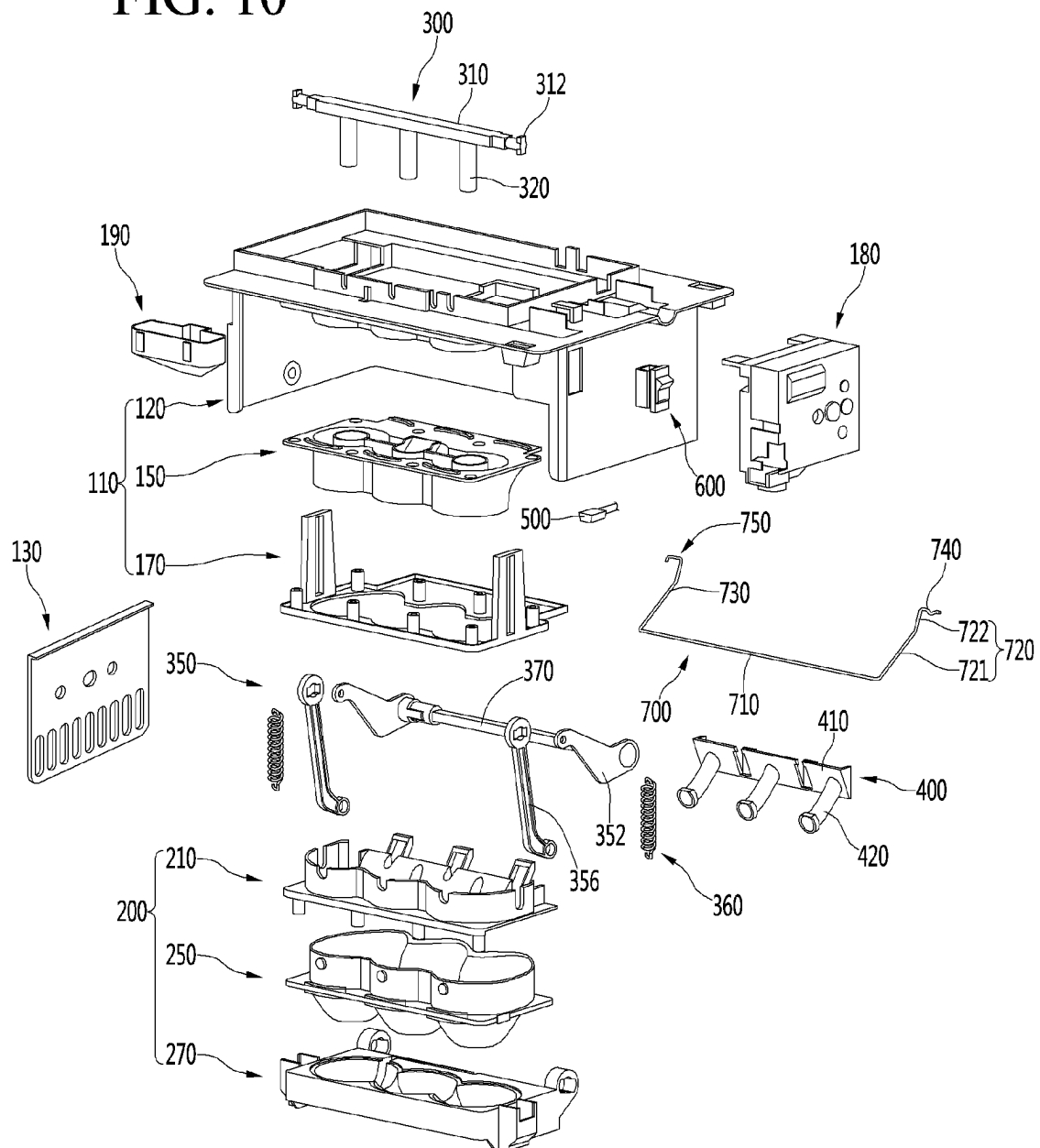


FIG. 11

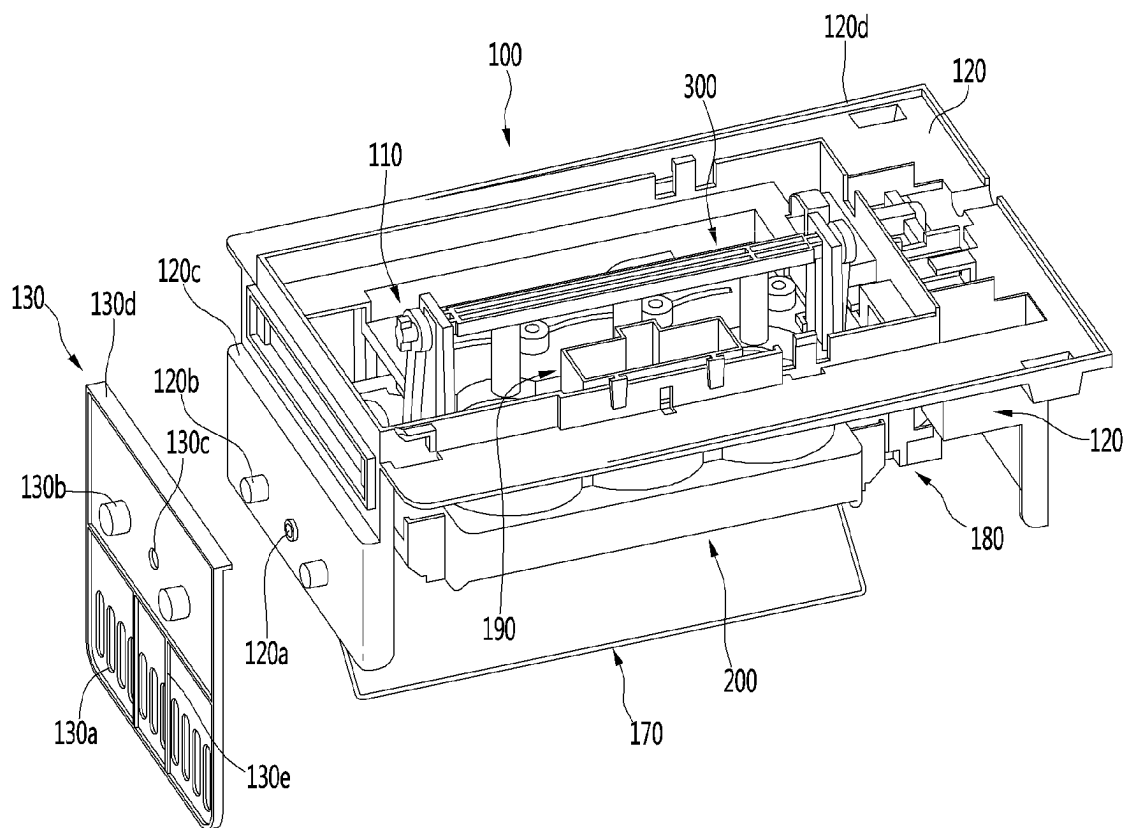


FIG. 12

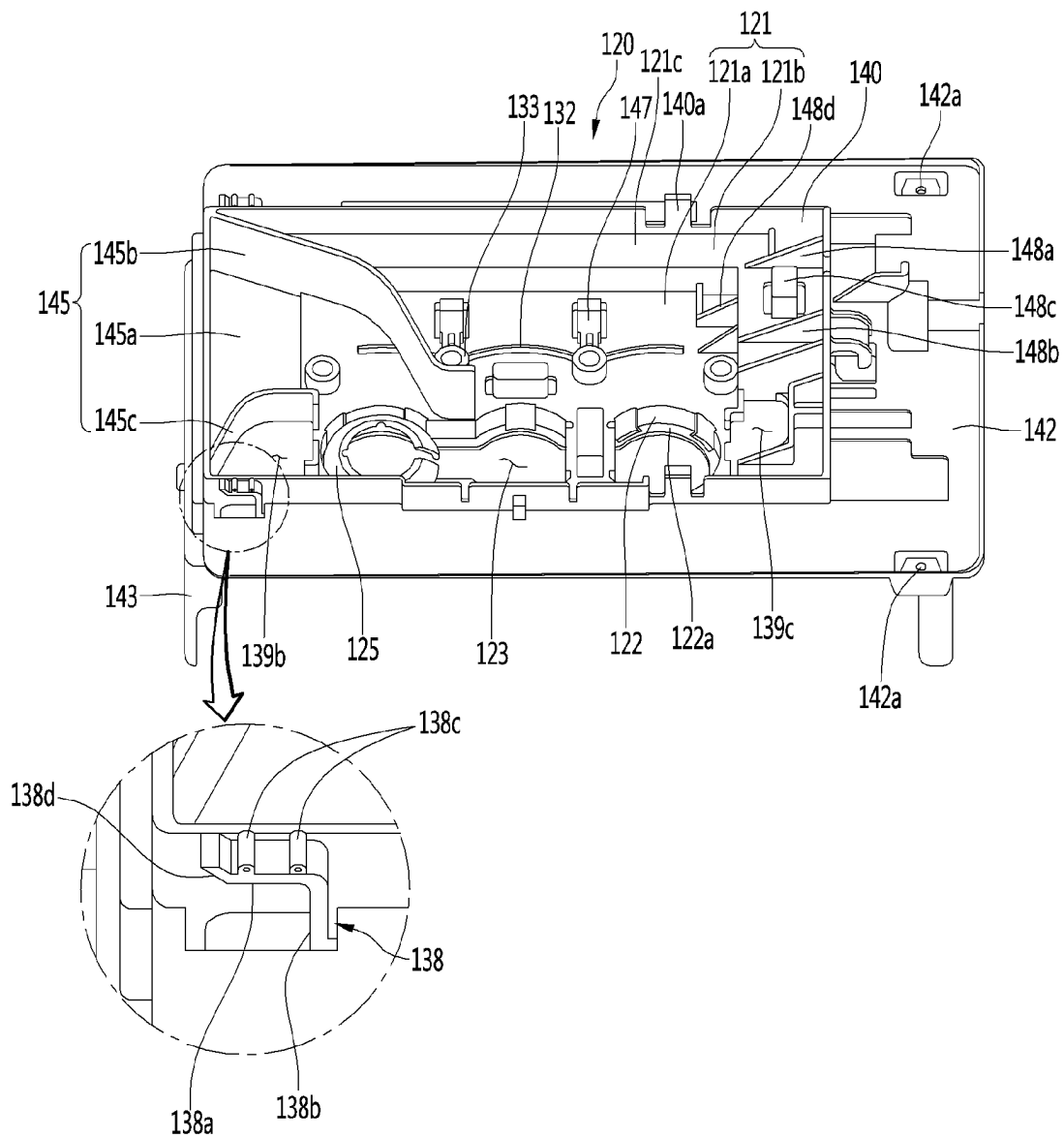


FIG. 13

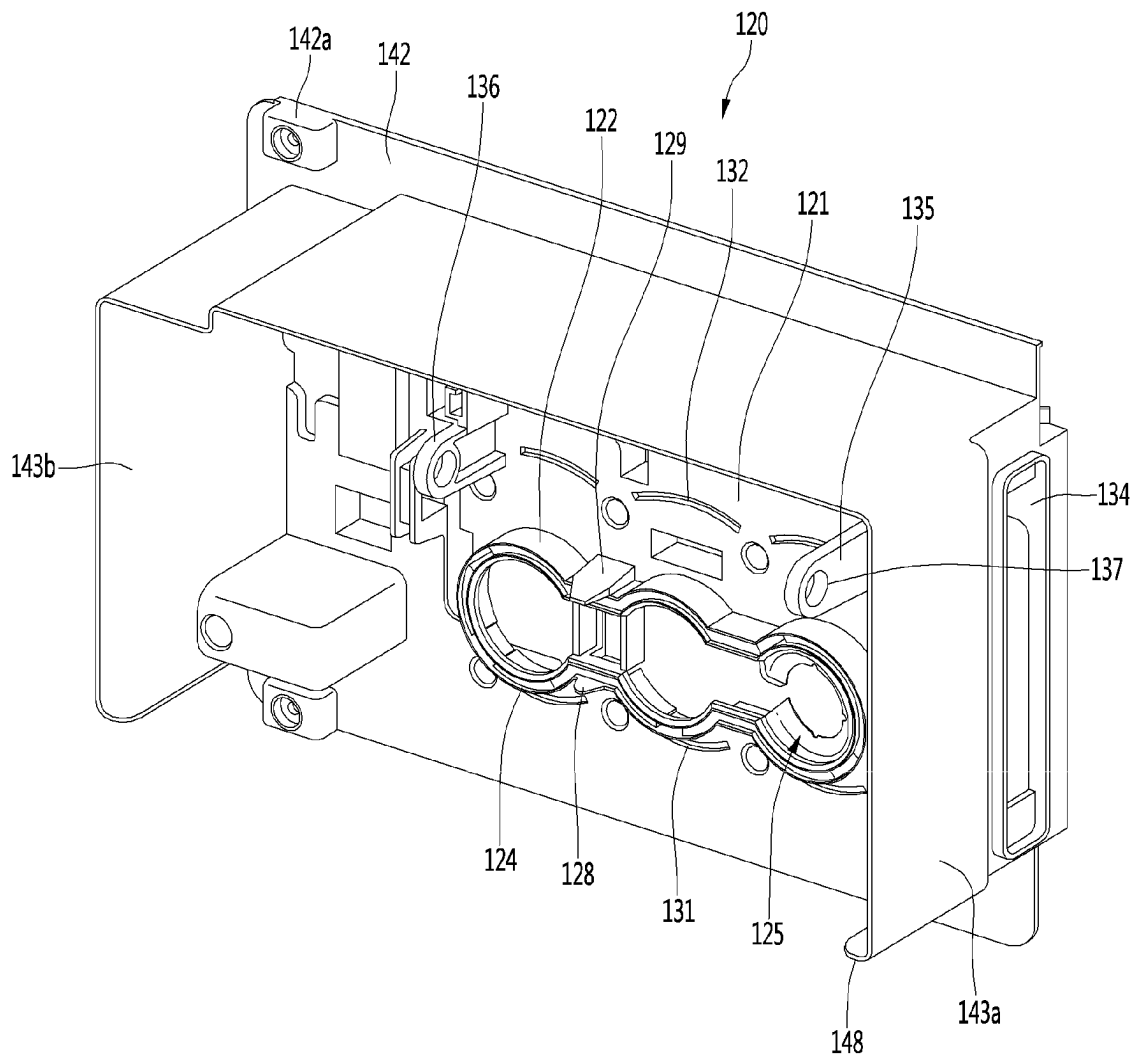


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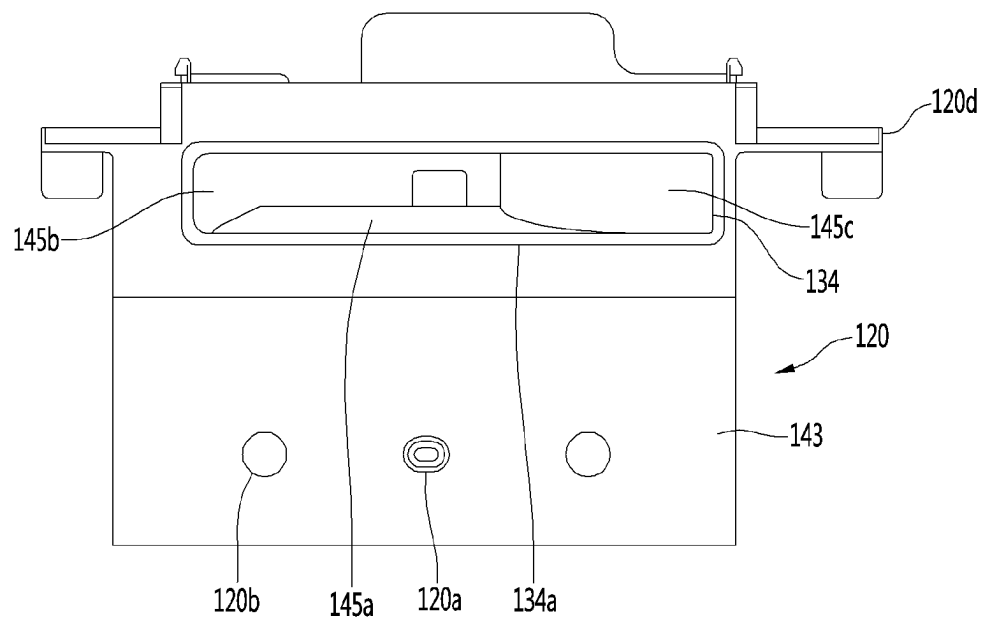


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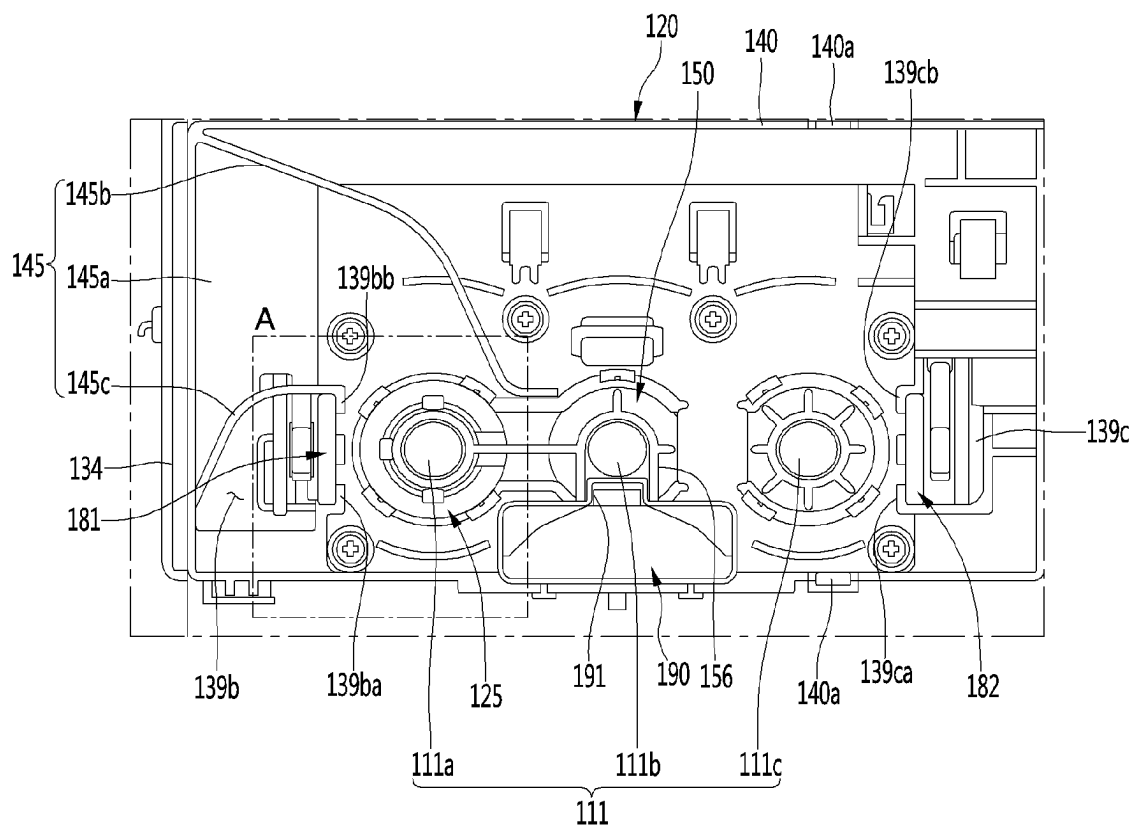


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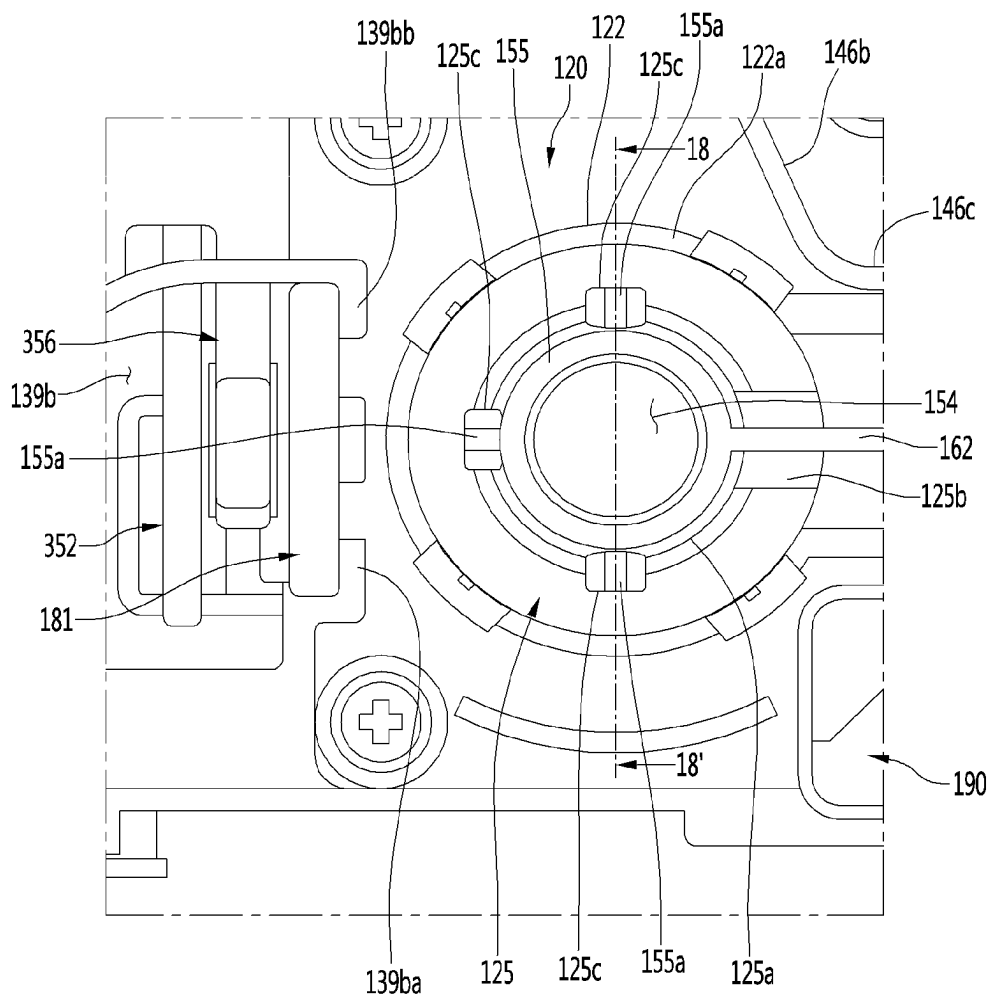


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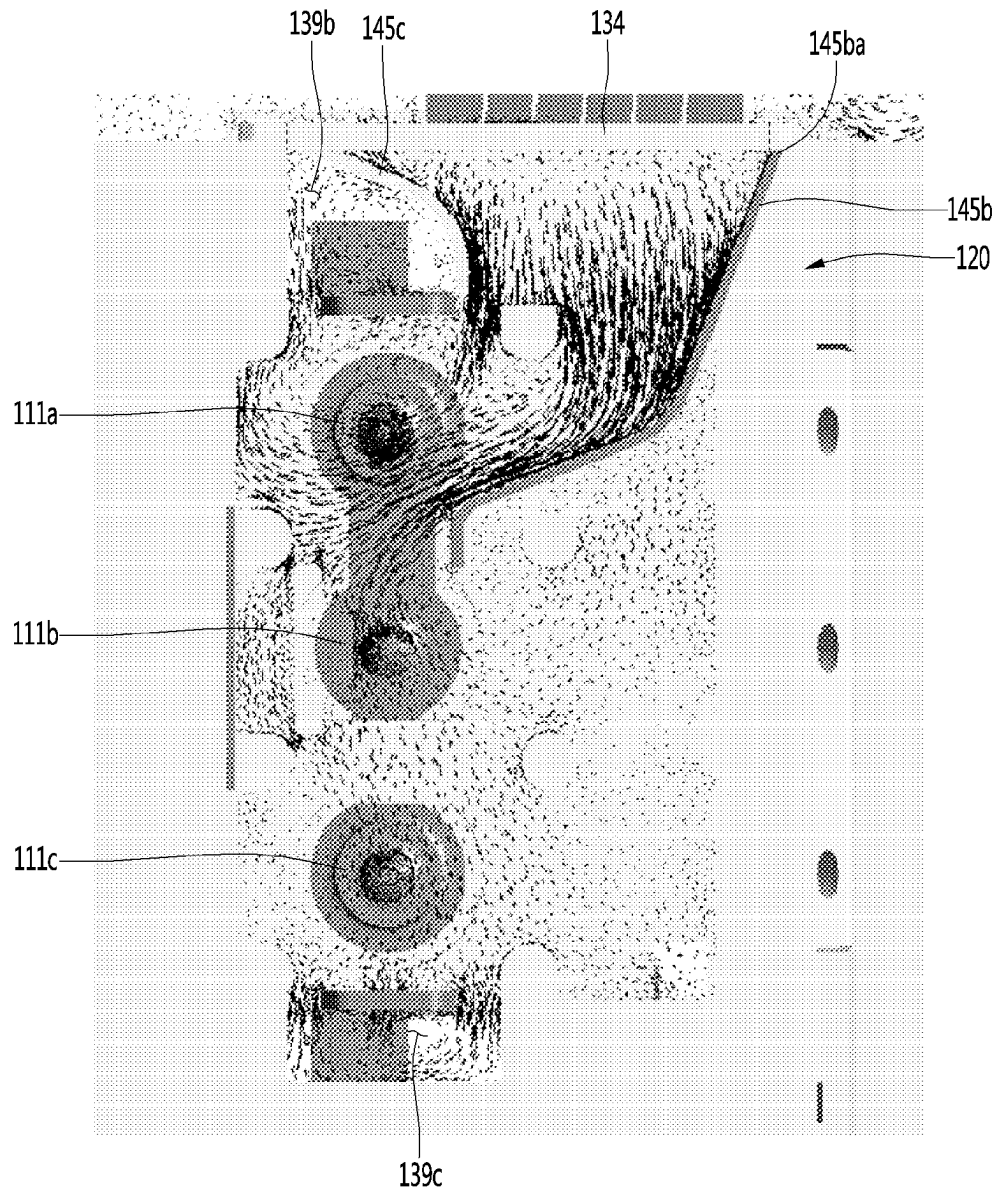


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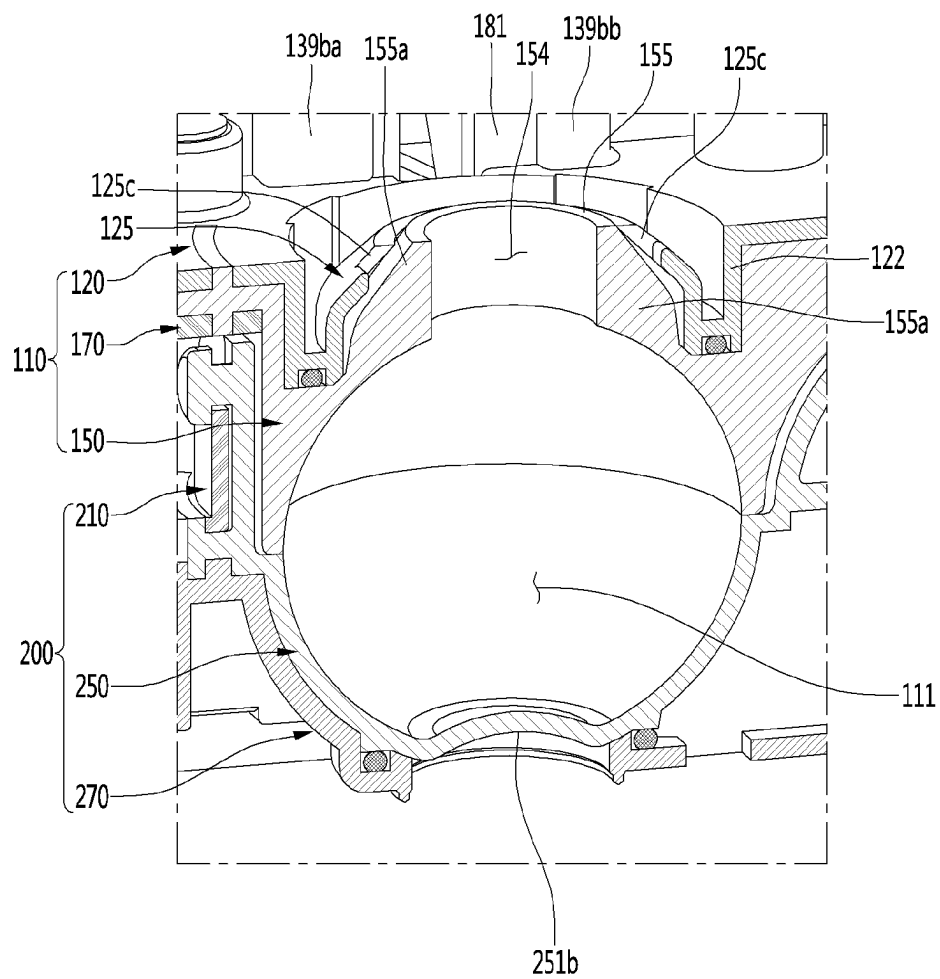


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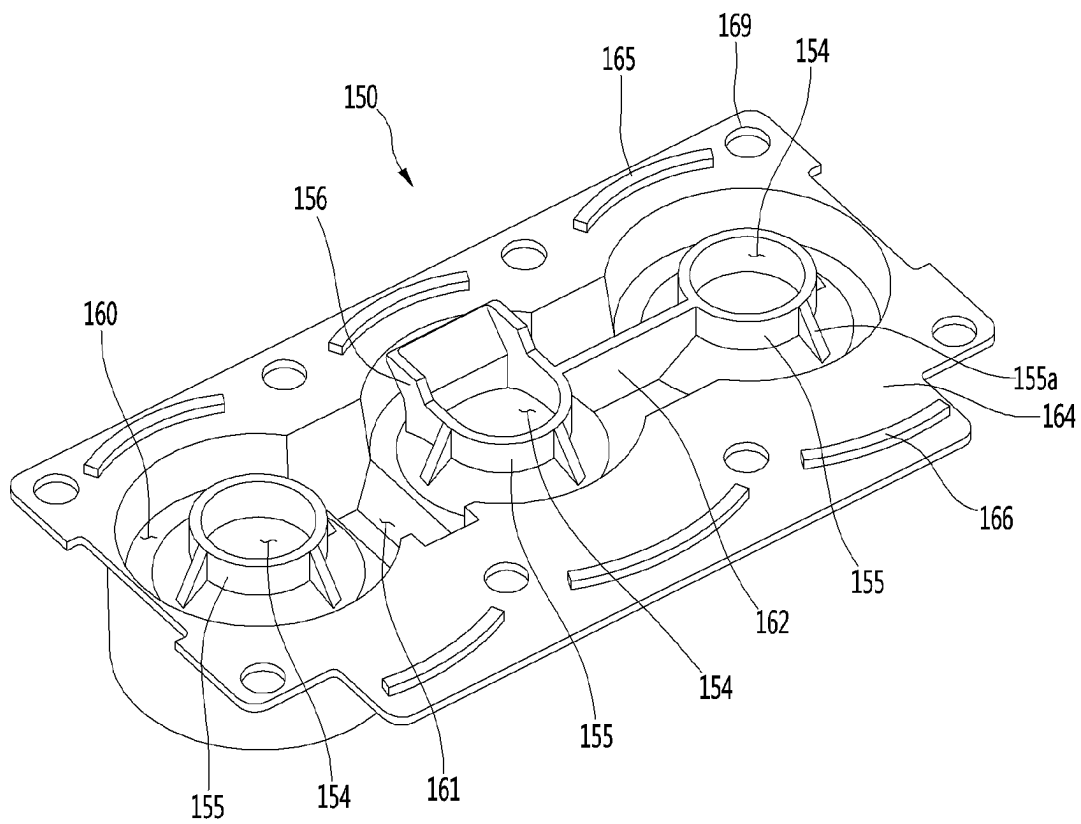


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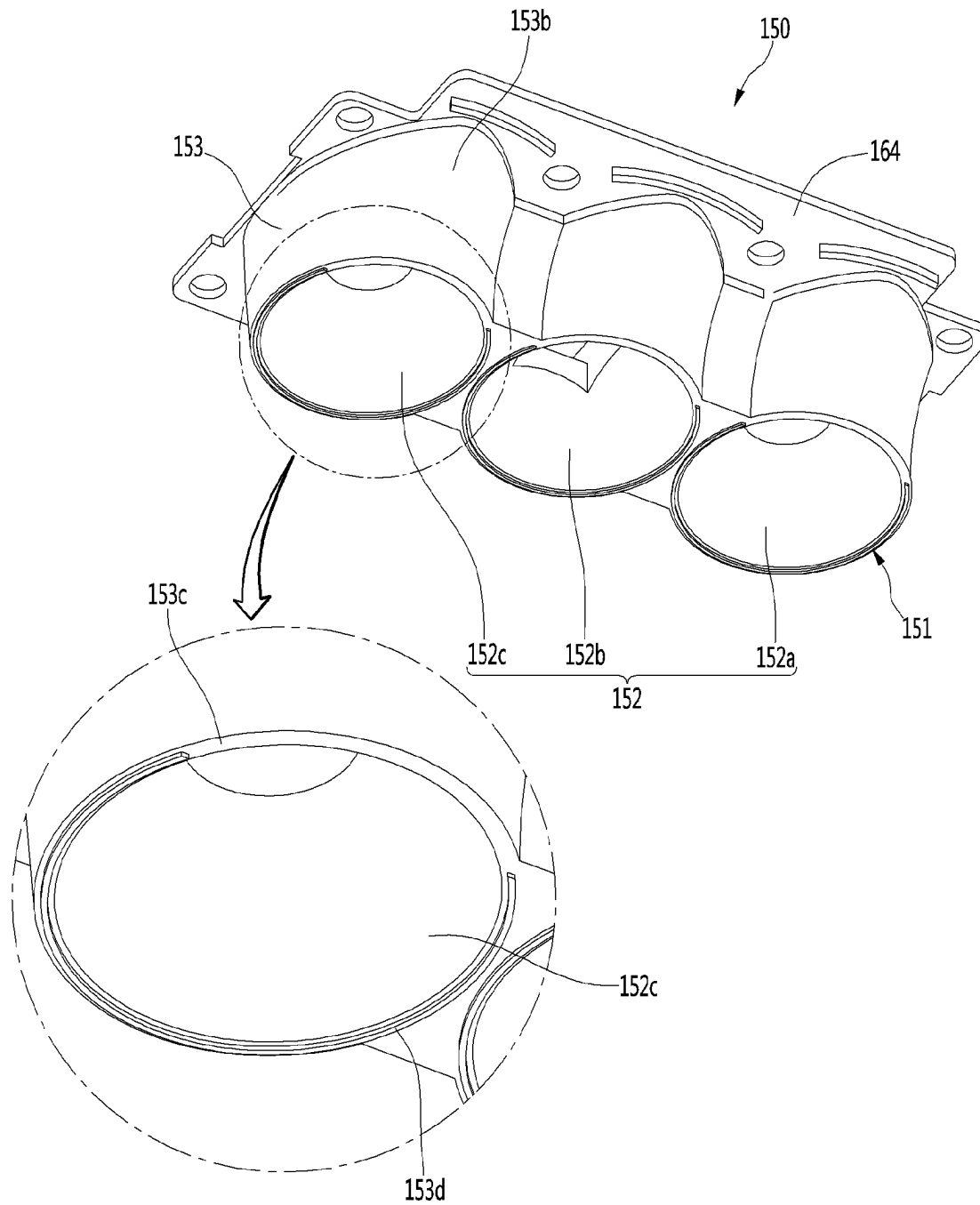


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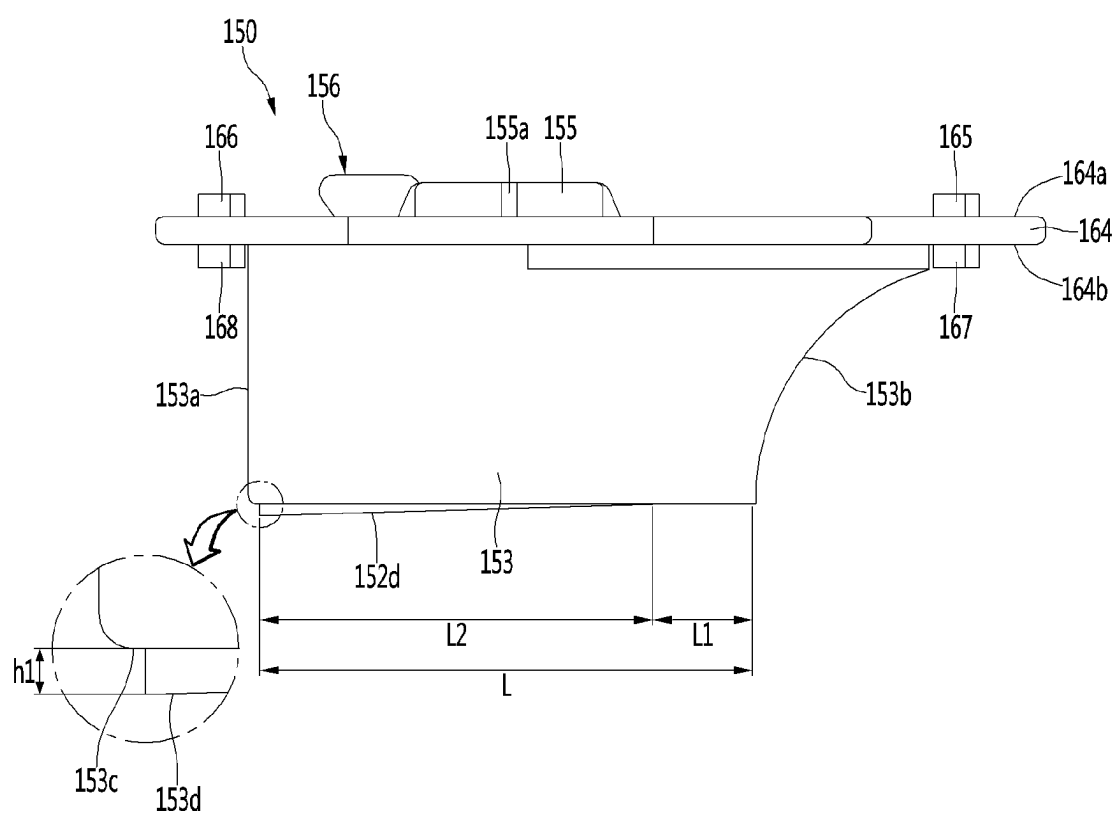


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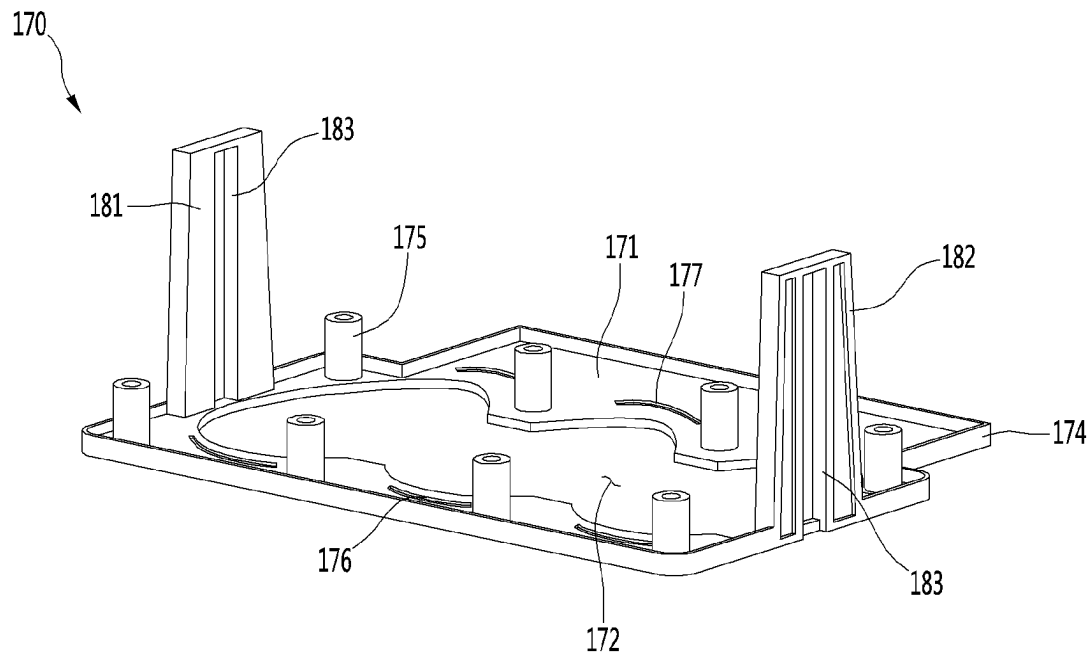


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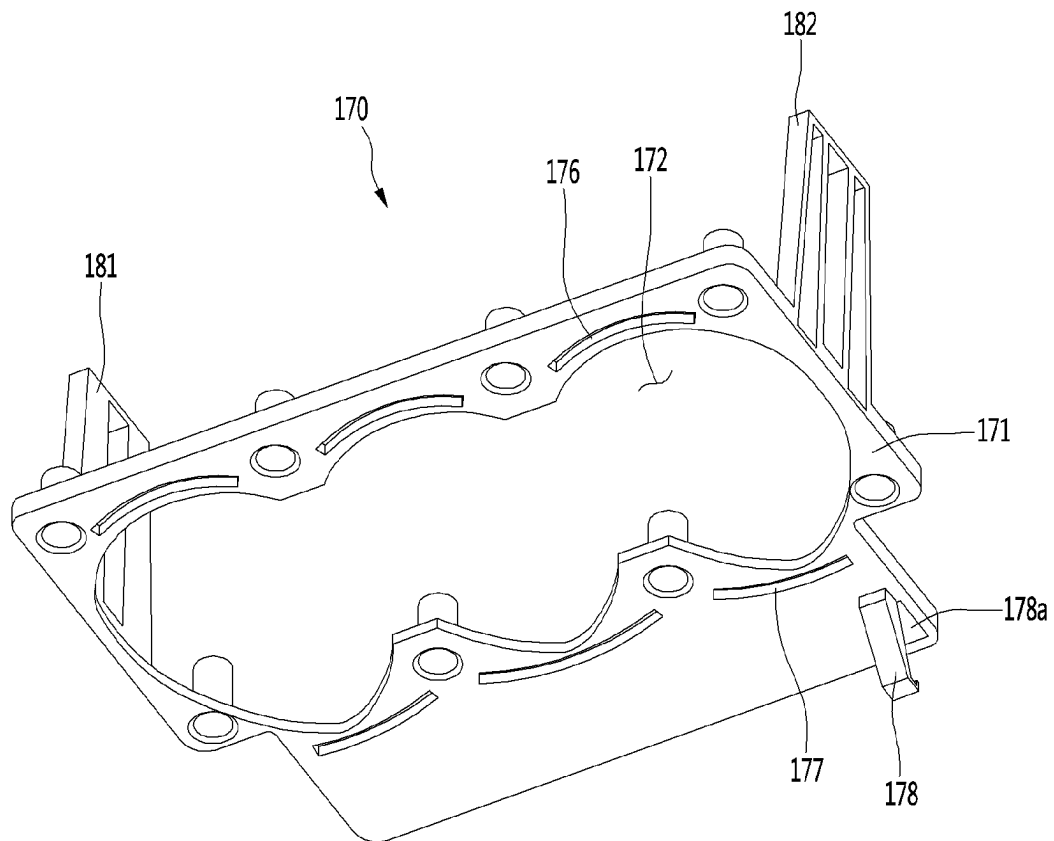


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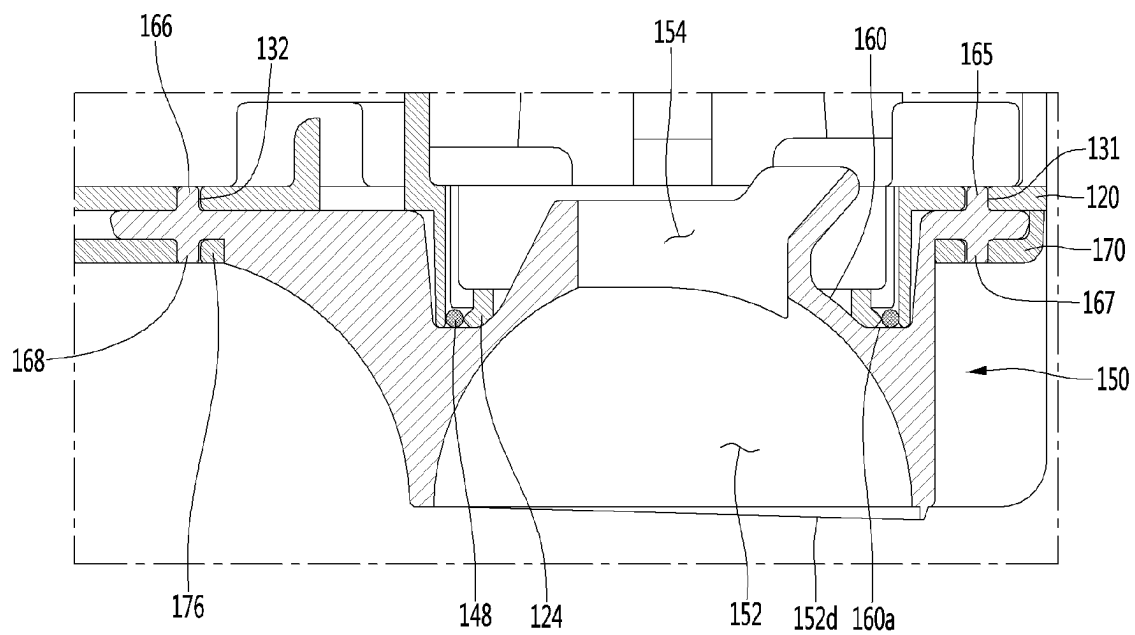


FIG. 25

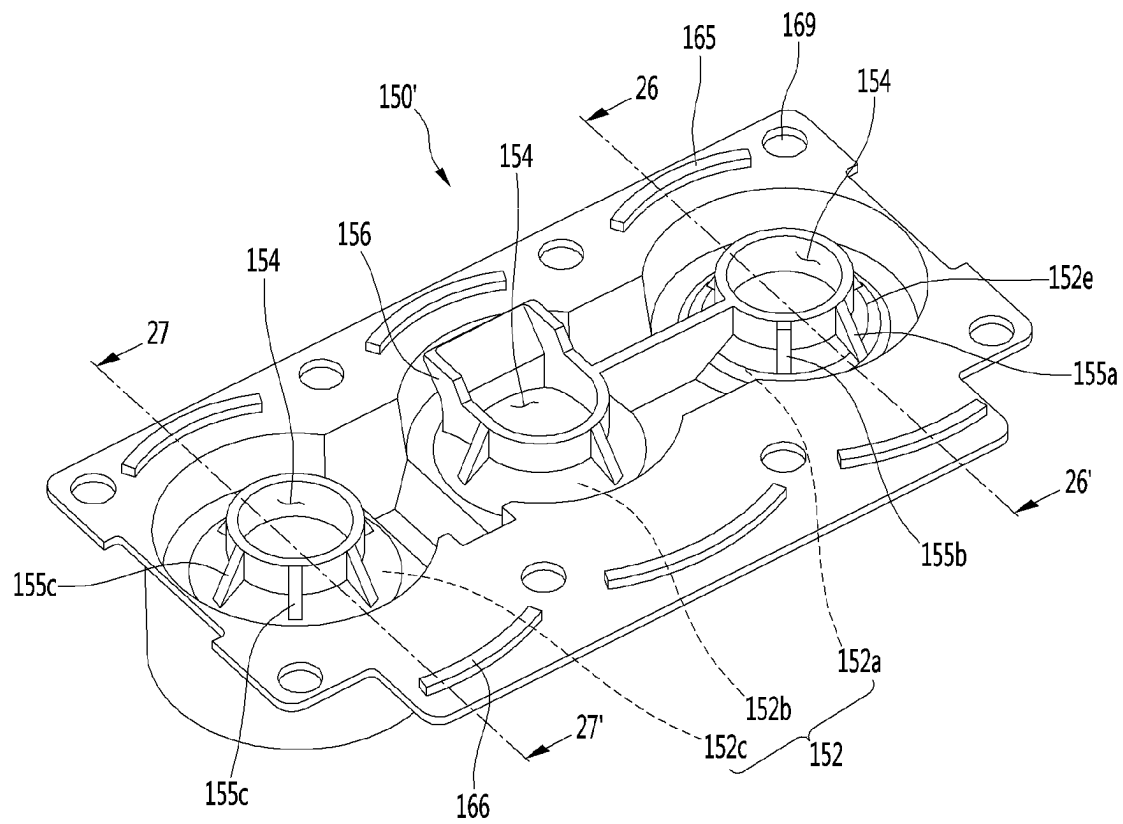


FIG. 26

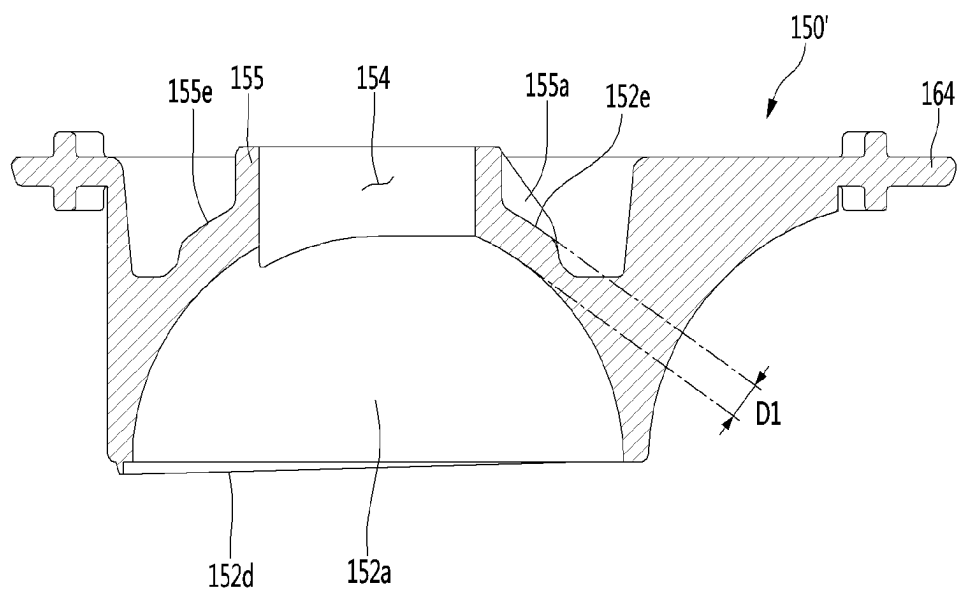


FIG. 27

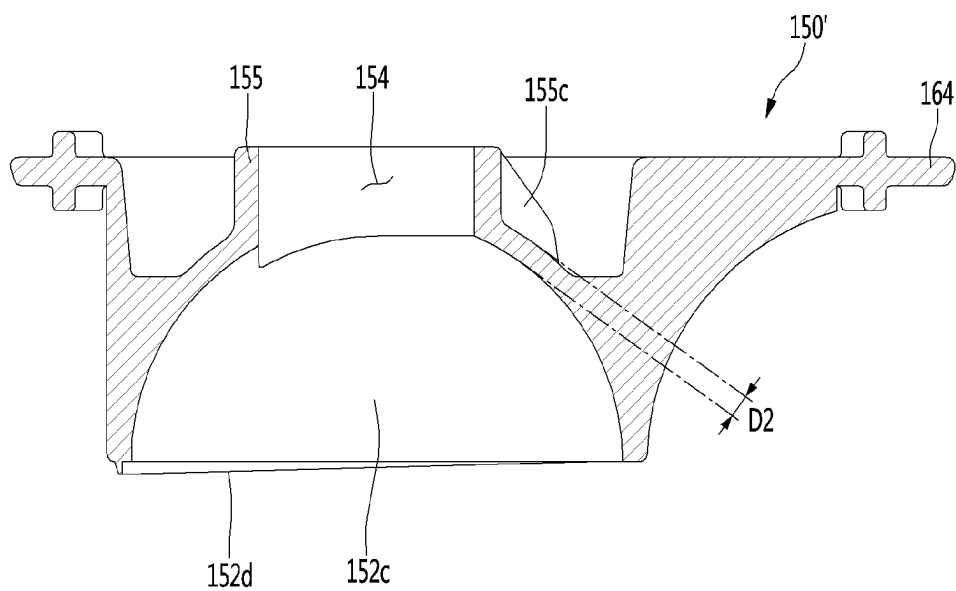


FIG. 28

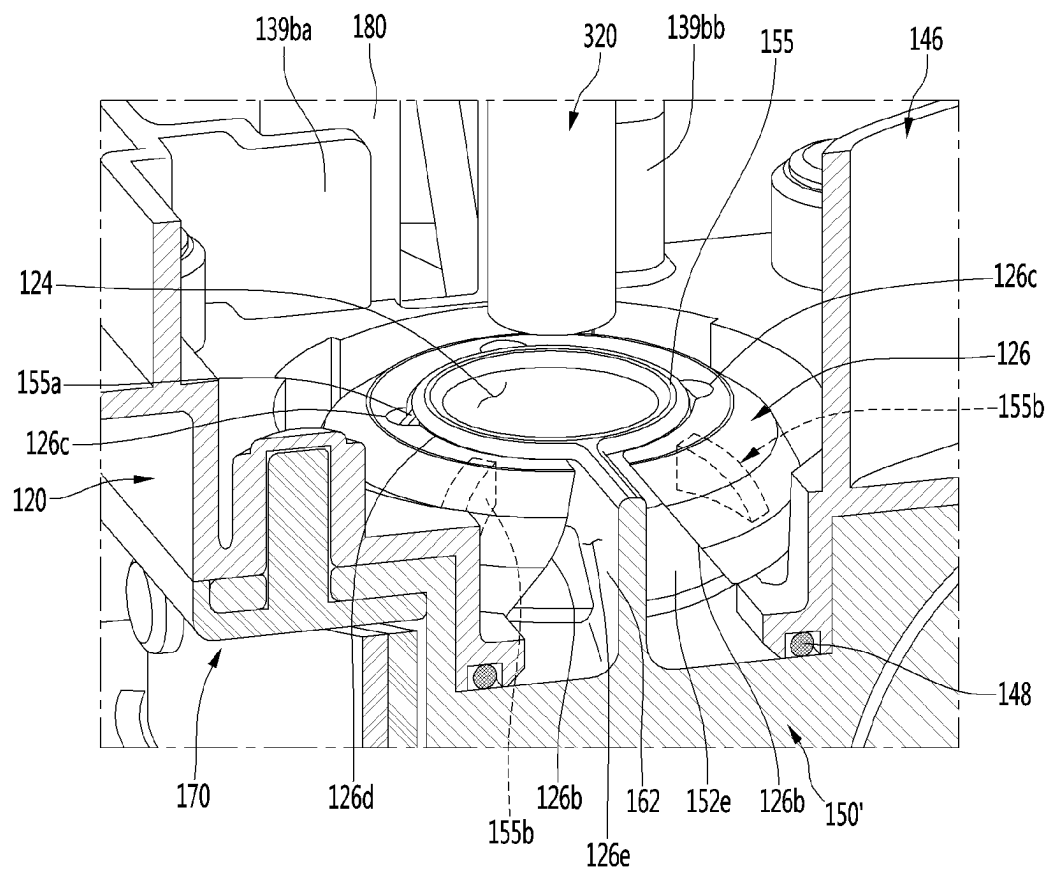


FIG. 29

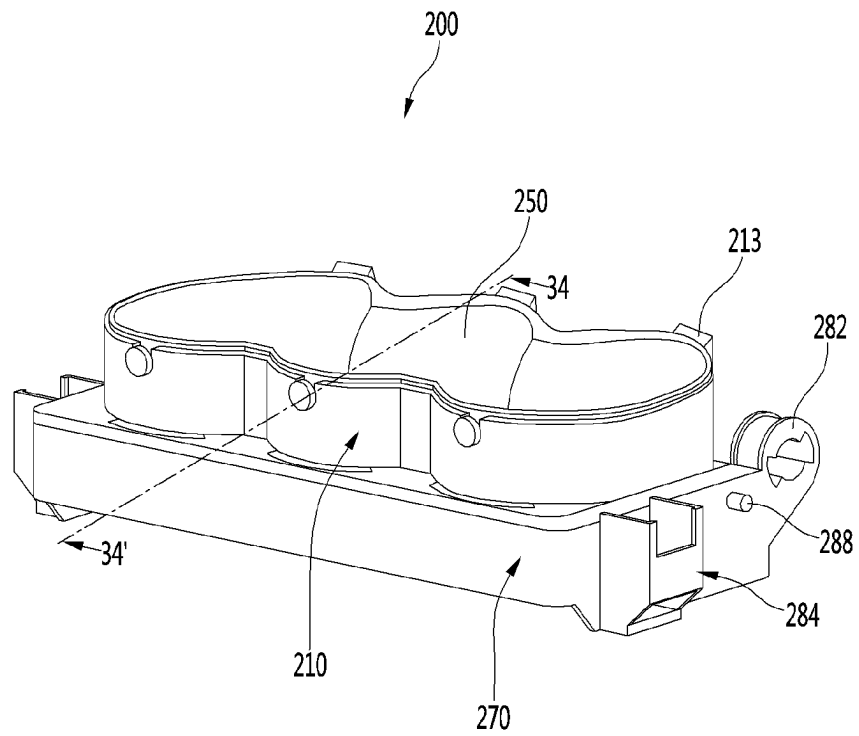


FIG. 30

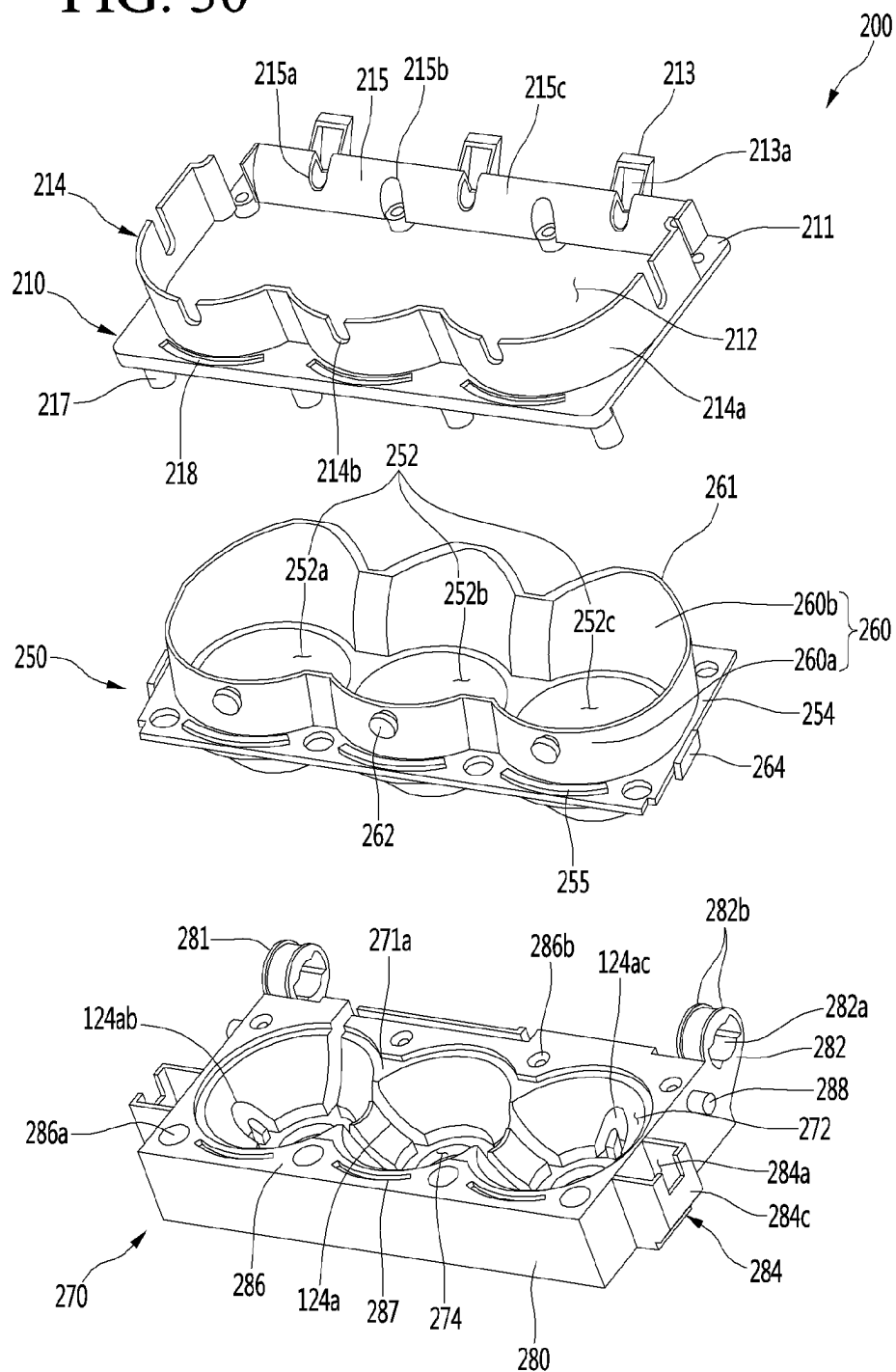


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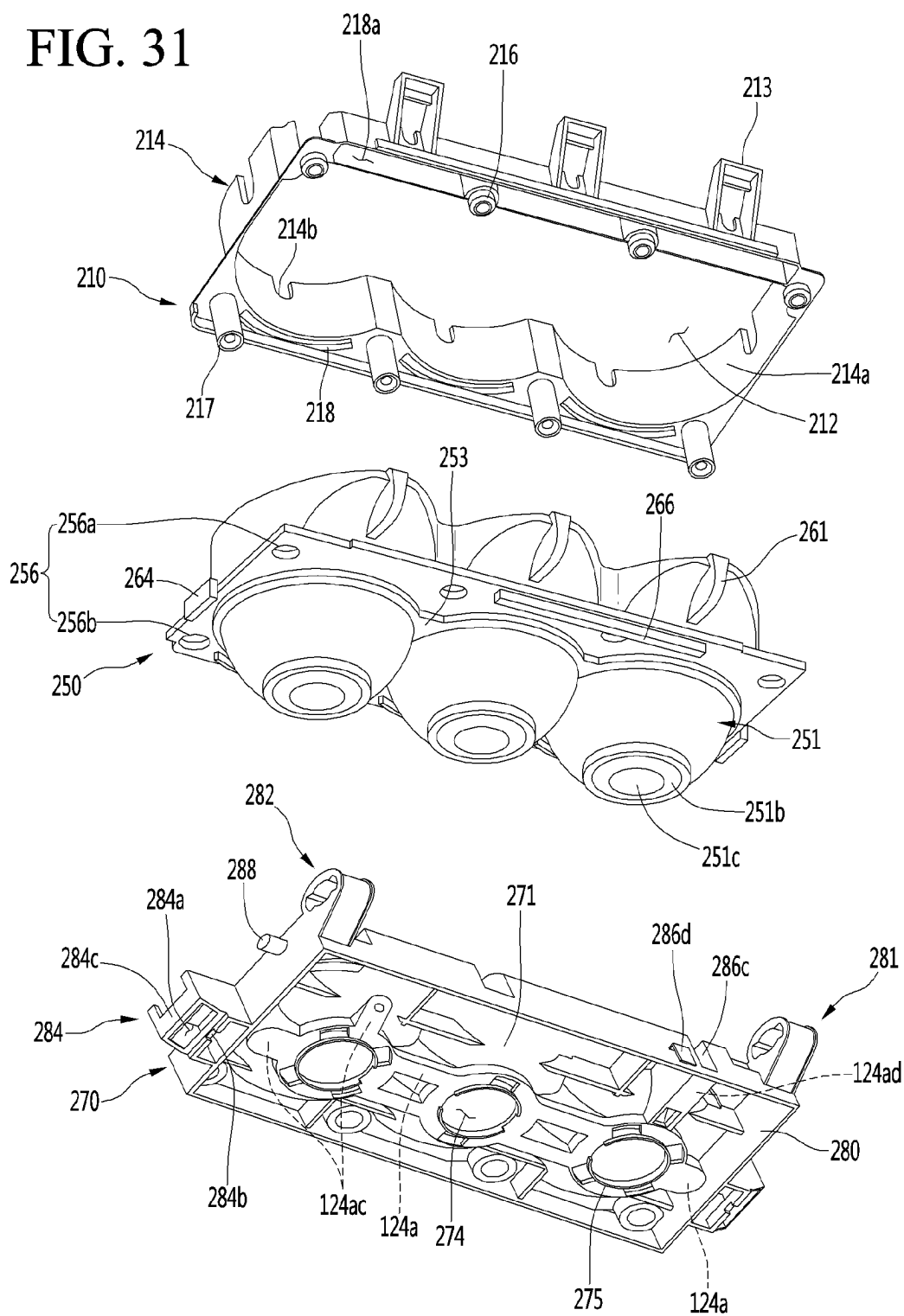


FIG. 32

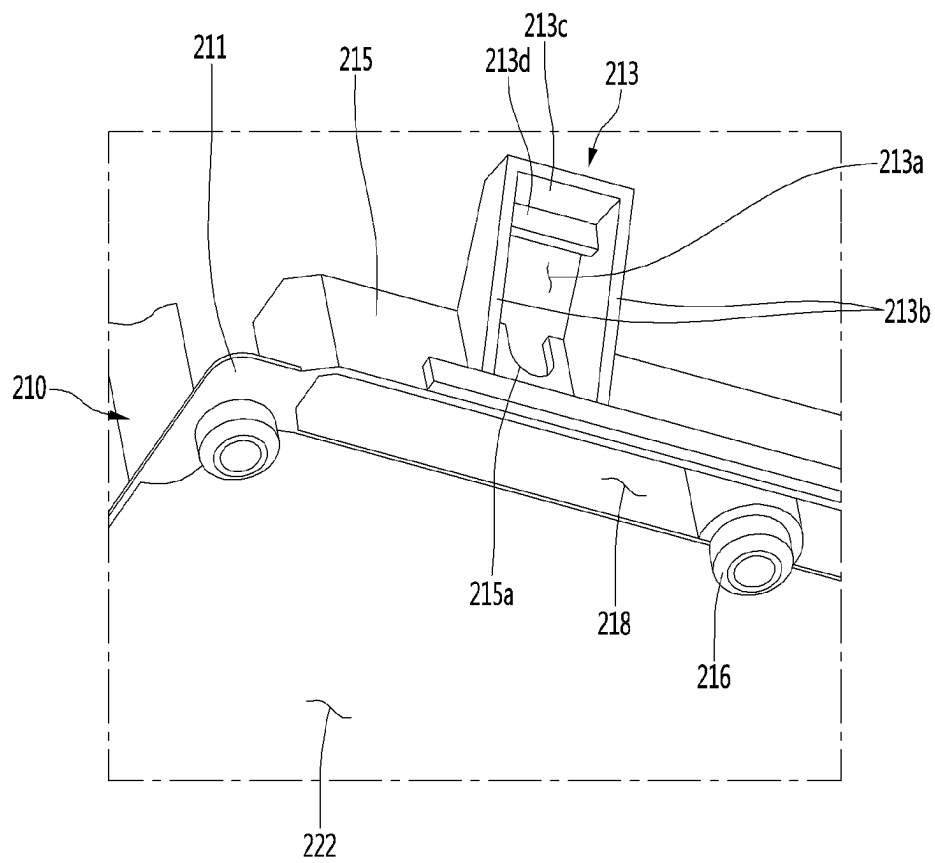


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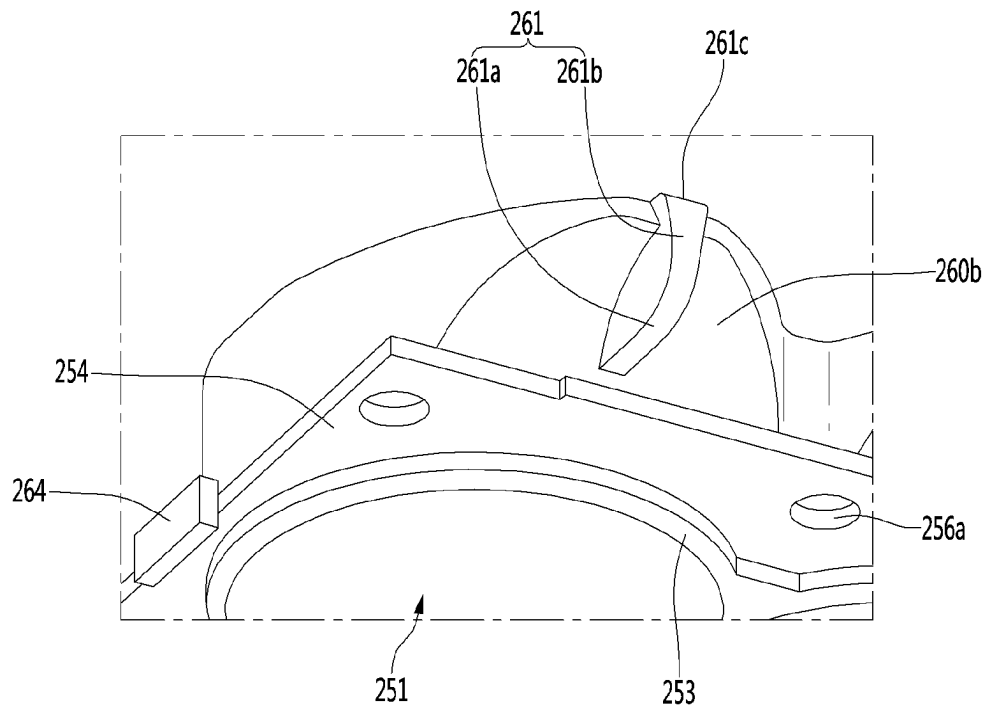


FIG. 34

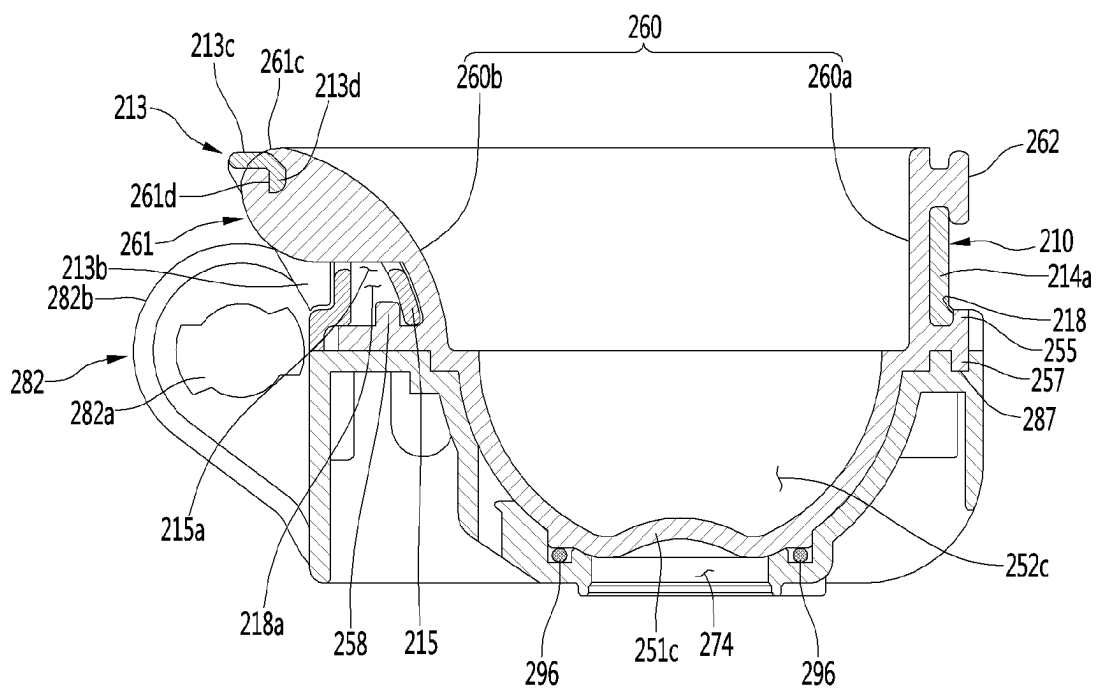


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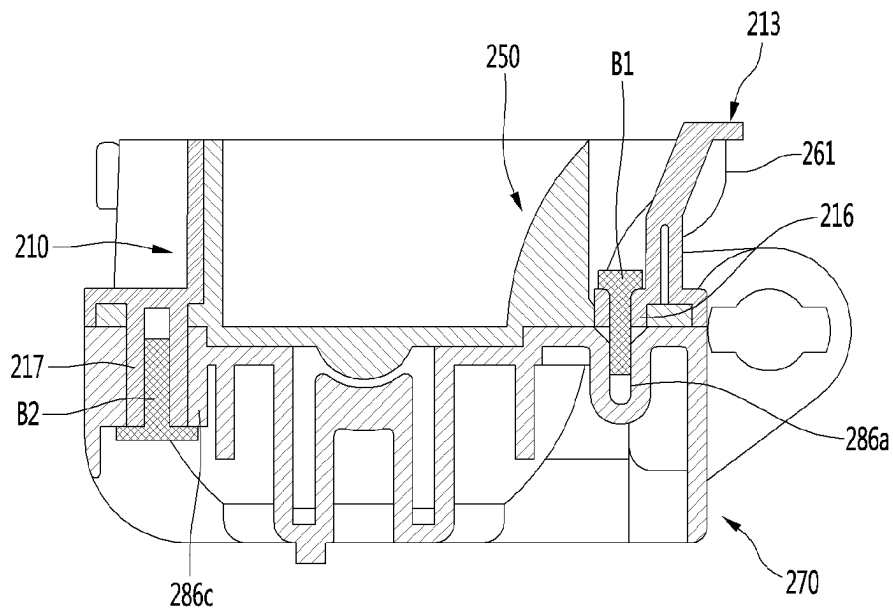


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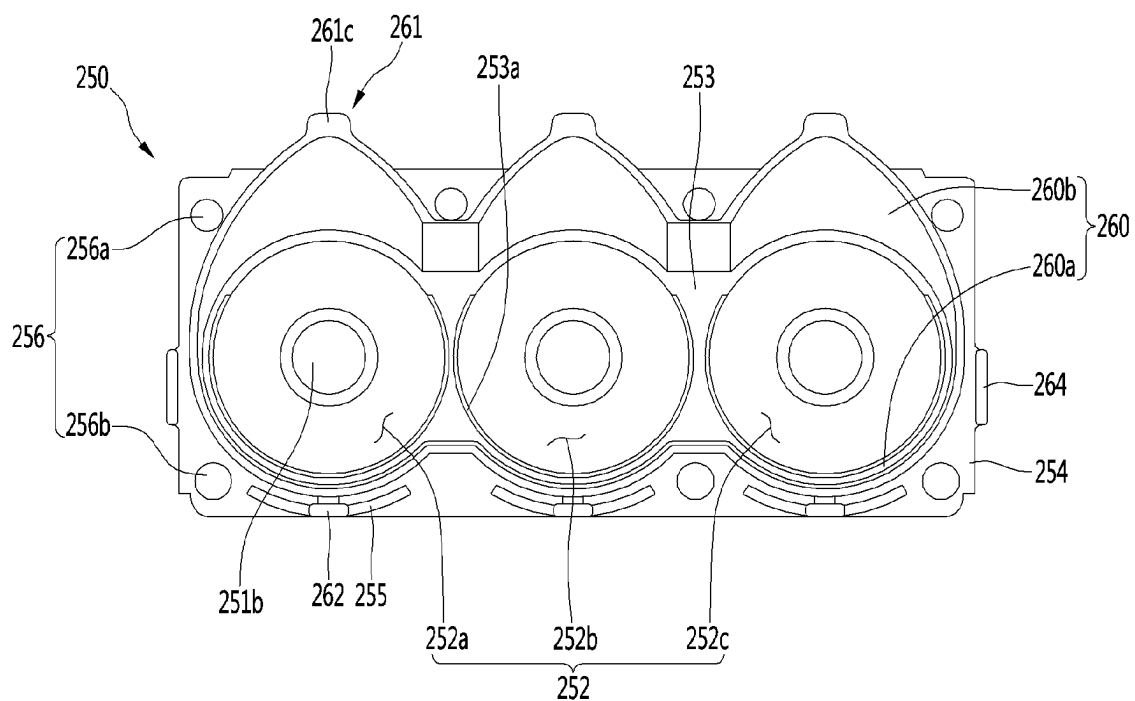


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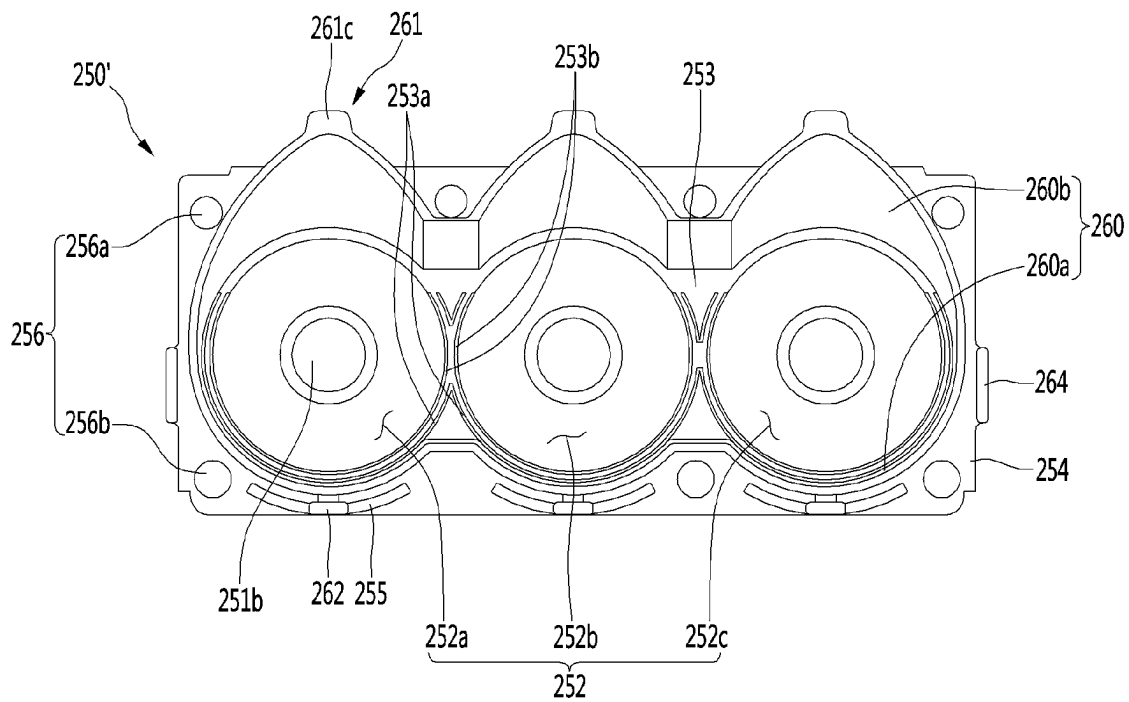


FIG. 38

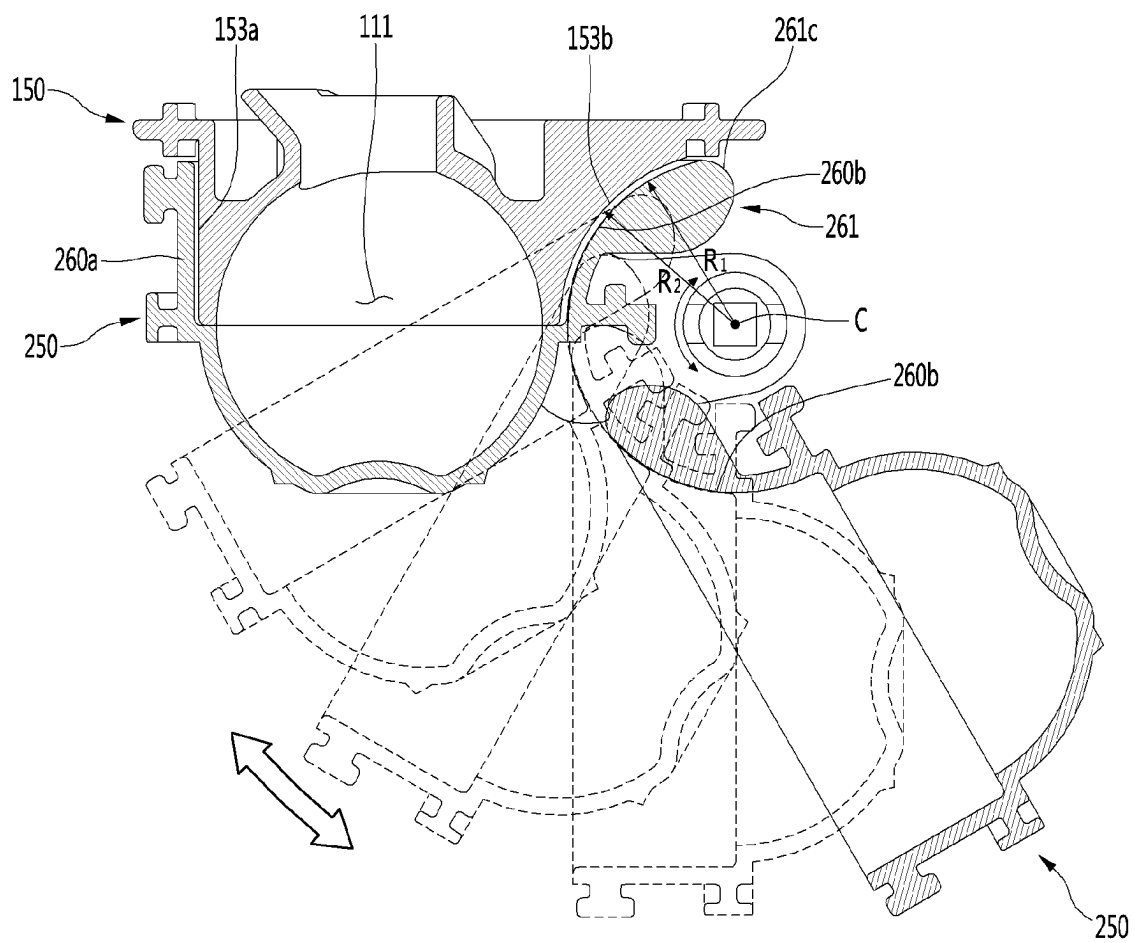


FIG. 39

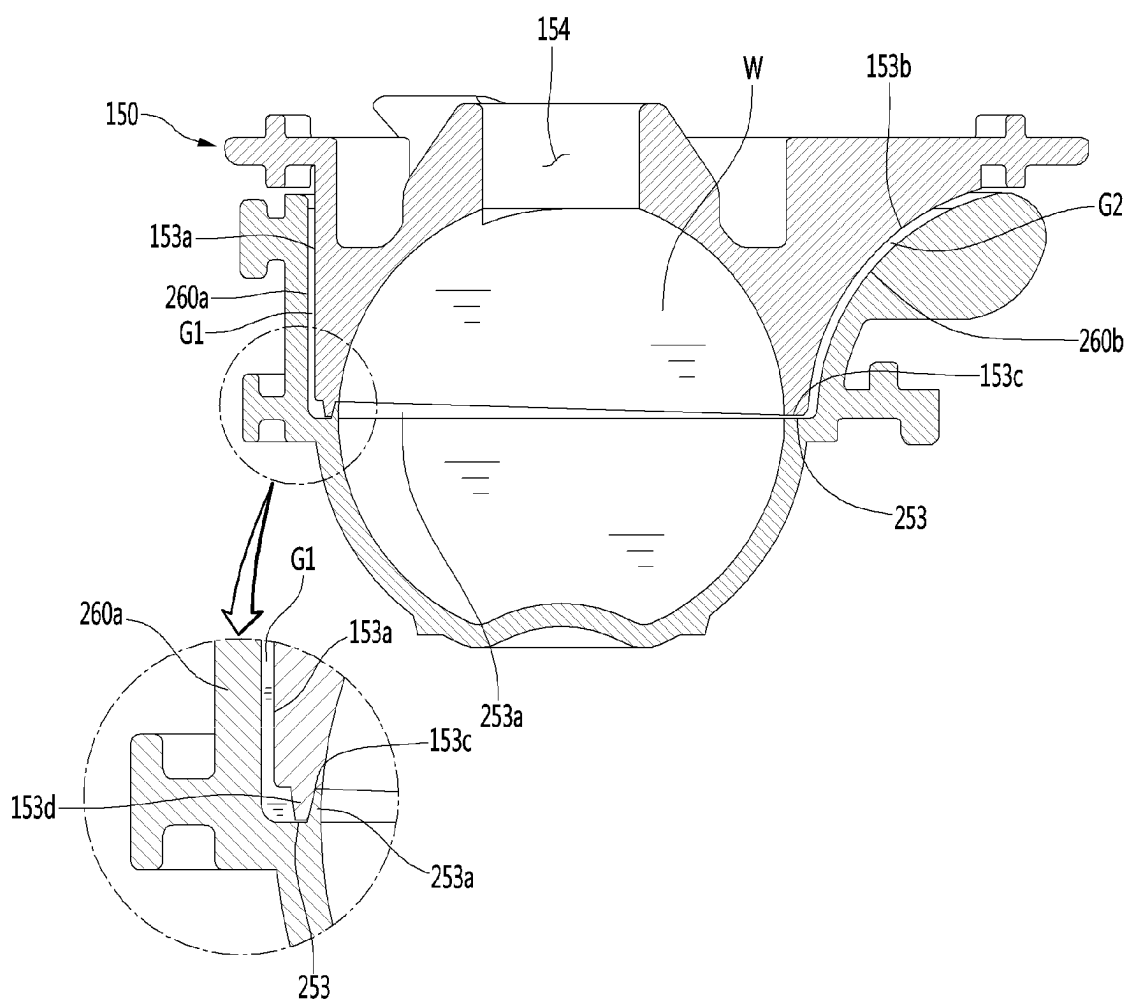


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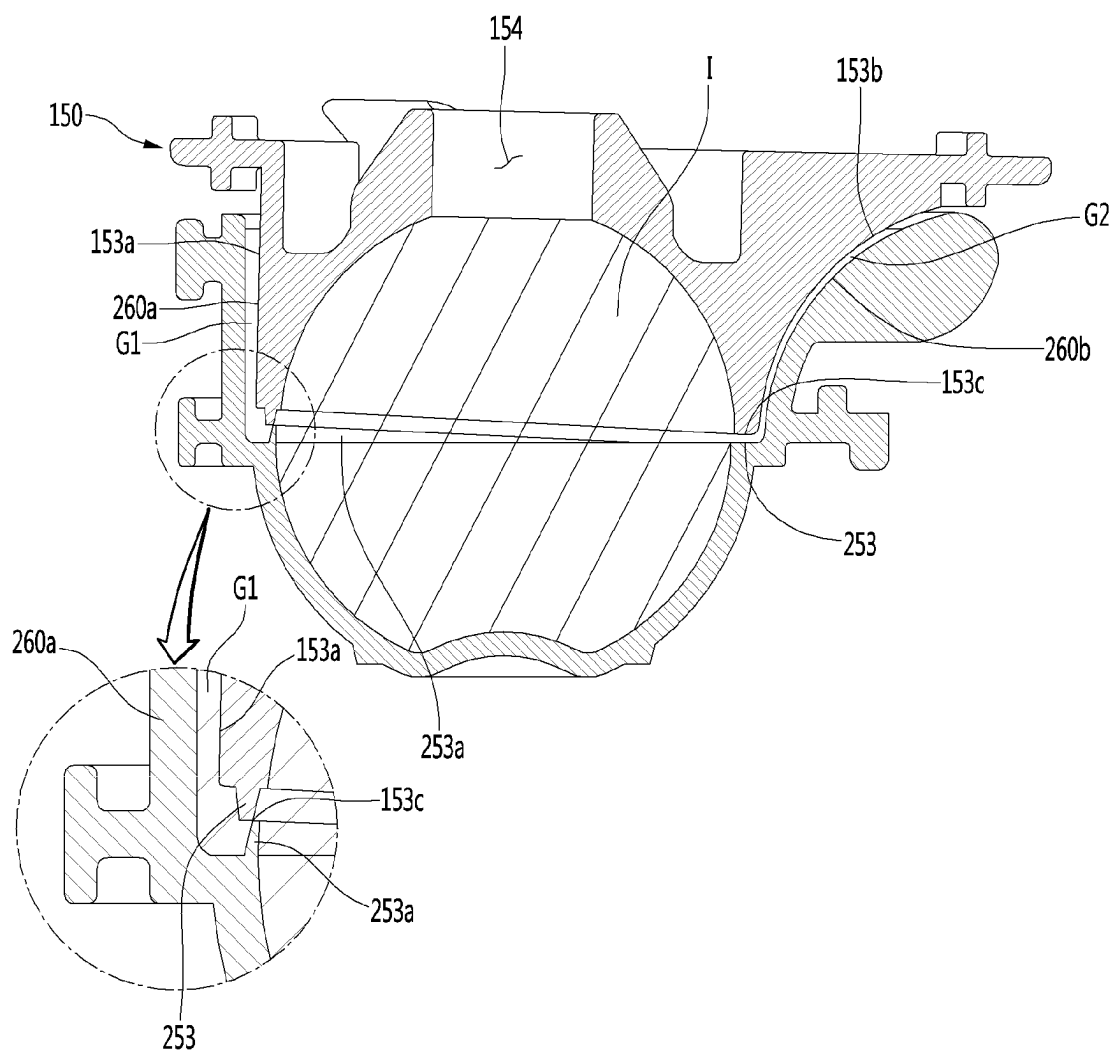


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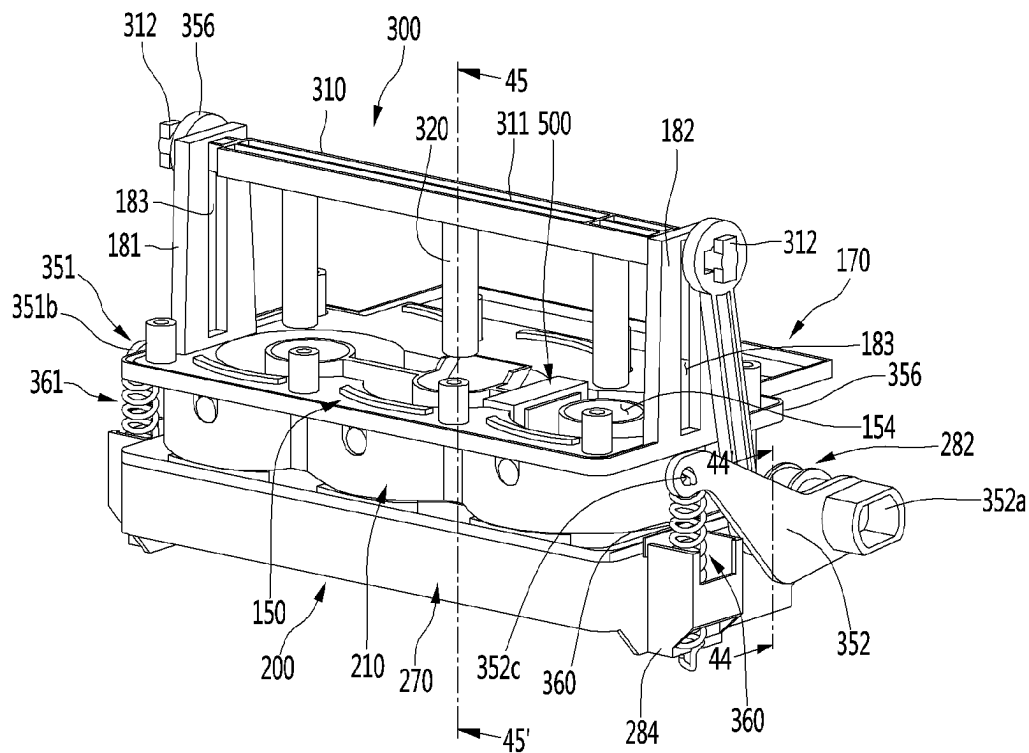


FIG. 42

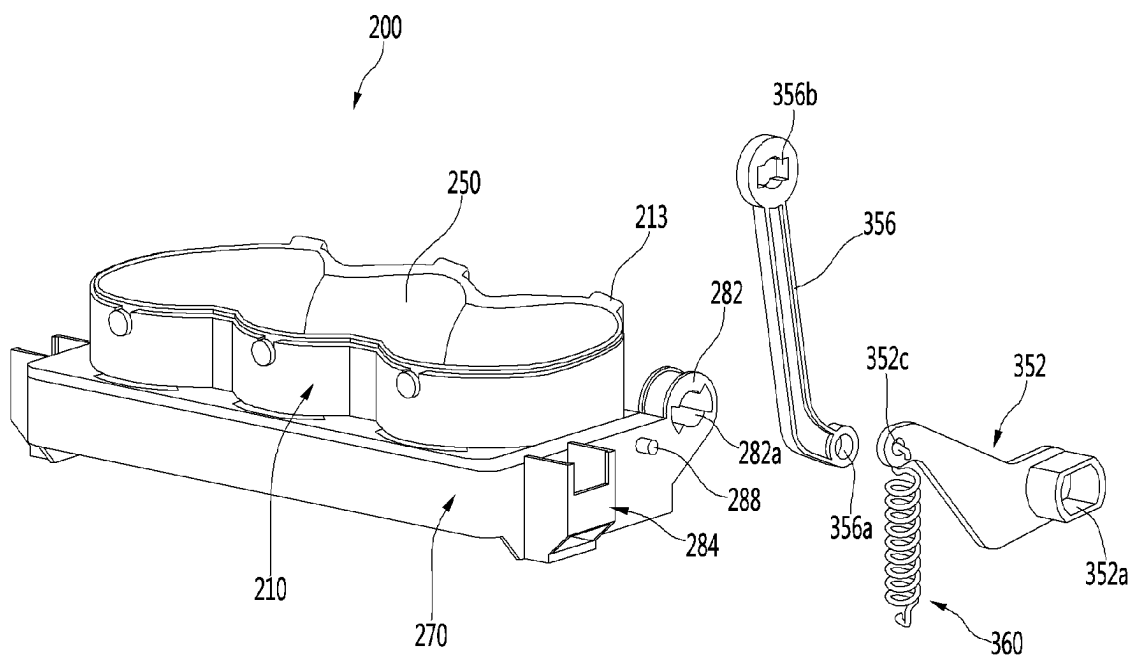


FIG. 43

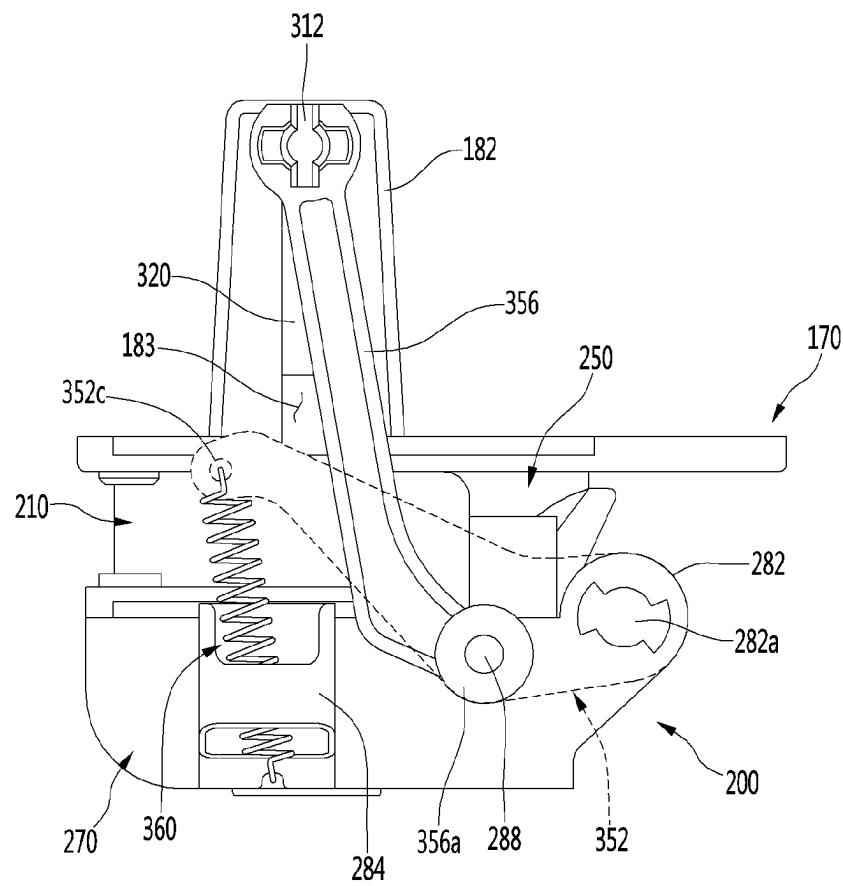


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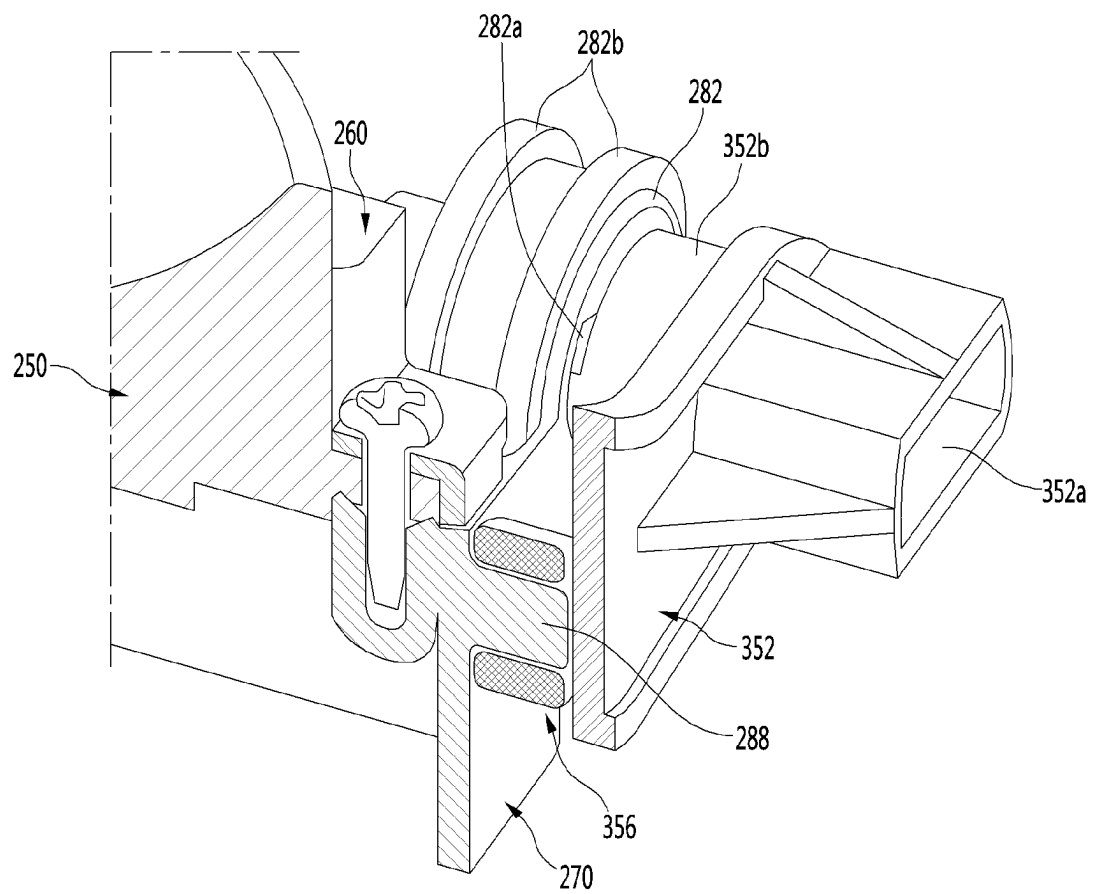


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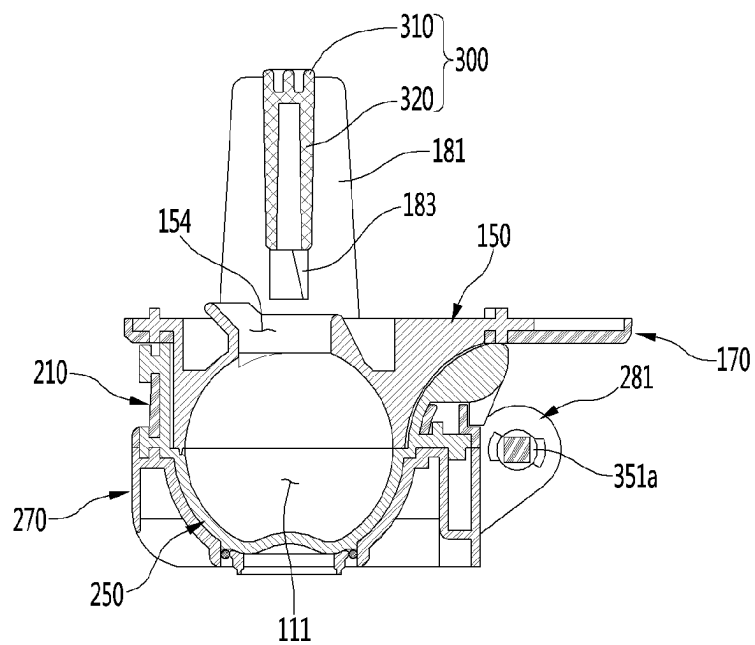


FIG. 46

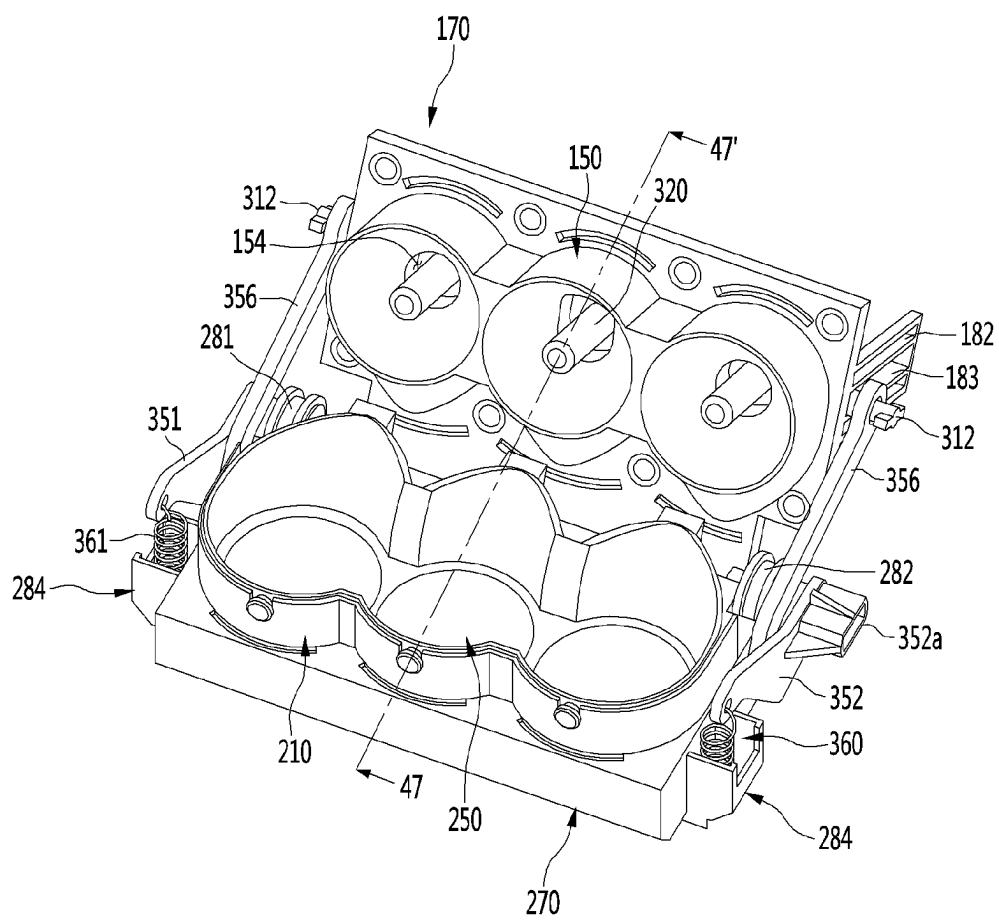


FIG. 47

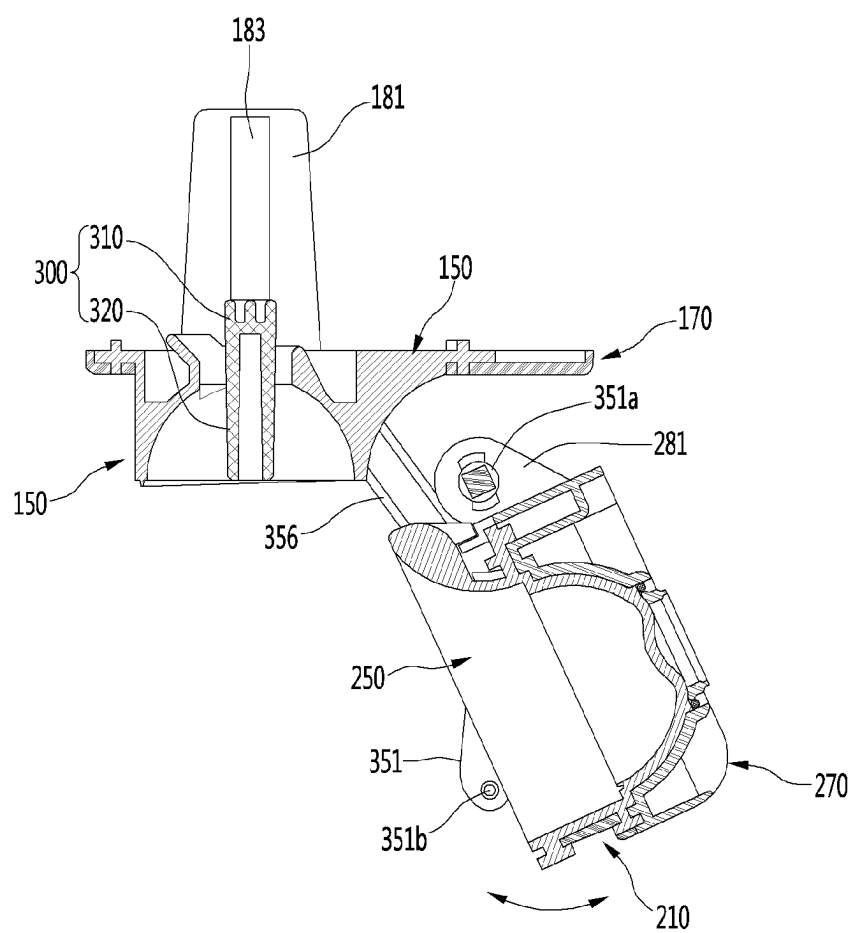


FIG. 48

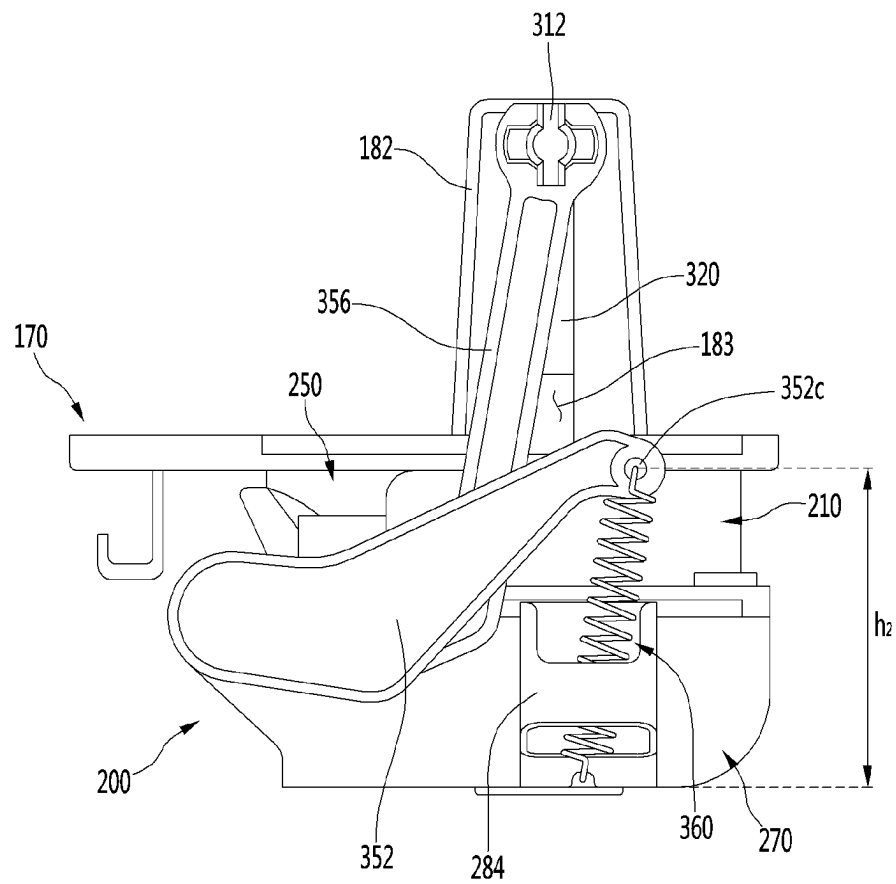


FIG. 49

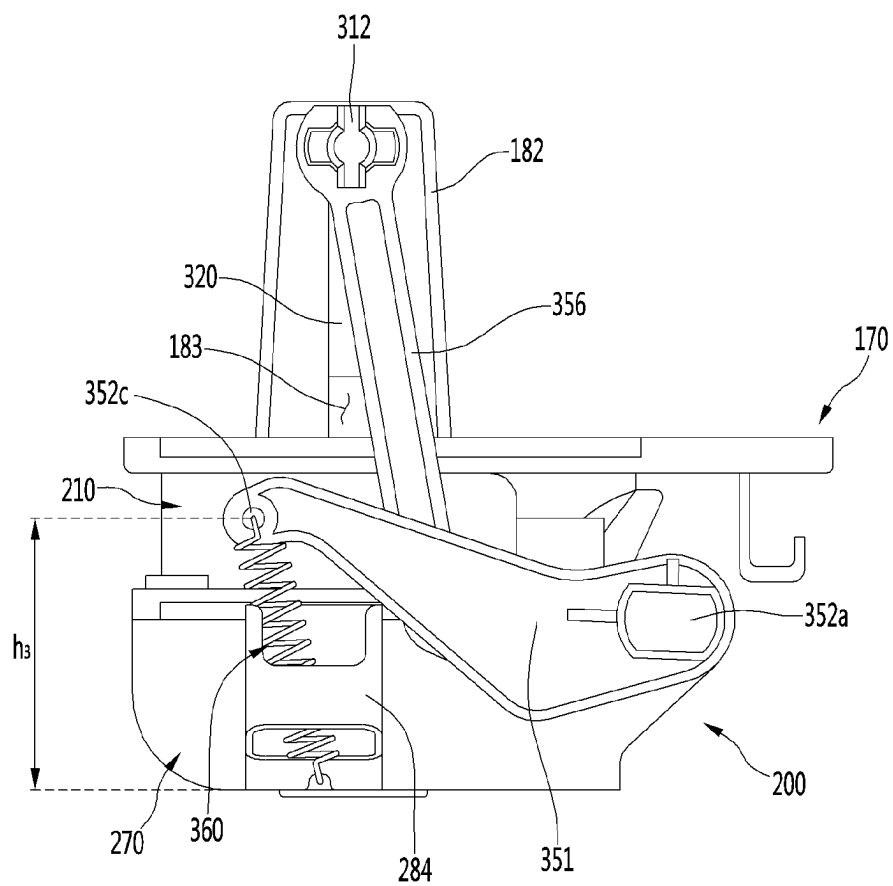


FIG. 50

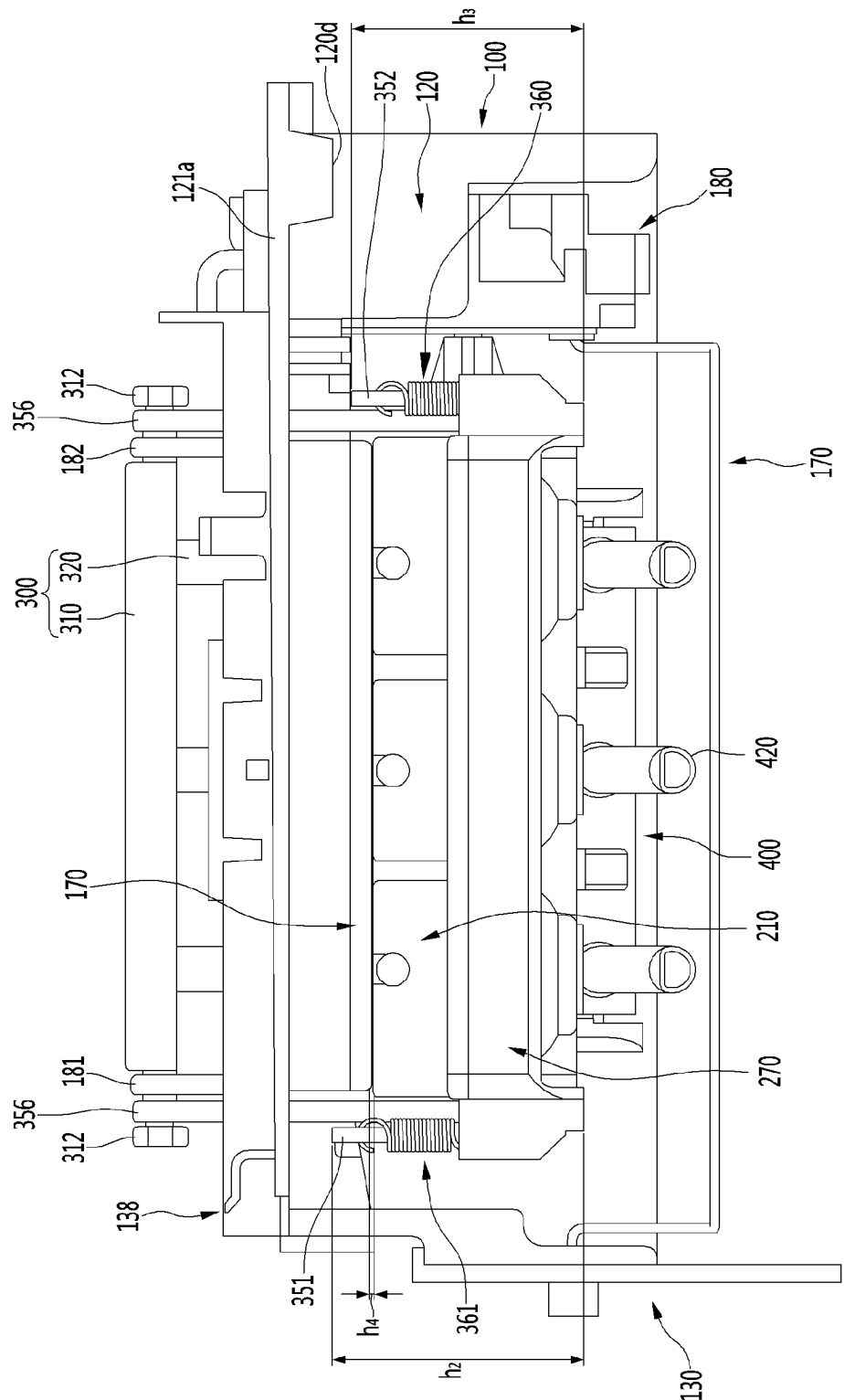


FIG. 51

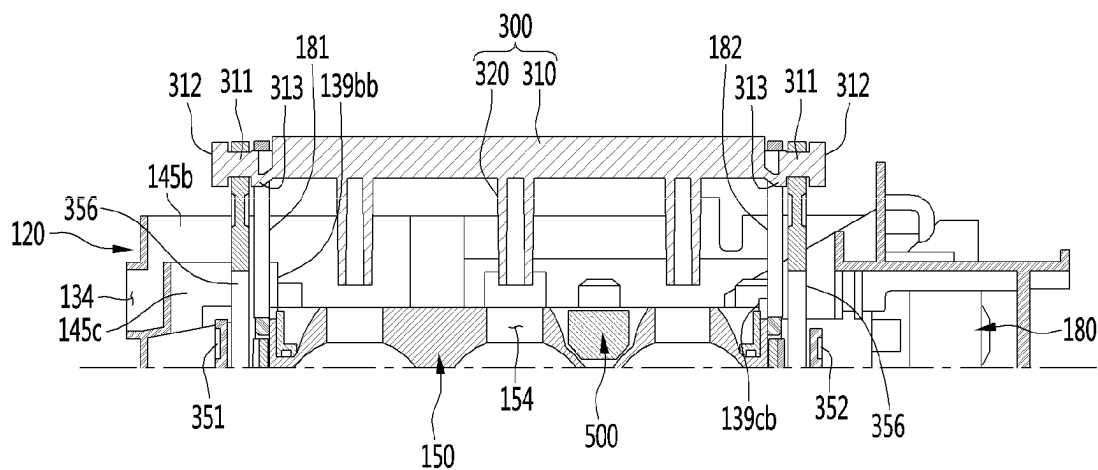


FIG. 52

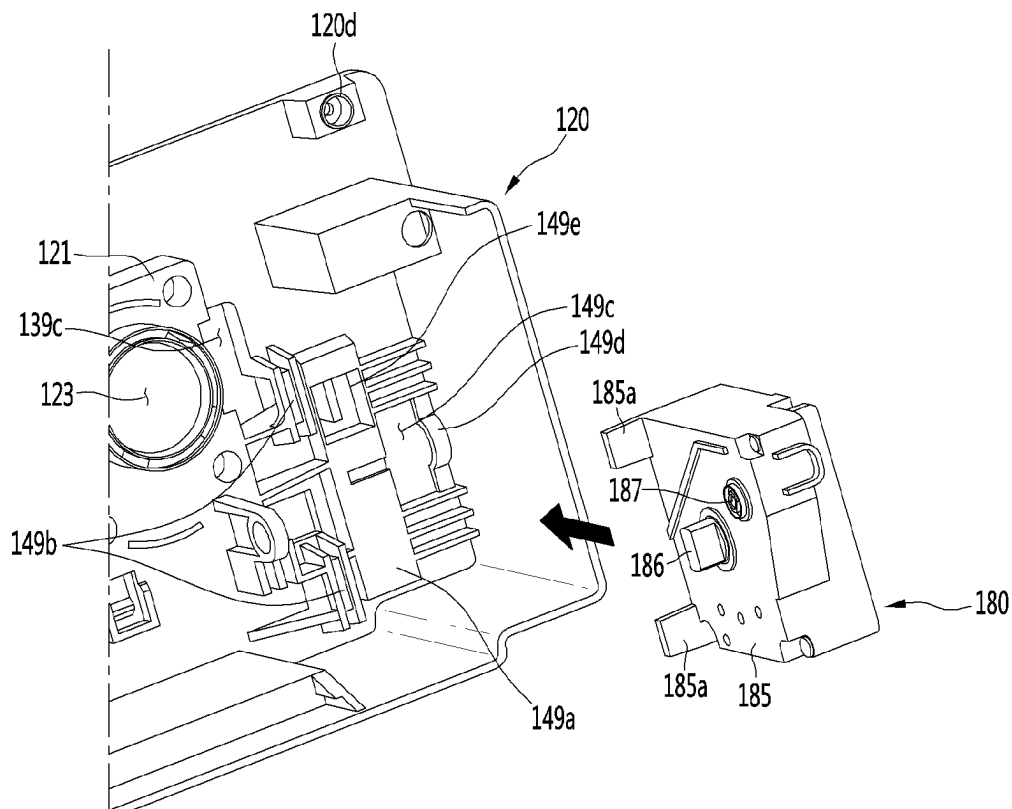


FIG. 53

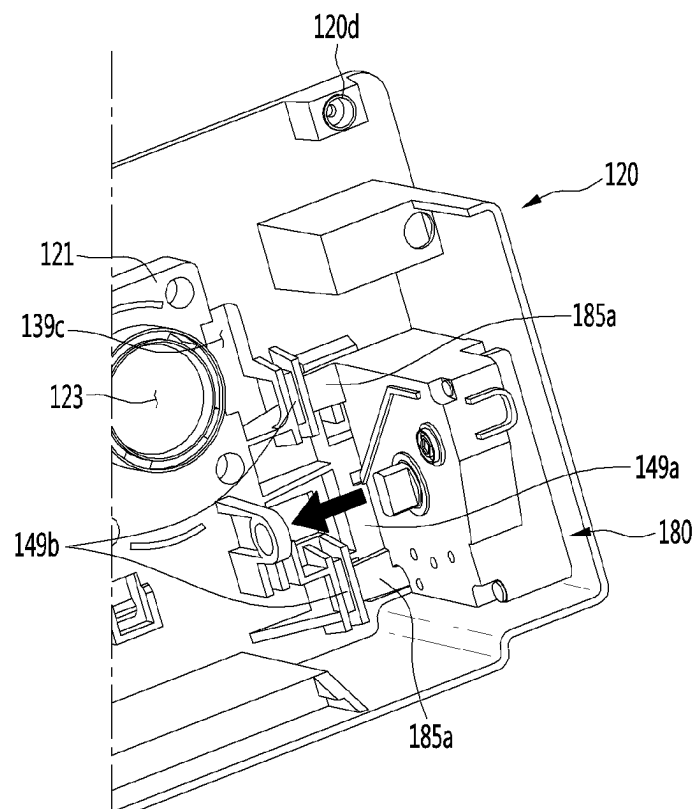


FIG. 54

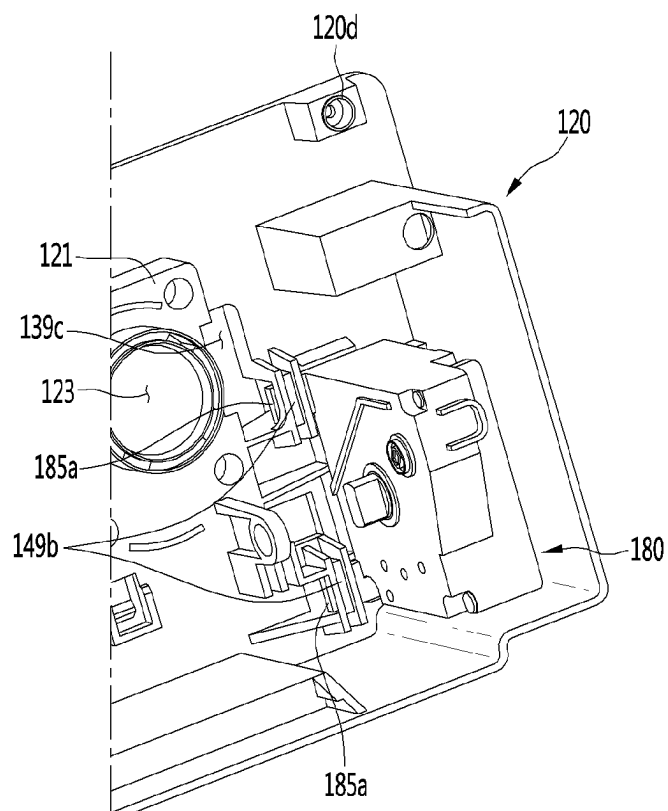


FIG. 55

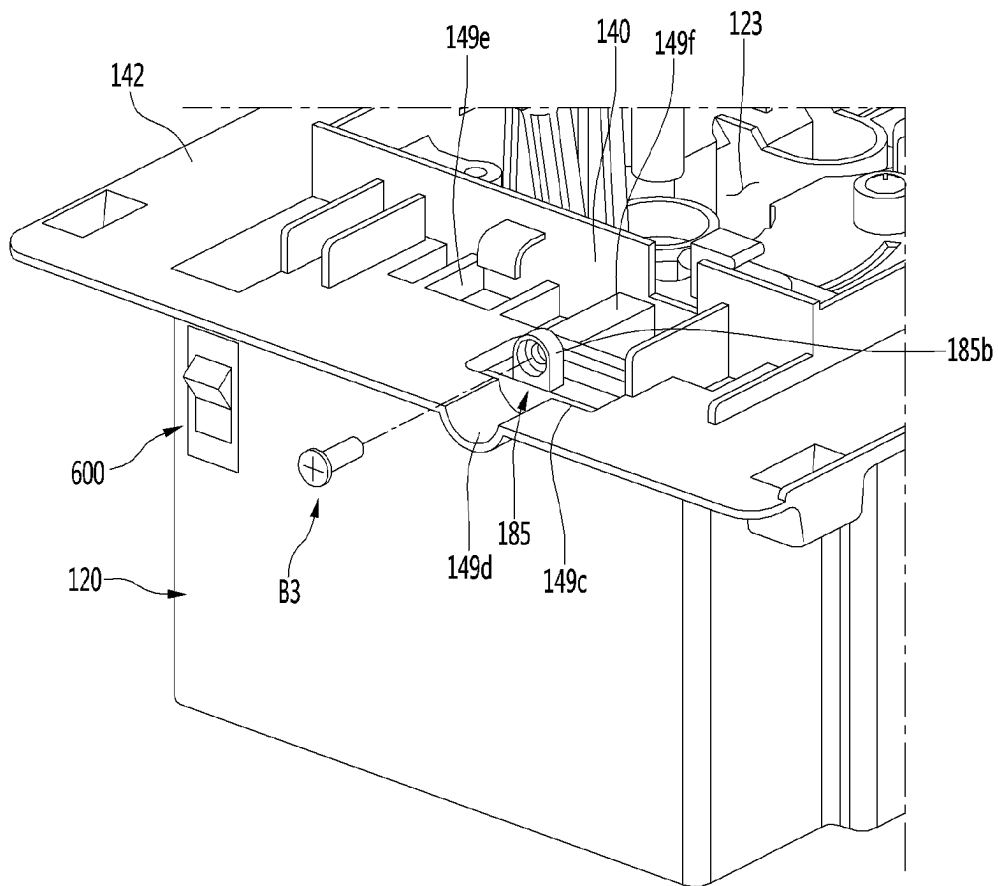


FIG. 56

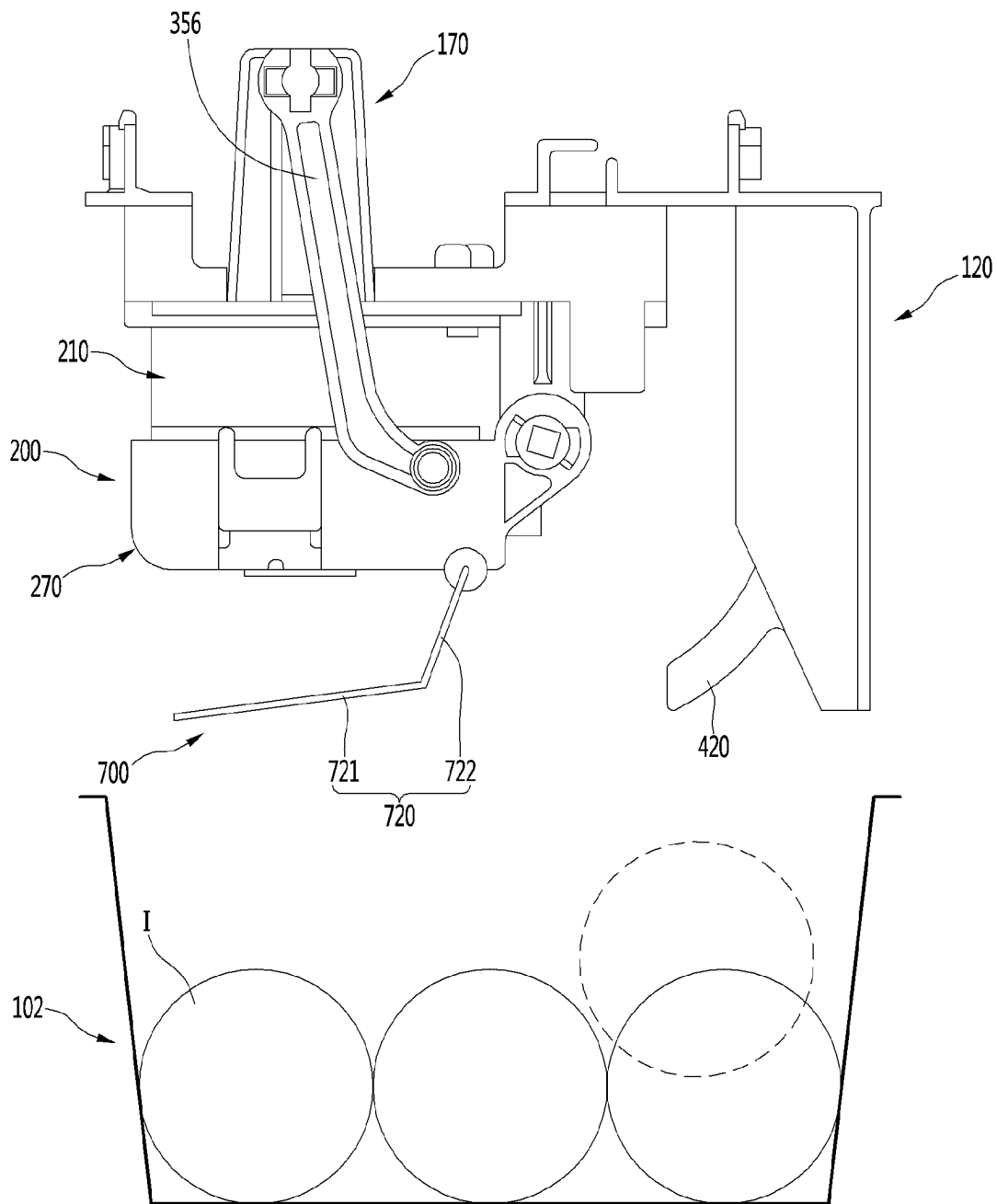


FIG. 57

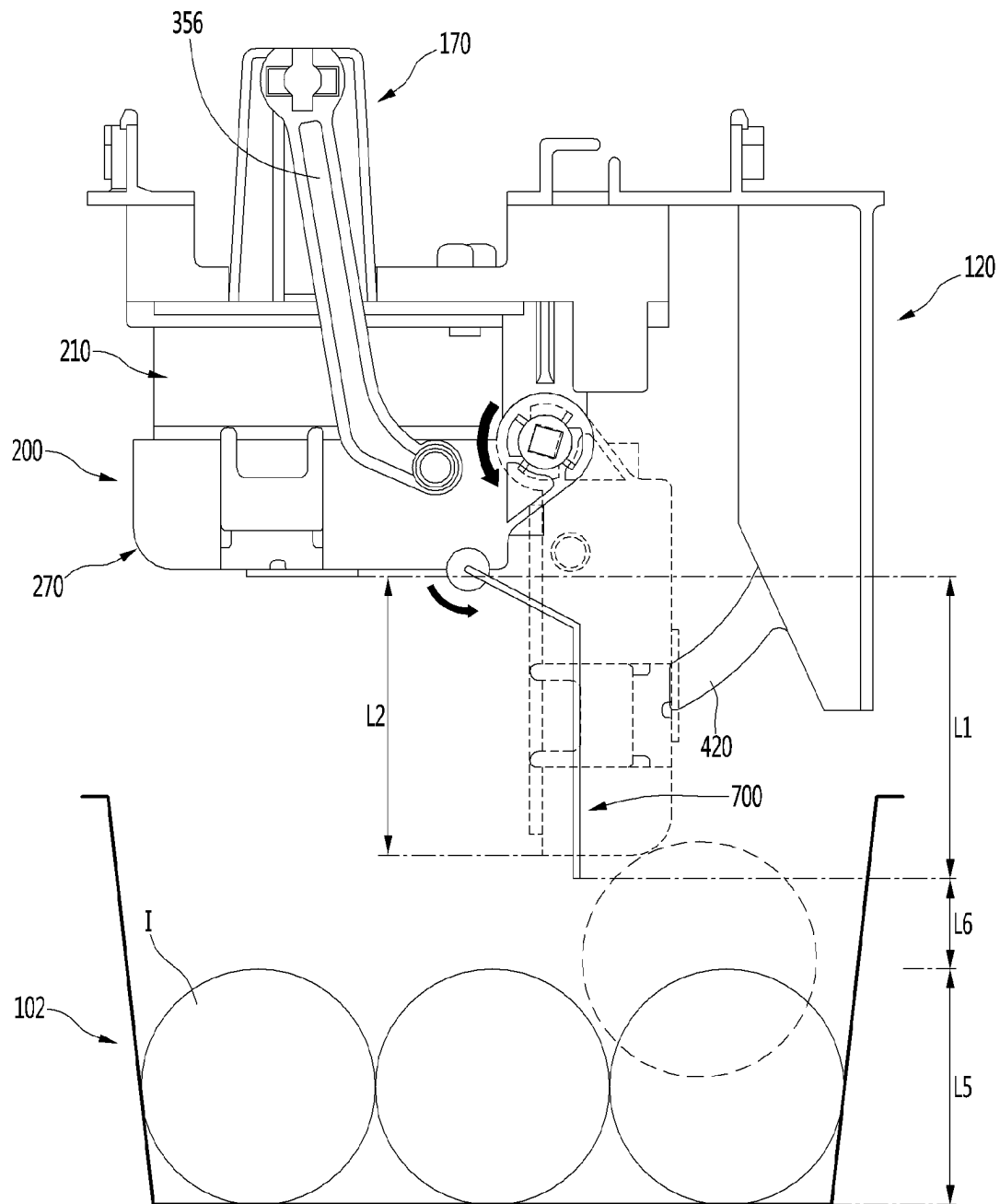


FIG. 58

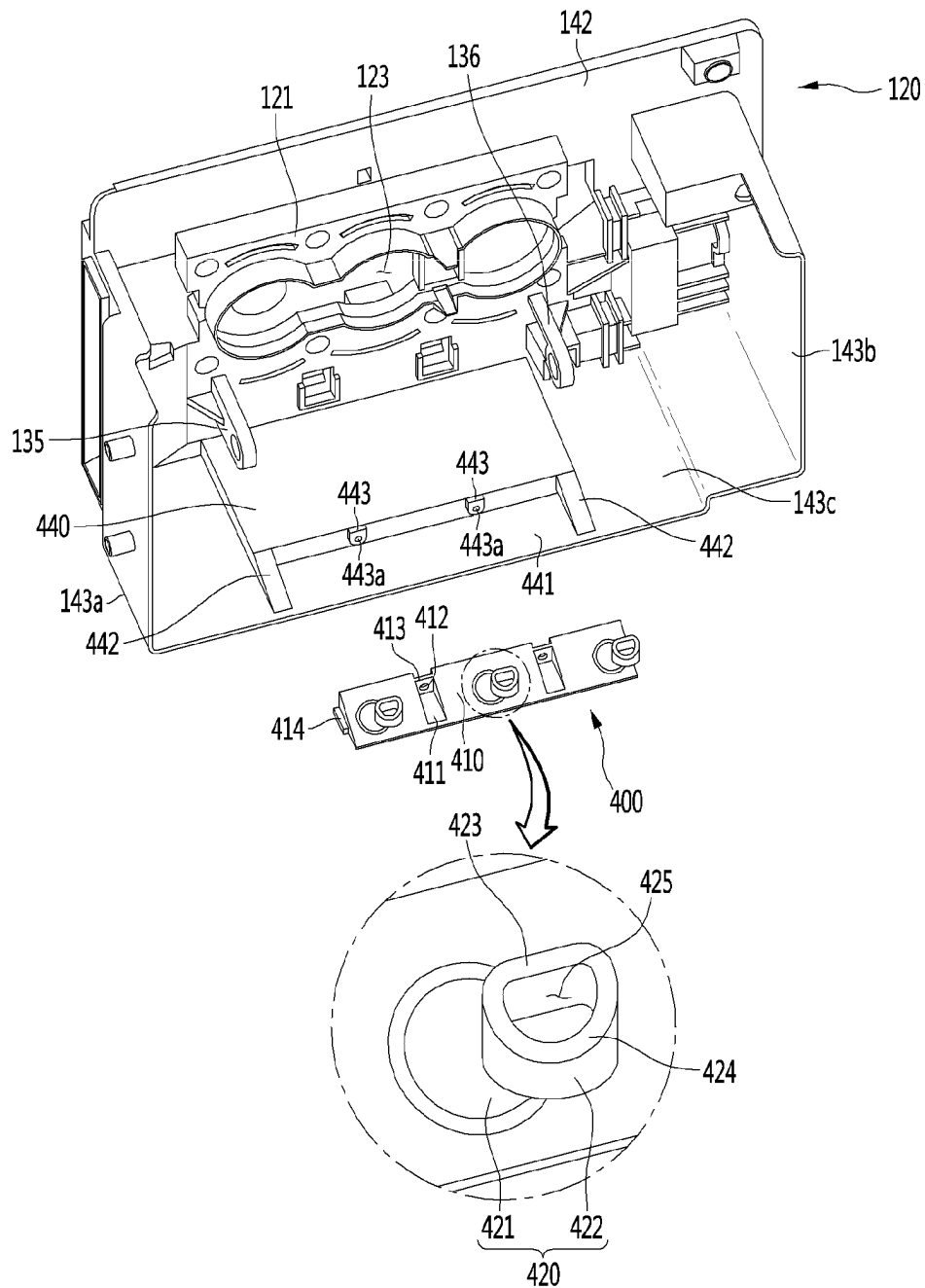


FIG. 59

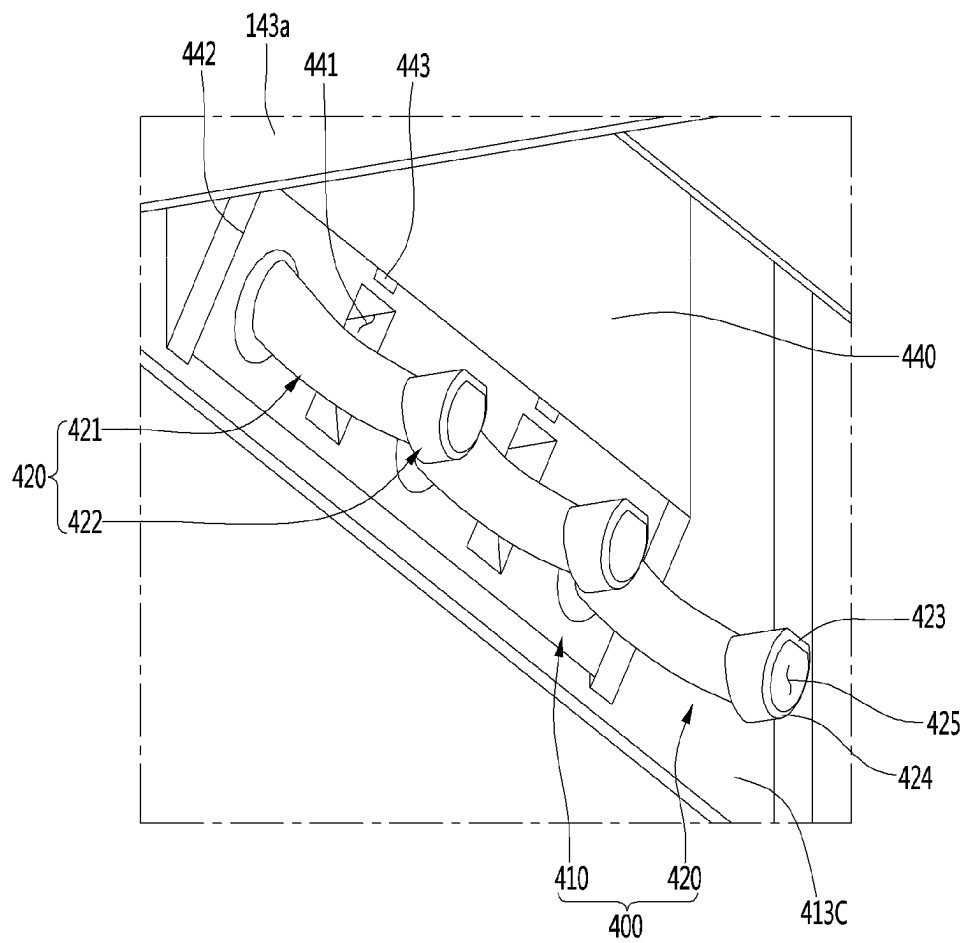


FIG. 60

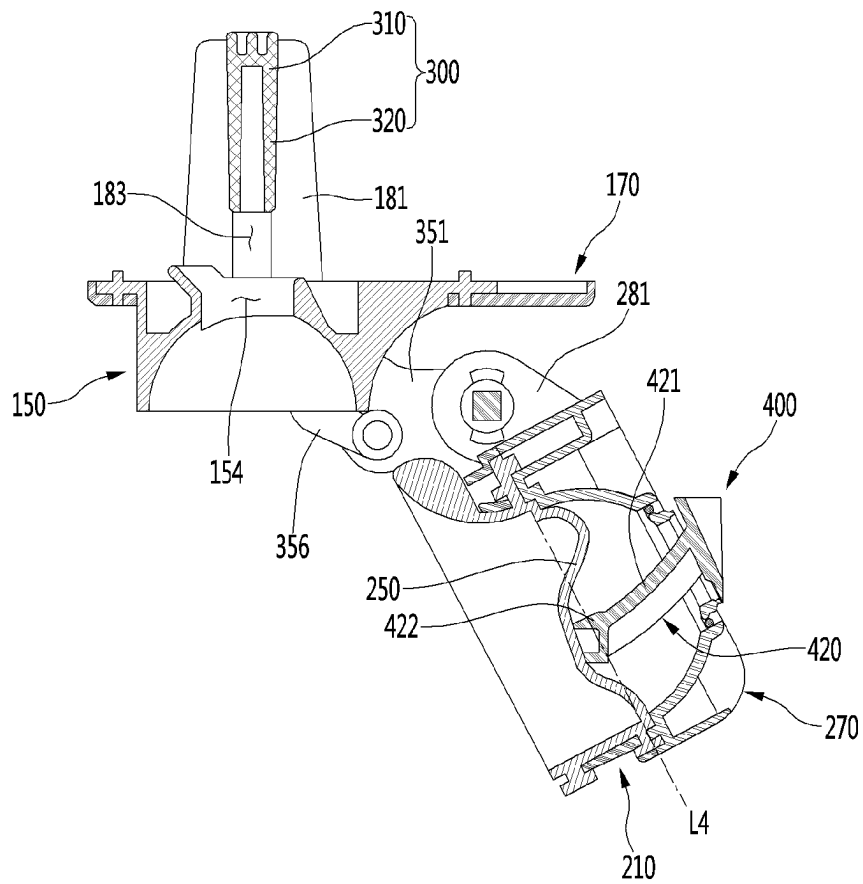


FIG. 61

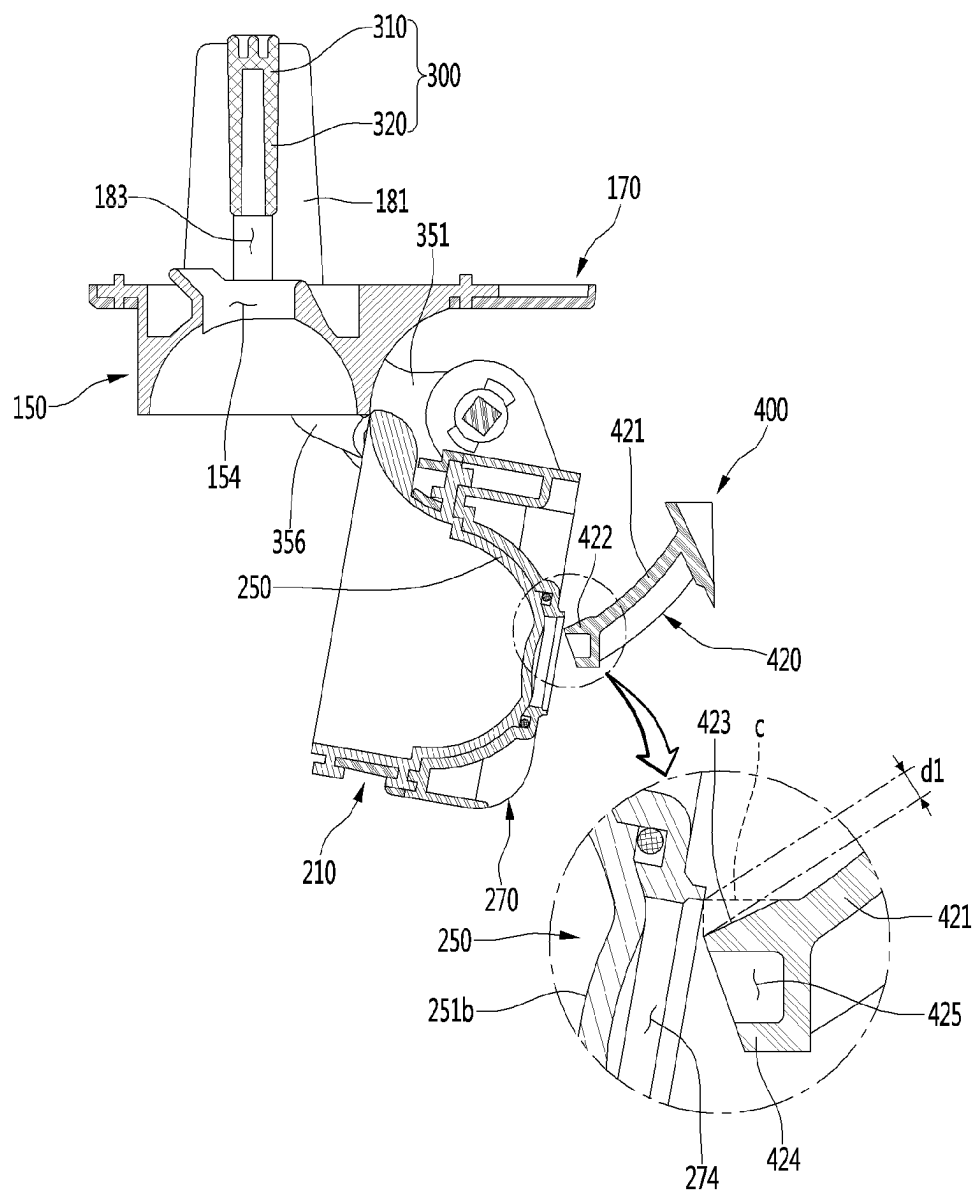


FIG. 62

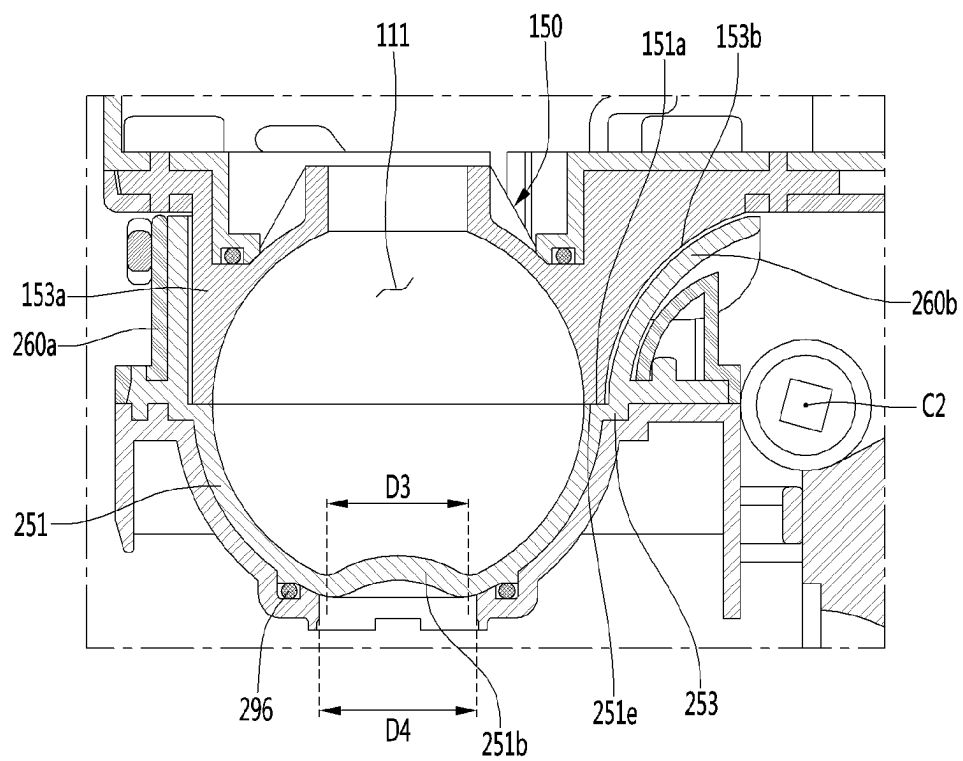


FIG. 63

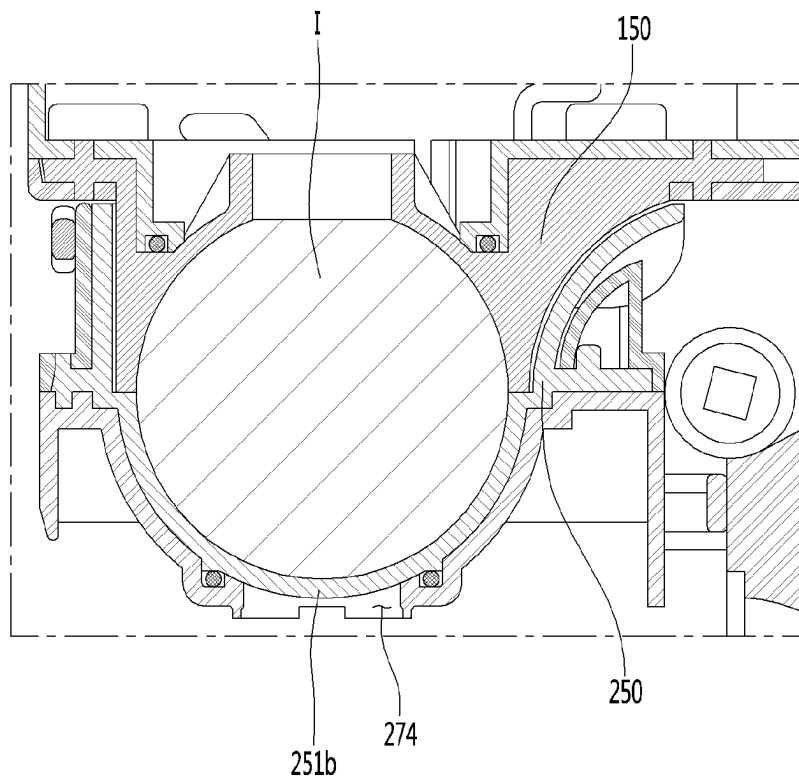


FIG. 64

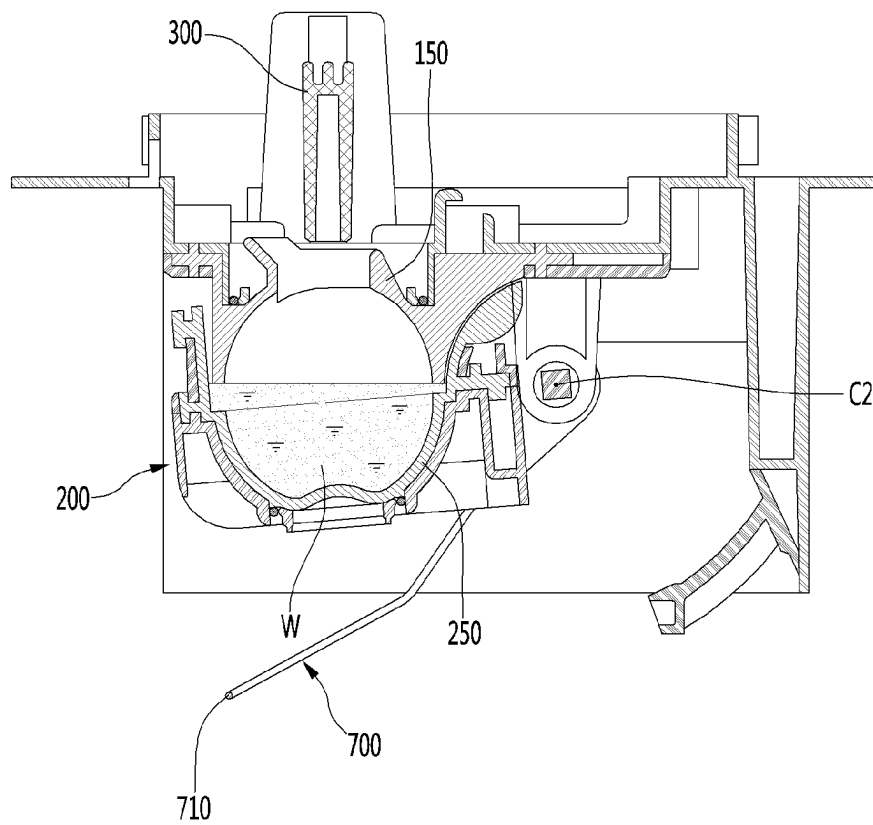


FIG. 65

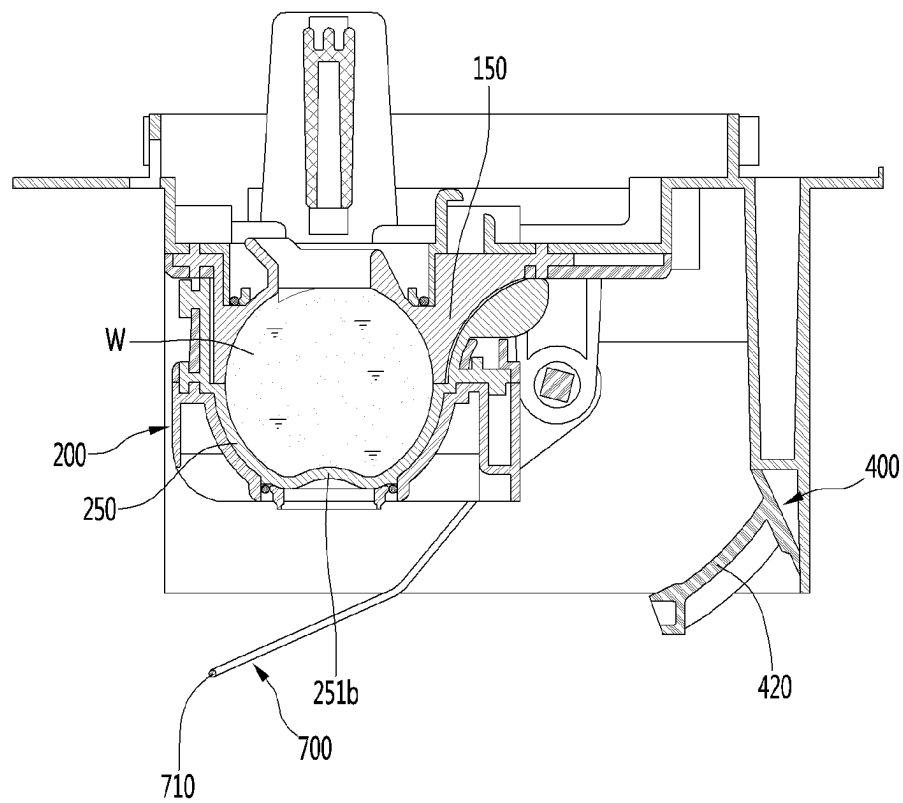


FIG. 66

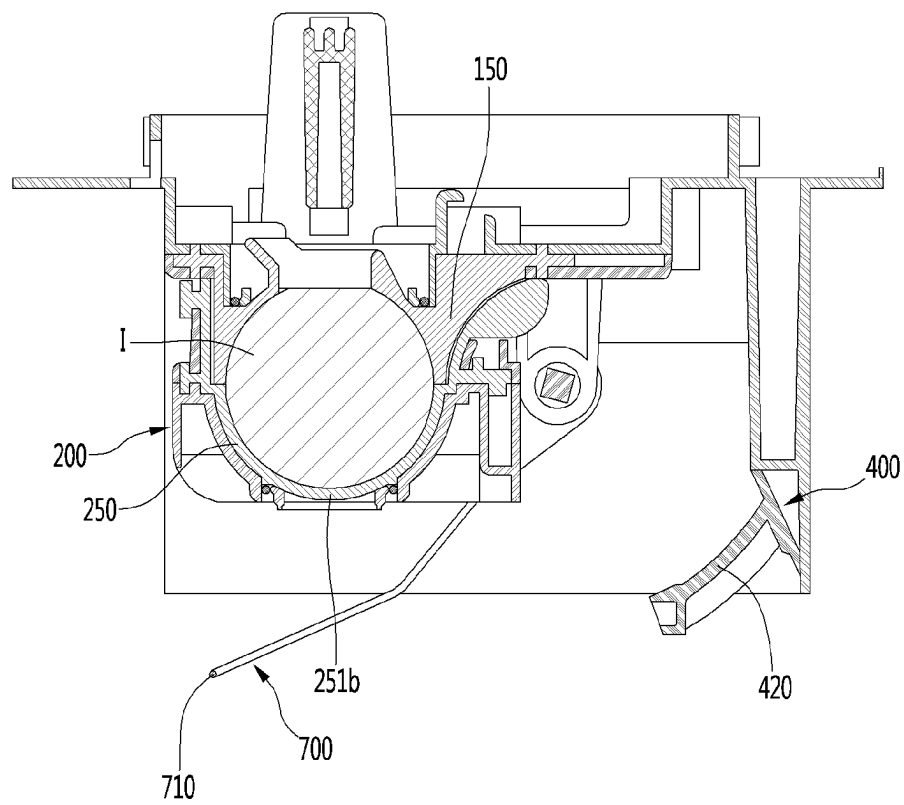


FIG. 67

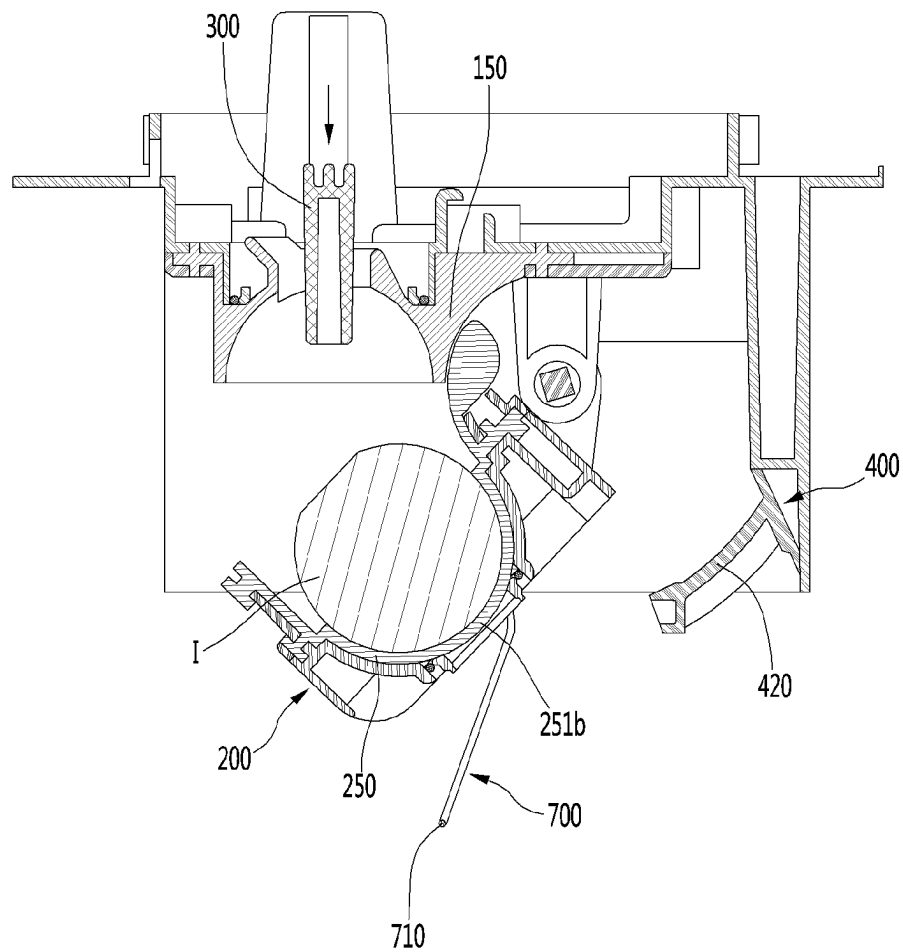


FIG. 68

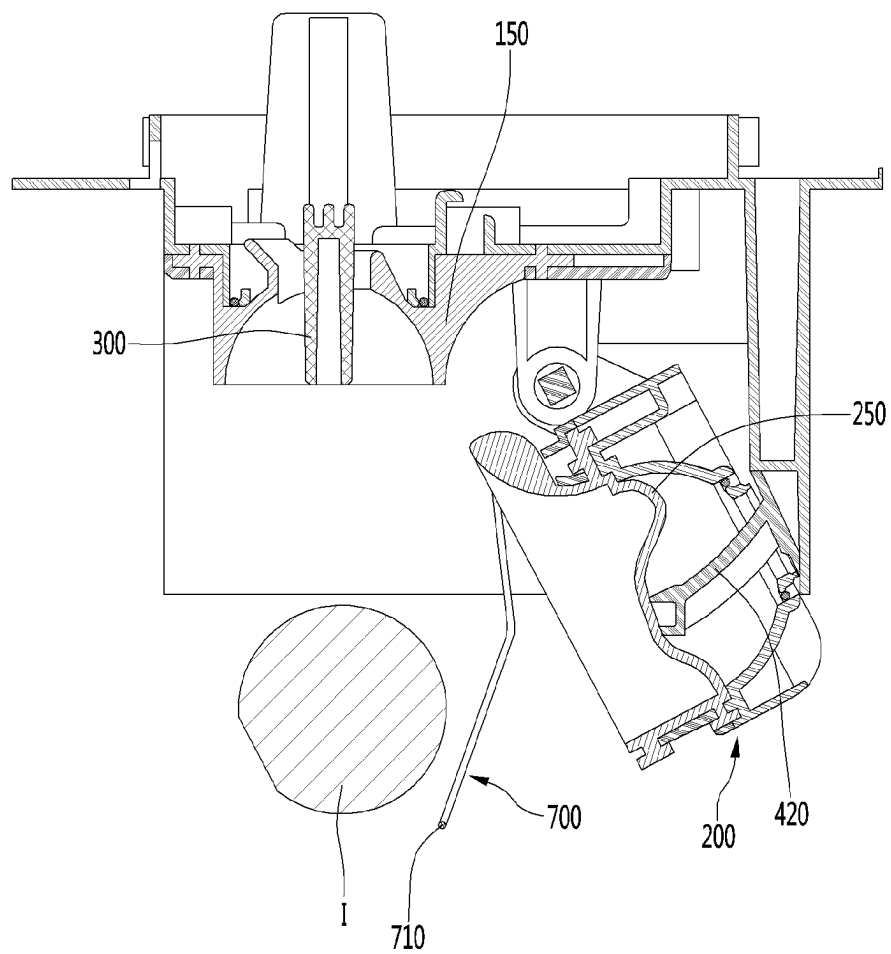


FIG. 69

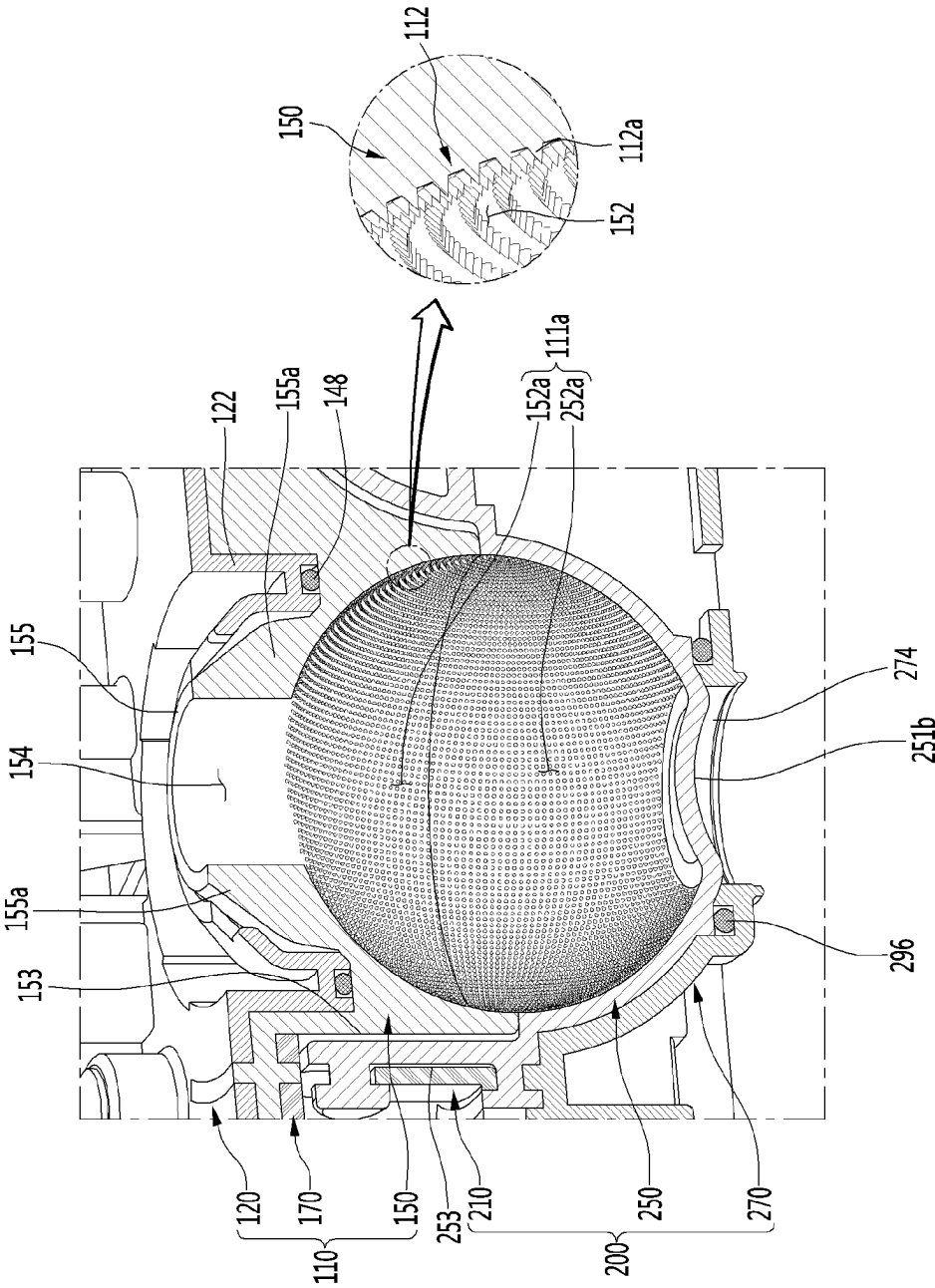


FIG. 70

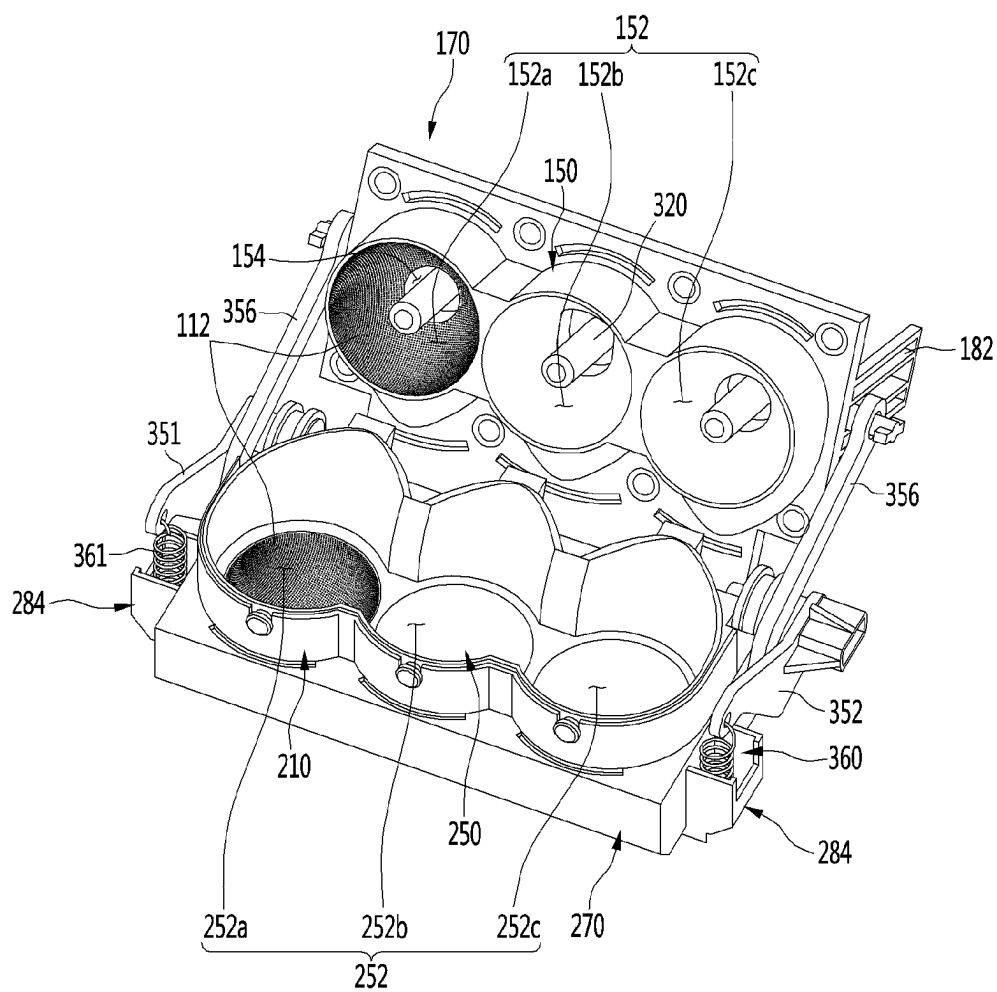


FIG. 71

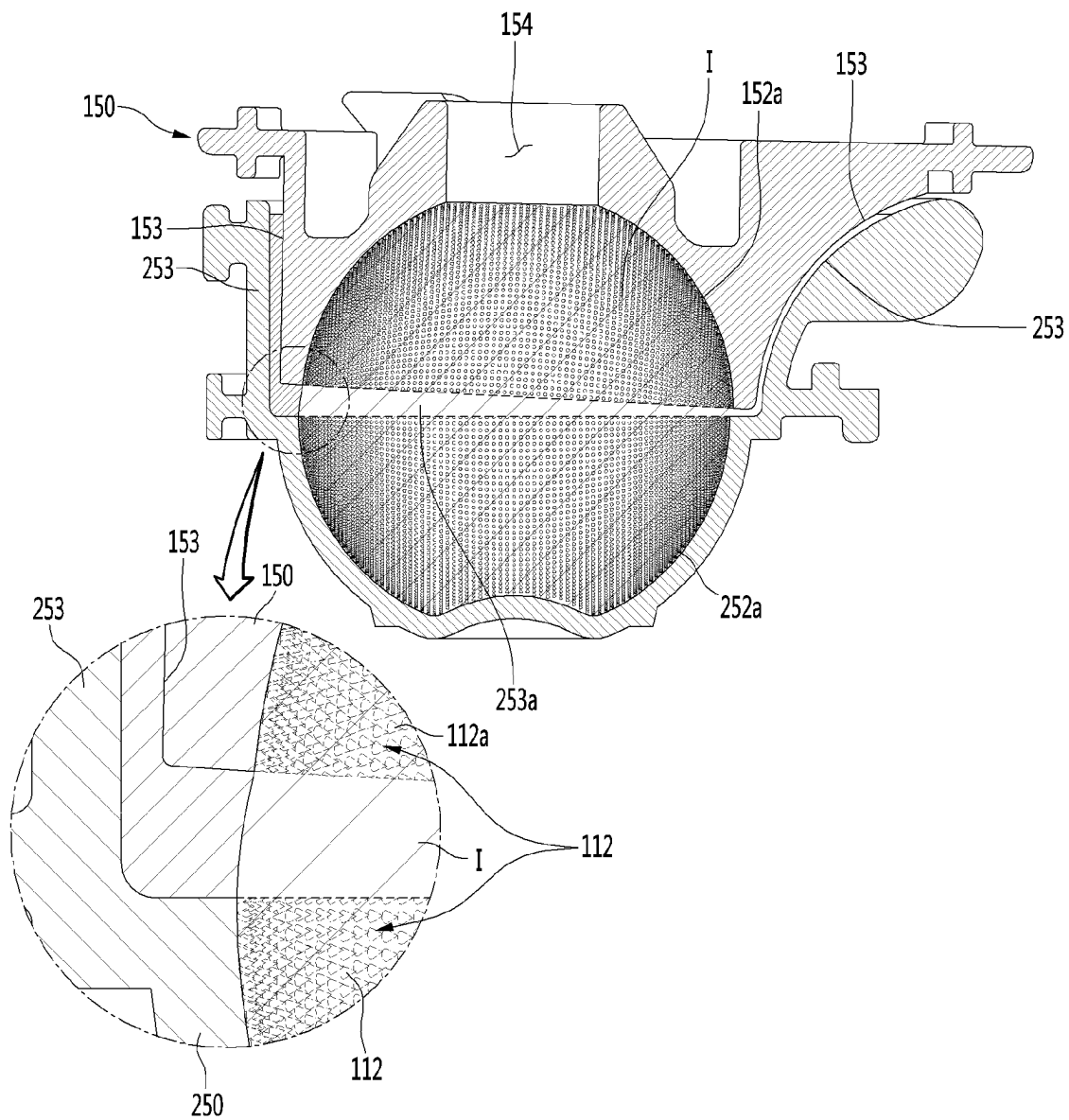
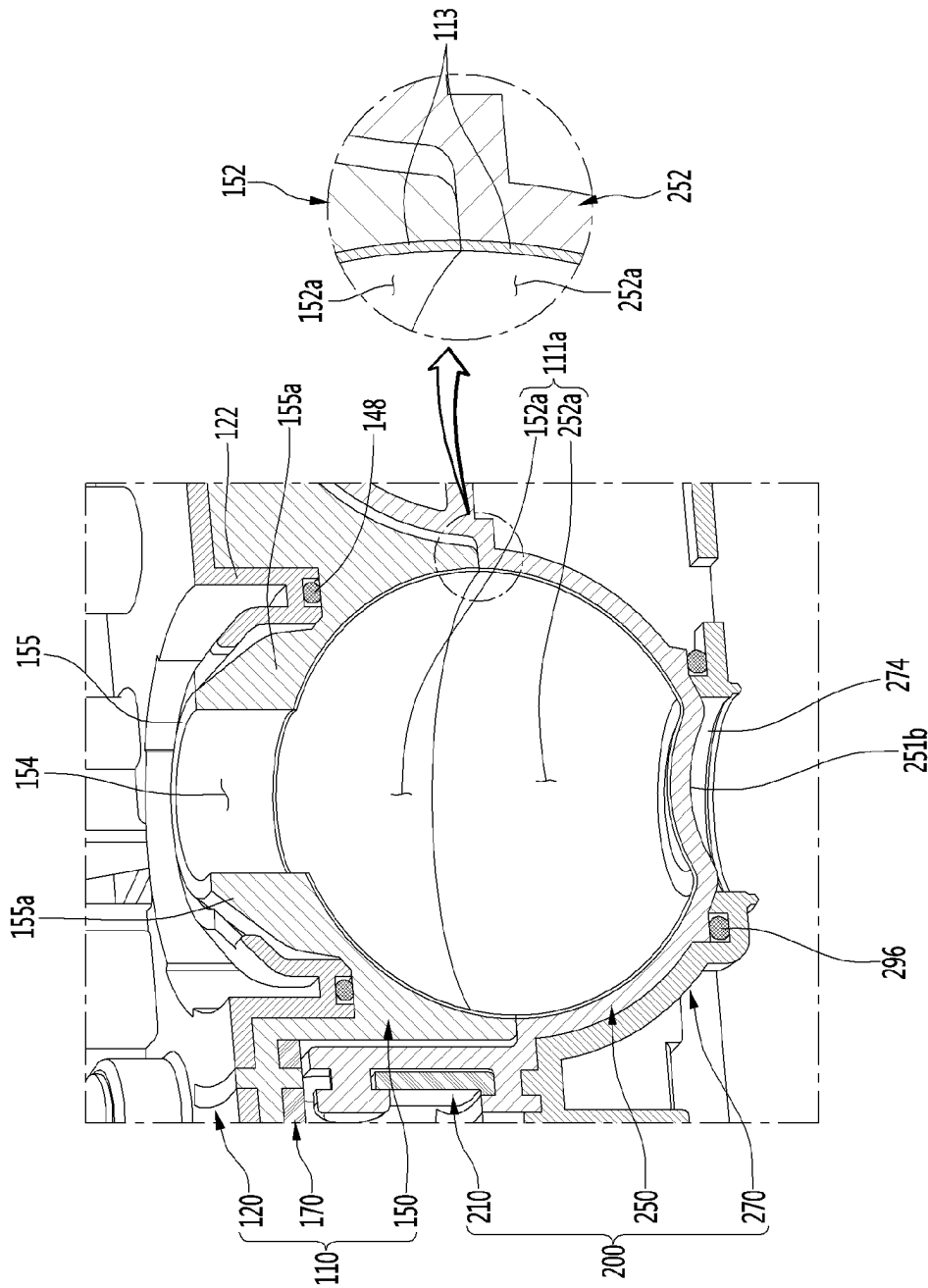


FIG. 72



ICE MAKER AND REFRIGERATOR**CROSS-REFERENCE TO RELATED APPLICATIONS**

This application is a continuation of U.S. application Ser. No. 17/950,626, filed on Sep. 22, 2022, which is a continuation of U.S. application Ser. No. 16/685,793, filed on Nov. 15, 2019, which claims priority under 35 U.S.C. § 119 and 35 U.S.C. § 365 to Korean Patent Application No. 10-2018-0142079, filed in Korea on Nov. 16, 2018, Korean Patent Application No. 10-2019-0081737, filed in Korea on Jul. 6, 2019, and Korean Patent Application No. 10-2019-0110813, filed in Korea on Sep. 6, 2019, whose entire disclosures are hereby incorporated by reference.

BACKGROUND**Field of the Disclosure**

The present disclosure relates to an ice maker and a refrigerator.

Discussion of the Related Art

In general, a refrigerator is a home appliance for storing foods at a low temperature by low temperature air.

The refrigerator uses cold-air to cool inside of a storage space, so that the stored food may be stored in a refrigerated or frozen state.

Typically, an ice maker for making ice is provided inside the refrigerator.

The ice maker is configured to receive water from a water source or a water tank in a tray to make ice.

Further, the ice maker is configured to remove the ice from the ice tray in a heating or twisting manner after the ice-making is completed.

As such, the ice maker, which automatically receives the water and removes the ice, has an open top to scoop molded ice.

As described above, the ice made in the ice maker having a structure as described above may have at least one flat surface such as crescent or cubic shape.

When the ice has a spherical shape, it is more convenient to ice the ice, and also, it is possible to provide different feeling of use to a user. Also, even when the made ice is stored, a contact area between the ice cubes may be minimized to minimize a mat of the ice cubes.

Korean Patent Registration No. 10-1850918 as Prior Art document discloses an ice maker.

The ice maker of Prior Art document includes an upper tray in which a plurality of upper cells of a hemispherical shape are arranged and a pair of link guides extending upwardly from both sides are disposed, a lower tray in which a plurality of lower cells of a hemispherical shape are arranged and which is pivotally connected to the upper tray, a pivoting shaft connected to rear ends of the lower tray and the upper tray to allow the lower tray to pivot relative to the upper tray, a pair of links having one end thereof connected to the lower tray and the other end thereof connected to the link guide, and an ejecting pin assembly having both ends thereof respectively connected to the pair of links while being respectively inserted into the link guides, wherein the ejecting pin assembly ascends and descends together with the link.

In the Prior Art document, there is formed a guide sleeve extending upward from each cell of the upper tray, but the

guide sleeve is to serve only as a guide of the ejecting pin. Therefore, a height of the guide sleeve is excessively large, so that flow of the cold-air may be inhibited. In addition, when a diameter of the guide sleeve is excessively large, the planar shape of the ice made in the cell becomes excessively large, and the ice does not appear as spherical ice.

Further, when the movement of the ejecting pin is generated, the ejecting pin may be caught to the upper tray and the guide sleeve of the metal material, and then broken.

SUMMARY OF THE DISCLOSURE

A purpose of an embodiment of the present disclosure is to provide an ice maker and a refrigerator that may prevent water from overflowing even when an amount of water supplied into a cell changes.

Another purpose of an embodiment of the present disclosure is to provide an ice maker and a refrigerator that prevent water from overflowing even when an amount of water supplied into a cell changes after water supply.

Another purpose of an embodiment of the present disclosure is to provide an ice maker and a refrigerator that smooth flow of cold-air passing through an upper tray to improve an ice making performance.

Another purpose of an embodiment of the present disclosure is to provide an ice maker and a refrigerator that prevent damage and deformation of an ejector and an upper tray even when the ejector and the upper tray interfere with each other.

Another purpose of an embodiment of the present disclosure is to provide an ice maker that prevents formation of a protrusion on one side of ice during ice-making of spherical ice.

Another purpose of an embodiment of the present disclosure is to provide an ice maker in which friction between a surface of ice made during spherical ice-making and an inner face of a tray is increased.

In a first aspect of the present disclosure, there is provided an ice maker including an upper tray made of an elastic material, wherein a plurality of hemispherical upper chambers are defined in the upper tray, each ejector-receiving opening opened to be in communication with each of the plurality of upper chambers, each opening-defining wall extending upward along a circumference of each ejector-receiving opening, an upper ejector disposed above the upper tray, wherein the upper ejector is configured to pass through the ejector-defining wall and vertically move to remove each ice from each ice chamber, a lower tray made of an elastic material, and having a plurality of lower chambers defined therein in contact with the plurality of upper chambers by pivoting to define a plurality of spherical ice chambers, respectively, and a driver for pivoting the lower tray to open and close the upper tray and the lower tray, wherein the upper tray includes a horizontal extension forming a top face of the upper tray, wherein the upper chamber is positioned below the horizontal extension such that the opening-defining wall is formed at a vertical level equal to or lower than a vertical level of the opening-defining wall.

In one embodiment, a volume of a space inside the opening-defining wall may be within 2 to 4% of a total volume of the ice chamber.

In one embodiment, the ice maker may further include an upper casing fixedly mounted on the upper tray such that an upper portion of the upper tray is exposed, and wherein a cold-air hole for supplying cold-air to the top face of the upper tray may be defined in the upper casing.

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In one embodiment, a cold-air discharge hole for discharging the cold-air may be defined at one side of the upper tray spaced apart from the cold-air hole, and the plurality of ice chambers may be arranged in line between the cold-air hole and the cold-air discharge hole.

In one embodiment, a cold-air guide for guiding the supplied cold-air to the top face of the upper tray may be formed on the upper casing.

In one embodiment, a rib for connecting an outer face of the opening-defining wall with a top face of the upper chamber may be formed in the upper tray.

In one embodiment, the rib may further protrude outwards downwardly.

In one embodiment, the rib may include a plurality of ribs formed along a circumference of the opening-defining wall.

In one embodiment, a rib extending for connecting outer portions of neighboring opening-defining walls with each other may be formed between the neighboring opening-defining walls.

In one embodiment, a water-supply guide extending obliquely to guide water into the upper chamber may be formed on one of the opening-defining walls respectively formed on the plurality of upper chambers.

In one embodiment, a receiving portion recessed downward may be defined in the horizontal extension, and the plurality of upper chambers may be sequentially arranged inside the receiving portion.

In a second aspect of the present disclosure, there is provided a refrigerator including a cabinet having a refrigerating compartment and a freezing compartment defined therein, wherein the refrigerating compartment and the freezing compartment are divided vertically and an ice maker disposed in the freezing compartment to make ice, wherein the ice maker includes an upper casing mounted on a top face of the freezing compartment and having a cold-air hole defined therein, wherein cold-air flows into the upper casing through the cold-air hole, an upper tray made of an elastic material, wherein the upper tray is mounted below the upper casing such that at least a portion of a top face of the upper tray is exposed, and wherein the upper tray includes a plurality of hemispherical upper chambers defined therein, each ejector-receiving opening opened to be in communication with each of the plurality of upper chambers, each opening-defining wall extending upward along a circumference of each ejector-receiving opening, an upper ejector disposed above the upper tray, wherein the upper ejector is configured to pass through the ejector-defining wall and vertically move to remove each ice from each ice chamber, a lower tray made of an elastic material, and having a plurality of lower chambers defined therein in contact with the plurality of upper chambers by pivoting to define a plurality of spherical ice chambers, respectively, and a driver for pivoting the lower tray to open and close the upper tray and the lower tray, wherein the upper tray includes a horizontal extension forming a top face of the upper tray, wherein the upper chamber is positioned below the horizontal extension such that the opening-defining wall is formed at a vertical level equal to or lower than a vertical level of the opening-defining wall.

In one embodiment, the refrigerator may further include an upper support penetrated by a portion of the upper chamber, wherein the upper support supports the horizontal extension from below, and a guide extending upward of the upper tray may be formed on the upper support, wherein the upper ejector may be connected to the guide in a vertically movable manner.

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In one embodiment, the horizontal extension may be disposed between the upper tray and the upper casing, and top and bottom faces of the horizontal extension may be restrained by the upper tray and the upper casing, respectively.

In one embodiment, the upper casing may have a cavity defined therein recessed downward, and a tray opening for exposing a portion of the top face of the upper tray including the opening-defining wall formed thereon may be defined in the cavity.

In one embodiment, a vertical level of the opening-defining wall may be lower than a vertical level of the cavity.

In one embodiment, a volume of an internal space of the opening-defining wall may correspond to a volume of ice debris that is able to be attached to a perimeter of the upper tray.

In one embodiment, a surface-treatment portion for increasing an attaching force between an inner face of the ice chamber and a surface of the ice during ice-removal may be formed on at least a portion of the inner face of the ice chamber.

In one embodiment, the surface-treatment portion may have a bump or a pattern formed thereon to increase a roughness of a surface of the ice chamber.

In one embodiment, the refrigerator may further include a refrigerator door for opening and closing the refrigerating compartment, and the refrigerator door may further include a dispenser for dispensing water or ice and a main ice maker in communication with the dispenser, wherein the main ice maker makes ice and stores the made ice.

The ice maker and the refrigerator according to the embodiments of the present disclosure have following effects.

According to the present embodiment, the cold-air flowing into the ice maker through the cold-air hole passes above the ice chambers by the cold-air guide, so that the ice formation speed may become uniform and the ice may maintain in the spherical shape.

Further, the opening-defining wall formed on the upper tray does not protrude excessively, so that the cold-air flowing along the top face of the upper tray may be evenly transferred to the entirety of the upper tray and may flow smoothly, thereby increasing the ice-removal efficiency.

Further, according to the present embodiment, the ice formation speed is delayed by the lower heater for supplying the heat to the ice chamber, so that the bubbles may move from the portion where the ice is formed toward the water, thereby producing the transparent ice.

Further, according to the present embodiment, regardless of the type of the refrigerator in which the ice maker is mounted, the cold-air passed through the cold-air hole flows along the cold-air guide, so that the flow patterns of the cold-air become almost the same. Therefore, the transparency of the ice may be uniform regardless of the type of refrigerator.

Further, according to the present embodiment, the ejector-receiving opening in communication with the upper chamber may be defined in the top face of the upper tray, and the opening-defining wall may be formed along the circumference of the ejector-receiving opening. The ejector-receiving opening not only guides the ejector, but also prevent overflowing of the water by receiving the water in the internal space of the opening-defining wall even when the amount of water inside the ice chamber increases due to the ice debris in the ice chamber or the water flowed from another chamber. Thus, problems caused by the overflowing of the water

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may be prevented, and the ice-removal may be performed without any problem even when the amount of water changes.

In addition, the volume of the space inside the opening-defining wall may be determined to be 2 to 4% of the total volume of the ice chamber in consideration of an error of a sensor occurring in the actual ice-removal, water inflow from the neighboring chamber, or inflow of the ice debris. Thus, while preventing water overflow, the internal space of the opening-defining wall may not be defined excessively large, so that the ice of substantially spherical shape may be made.

Further, according to the present embodiment, the ribs are formed along the circumference of the opening-defining wall, so that the deformation of the opening-defining wall may be prevented even when the interference occurs during the movement of the ejector, and even when the opening-defining wall is deformed, the opening-defining wall may be returned to its original position to perform the normal ice-removal.

Further, according to the present embodiment, the inner face of the ice chamber to which the water is supplied and the ice is made in the ice maker may be surface-treated such that friction between the inner face of the ice chamber and the surface of the ice is increased.

Therefore, even when the volume of the ice increases due to the water phase-change during the ice-removal process and the gap between the upper tray and the lower tray increases, the ice and the ice chamber are attached to each other, so that the formation of the protrusion on the surface of the ice due to the flow of the water to the portion between the ice chamber and the surface of the ice may be prevented.

In particular, even when the plurality of ice chambers are formed sequentially, and a freezing speed in one ice chamber is different from a freezing speed in a neighboring ice chamber, the formation of the protrusion on the surface of the ice caused by the water introduced from the neighboring ice chamber may be prevented.

Thus, the removed ice may have a generally spherical shape.

In addition, the ice cubes in the plurality of ice chambers may have a uniform size and a spherical shape.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a perspective view of a refrigerator according to an embodiment of the present disclosure.

FIG. 2 is a view showing a state in which a door is opened.

FIG. 3 is a partial enlarged view illustrating a state in which an ice maker is mounted according to an embodiment of the present disclosure.

FIG. 4 is a partial perspective view illustrating an interior of a freezing compartment according to an embodiment of the present disclosure.

FIG. 5 is an exploded perspective view of a grill pan and an ice duct according to an embodiment of the present disclosure.

FIG. 6 is a cross-sectional side view of a freezing compartment in a state in which a freezing compartment drawer and an ice bin are retracted therein, according to an embodiment of the present disclosure.

FIG. 7 is a partially-cut perspective view of a freezing compartment in a state in which a freezing compartment drawer and an ice bin are extended therefrom.

FIG. 8 is a perspective view of an ice maker viewed from above.

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FIG. 9 is a perspective view of a lower portion of an ice maker viewed from one side.

FIG. 10 is an exploded perspective view of an ice maker.

FIG. 11 is an exploded perspective view showing a coupling structure of an ice maker and a cover plate.

FIG. 12 is a perspective view of an upper casing according to an embodiment of the present disclosure viewed from above.

FIG. 13 is a perspective view of an upper casing viewed from below.

FIG. 14 is a side view of an upper casing.

FIG. 15 is a partial plan view of an ice maker viewed from above.

FIG. 16 is an enlarged view of a portion A of FIG. 15.

FIG. 17 shows flow of cold-air on a top face of an ice maker.

FIG. 18 is a perspective view of FIG. 16 taken along a line 18-18'.

FIG. 19 is a perspective view of an upper tray according to an embodiment of the present disclosure viewed from above.

FIG. 20 is a perspective view of an upper tray viewed from below.

FIG. 21 is a side view of an upper tray.

FIG. 22 is a perspective view of an upper support according to an embodiment of the present disclosure viewed from above.

FIG. 23 is a perspective view of an upper support viewed from below.

FIG. 24 is a cross-sectional view showing a coupling structure of an upper assembly according to an embodiment of the present disclosure.

FIG. 25 is a perspective view of an upper tray according to another embodiment of the present disclosure viewed from above.

FIG. 26 is a cross-sectional view of FIG. 25 taken along a line 26-26'.

FIG. 27 is a cross-sectional view of FIG. 25 taken along a line 27-27'.

FIG. 28 is a partially-cut perspective view showing a structure of a shield of an upper casing according to another embodiment of the present disclosure.

FIG. 29 is a perspective view of a lower assembly according to an embodiment of the present disclosure.

FIG. 30 is an exploded perspective view of a lower assembly viewed from above.

FIG. 31 is an exploded perspective view of a lower assembly viewed from below.

FIG. 32 is a partial perspective view illustrating a protruding confiner of a lower casing according to an embodiment of the present disclosure.

FIG. 33 is a partial perspective view illustrating a coupling protrusion of a lower tray according to an embodiment of the present disclosure.

FIG. 34 is a cross-sectional view of a lower assembly.

FIG. 35 is a cross-sectional view of FIG. 27 taken along a line 35-35'.

FIG. 36 is a plan view of a lower tray.

FIG. 37 is a perspective view of a lower tray according to another embodiment of the present disclosure.

FIG. 38 is a cross-sectional view that sequentially illustrates a pivoting state of a lower tray.

FIG. 39 is a cross-sectional view showing states of an upper tray and a lower tray immediately before or during ice-making.

FIG. 40 shows states of upper and lower trays upon completion of ice-making.

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FIG. 41 is a perspective view showing a state in which an upper assembly and a lower assembly are closed, according to an embodiment of the present disclosure.

FIG. 42 is an exploded perspective view showing a coupling structure of a connector according to an embodiment of the present disclosure.

FIG. 43 is a side view showing a disposition of a connector.

FIG. 44 is a cross-sectional view of FIG. 41 taken along a line 44-44'.

FIG. 45 is a cross-sectional view of FIG. 41 taken along a line 45-45'.

FIG. 46 is a perspective view showing a state in which upper and lower assemblies are open.

FIG. 47 is a cross-sectional view of FIG. 46 taken along a line 47-47'.

FIG. 48 is a side view showing a state of FIG. 41 viewed from one side.

FIG. 49 is a side view showing a state of FIG. 41 viewed from the other side.

FIG. 50 is a front view of an ice maker.

FIG. 51 is a partial cross-sectional view showing a coupling structure of an upper ejector.

FIG. 52 is an exploded perspective view of a driver according to an embodiment of the present disclosure.

FIG. 53 is a partial perspective view showing a driver being moved for provisional fixing of a driver.

FIG. 54 is a partial perspective view of a driver, which has been provisionally-fixed.

FIG. 55 is a partial perspective view for showing restraint and coupling of a driver.

FIG. 56 is a side view of an ice-full state detection lever positioned at a topmost position, which is an initial position, according to an embodiment of the present disclosure.

FIG. 57 is a side view of an ice-full state detection lever positioned at a bottommost position, which is a detection position.

FIG. 58 is an exploded perspective view showing a coupling structure of an upper casing and a lower ejector according to an embodiment of the present disclosure.

FIG. 59 is a partial perspective view showing a detailed structure of a lower ejector.

FIG. 60 shows a deformed state of a lower tray when the lower assembly is fully pivoted.

FIG. 61 shows a state just before a lower ejector passes through a lower tray.

FIG. 62 is a cutaway view taken along a line 62-62' of FIG. 8.

FIG. 63 is a view showing a state in which the ice generation is completed in FIG. 62.

FIG. 64 is a cross-sectional view taken along a line 62-62' of FIG. 8 in a water-supplied state.

FIG. 65 is a cross-sectional view taken along a line 62-62' of FIG. 8 in an ice-making process.

FIG. 66 is a cross-sectional view taken along a line 62-62' of FIG. 8 in a state in which the ice-making process is completed.

FIG. 67 is a cross-sectional view taken along a line 62-62' of FIG. 8 at an initial ice-removal state.

FIG. 68 is a cross-sectional view taken along a line 62-62' of FIG. 8 in a state in which an ice-removal process is completed.

FIG. 69 is a cut perspective view illustrating an interior of an ice chamber according to another embodiment of the present disclosure.

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FIG. 70 is a perspective view illustrating an open state of a lower assembly according to another embodiment of the present disclosure.

FIG. 71 is a cross-sectional view illustrating states of an upper tray and a lower tray when an ice volume is increased in an ice-removal state of another embodiment of the present disclosure.

FIG. 72 is a cut perspective view illustrating an interior of an ice chamber in another embodiment of the present disclosure.

DESCRIPTION OF SPECIFIC EMBODIMENTS

Hereinafter, some embodiments of the present disclosure will be described in detail with reference to the accompanying drawings. It should be noted that when components in the drawings are designated by reference numerals, the same components have the same reference numerals as far as possible even though the components are illustrated in different drawings. Further, in description of embodiments of the present disclosure, when it is determined that detailed descriptions of well-known configurations or functions disturb understanding of the embodiments of the present disclosure, the detailed descriptions will be omitted.

Also, in the description of the embodiments of the present disclosure, the terms such as first, second, A, B, (a) and (b) may be used. Each of the terms is merely used to distinguish the corresponding component from other components, and does not delimit an essence, an order or a sequence of the corresponding component. It should be understood that when one component is "connected", "coupled" or "joined" to another component, the former may be directly connected or joined to the latter or may be "connected", "coupled" or "joined" to the latter with a third component interposed therebetween.

FIG. 1 is a perspective view of a refrigerator according to an embodiment of the present disclosure. Further, FIG. 2 is a view showing a state in which a door is opened. Further, FIG. 3 is a partial enlarged view of an ice maker according to an embodiment of the present disclosure.

For convenience of description and understanding, directions will be defined. Hereinafter, based on a bottom face on which the refrigerator is installed, a direction toward the bottom face may be referred to as a downward direction, and a direction toward a top face of a cabinet 2, which is opposite to the bottom face, may be referred to as an upward direction. Further, when an undefined direction is described, the direction may be described by being defined based on each drawing.

Referring to FIGS. 1 to 3, a refrigerator 1 according to an embodiment of the present disclosure may include a cabinet 2 for defining a storage space therein, and a door for opening and closing the storage space.

In detail, the cabinet 2 defines the storage space vertically divided by a barrier. A refrigerating compartment 3 may be defined at an upper portion of the storage space, and a freezing compartment 4 may be defined at a lower portion of the storage space.

An accommodation member such as a drawer, a shelf, a basket, and the like may be disposed in each of the refrigerating compartment 3 and the freezing compartment 4.

The door may include a refrigerating compartment door 5 shielding the refrigerating compartment 3 and a freezing compartment door 6 shielding the freezing compartment 4.

The refrigerating compartment door 5 includes a pair of left and right doors, which may be opened and closed by

pivoting. Further, the freezing compartment door **6** may be disposed to be retractable or extendable like a drawer.

In another example, the arrangement of the refrigerating compartment **3** and the freezing compartment **4** and the shape of the door may be changed based on kinds of the refrigerators. However, the present disclosure may not be limited thereto, and may be applied to various kinds of refrigerators. For example, the freezing compartment **4** and the refrigerating compartment **3** may be arranged horizontally, or the freezing compartment **4** may be disposed above the refrigerating compartment **3**.

In one example, one of the pair of refrigerating compartment doors **5** on both sides may have an ice-making chamber **8** defined therein for receiving a main ice maker **81**. The ice-making chamber **8** may receive cold-air from an evaporator (not shown) in the cabinet **2** to allow ice to be made in the main ice maker **81**, and may define an insulated space together with the refrigerating compartment **3**. In another example, depending on a structure of the refrigerator, the ice-making chamber may be defined inside the refrigerating compartment **3** rather than the refrigerating compartment door **5**, and the main ice maker **81** may be disposed inside the ice-making chamber.

A dispenser **7** may be disposed on one side of the refrigerating compartment door **5**, which corresponds to a position of the ice-making chamber **8**. The dispenser **7** may be capable of dispensing water or ice, and may have a structure in communication with the ice-making chamber **8** to enable dispensing of ice made in the ice maker **81**.

In one example, the freezing compartment **4** may be equipped with an ice maker **100**. The ice maker **100**, which makes ice using water supplied, may produce ice in a spherical shape. The ice maker **100** may be referred to as an auxiliary ice maker because the ice maker **100** usually generates less ice than the main ice maker **81** or is used less than the main ice maker **81**.

The freezing compartment **4** may be equipped with a duct **44** for supplying cold-air to the freezing compartment **100**. Thus, a portion of the cold-air generated in the evaporator and supplied to the freezing compartment **4** may be flowed toward the ice maker **100** to make ice in an indirect cooling manner.

Further, an ice bin **102** in which the made ice is stored after being transferred from the ice maker **100** may be further provided below the ice maker **100**. Further, the ice bin **102** may be disposed in a freezing compartment drawer **41** which is extended from the freezing compartment **4**. Further, the ice bin **102** may be configured to be retracted and extended together with the freezing compartment drawer **41** to allow a user to take out the stored ice.

Thus, the ice maker **100** and the ice bin **102** may be viewed as at least a portion of which is received in the freezing compartment drawer **41**. Further, a large portion of the ice maker **100** and the ice bin **102** may be hidden when viewed from the outside. Further, the ice stored in the ice bin **102** may be easily taken out by the retraction and extension of the freezing compartment drawer **41**.

In another example, the ice made in the ice maker **100** or the ice stored in the ice bin **102** may be transferred to the dispenser **7** by transfer means and dispensed through the dispenser **7**.

In another example, the refrigerator **1** may not include the dispenser **7** and the main ice maker **81**, but include only the ice maker **1**. The ice maker **100** may be disposed in the ice-making chamber **8** in place of the main ice maker **81**.

Hereinafter, the mounting structure of the ice maker **100** will be described in detail with reference to the accompanying drawings.

Hereinafter, a mounting structure of the ice maker **100** will be described in detail with reference to the accompanying drawings.

FIG. **4** is a partial perspective view illustrating an interior of a freezing compartment according to an embodiment of the present disclosure. Further, FIG. **5** is an exploded perspective view of a grill pan and an ice duct according to an embodiment of the present disclosure.

As shown in FIGS. **4** and **5**, the storage space inside the cabinet **2** may be defined by an inner casing **21**. Further, the inner casing **21** defines the vertically divided storage space, that is, the refrigerating compartment **3** and freezing compartment **4**.

A portion of a top face of the freezing compartment **4** may be opened, and a mounting cover **43** may be formed at a position corresponding to a position where the ice maker **100** is mounted. The mounting cover **43** may be coupled and fixed to the inner casing **21**, and define a space further recessed upwardly from the top face of the freezing compartment **4** to secure a space in which the ice maker **100** is disposed. Further, the mounting cover **43** may include a structure for fixing and mounting the ice maker **100**.

Further, the mounting cover **43** may further include a cover recess **431** defined therein, which may be further recessed upwards to receive an upper ejector **300** to be described below. Since the upper ejector **300** has a structure that protrudes upward from the top face of the ice maker **100**, the upper ejector **300** may be received in the cover recess **431** to minimize a space used by the ice maker **100**.

Further, the mounting cover **43** may have a water-supply hole **432** defined therein for supplying water to the ice maker **100**. Although not shown, a pipe for supplying the water toward the ice maker **100** may penetrate the water-supply hole **432**. Further, an electrical-wire in connection with the ice maker **100** may pass through the mounting cover **43**. Further, because of the connector connected to the electrical-wire, the ice maker **100** may be in a state of being electrically connected and being able to be powered.

A rear wall face of the freezing compartment **4** may be formed by a grill pan **42**. The grill pan **42** may divide the space in the inner casing **21** horizontally, and may define, at rearward of the freezing compartment, a space for receiving an evaporator (not shown) that generates the cold-air and a blower fan (not shown) that circulates the cold-air therein.

The grill pan **42** may include cold-air ejectors **421** and **422** and a cold-air absorber **423**. Thus, the cold-air ejectors **421** and **422** and the cold-air absorber **423** may allow air circulation between the freezing compartment **4** and the space in which the evaporator is placed, and may cool the freezing compartment **4**. The cold-air ejectors **421** and **422** may be formed in a grill shape. The cold-air may be evenly discharged into the freezing compartment **4** through the upper cold-air ejector **421** and the lower cold-air ejector **422**.

In particular, the upper cold-air ejector **421** may be disposed at a top of the freezing compartment **4**. Further, the cold-air discharged from the upper cold-air ejector **421** may be used to cool the ice maker **100** and the ice bin **102** arranged at an upper portion of the freezing compartment **4**. In particular, the upper cold-air ejector **421** may include the cold-air duct **44** for supplying the cold-air to the ice maker **100**.

The cold-air duct **44** may connect the upper cold-air ejector **421** to the cold-air hole **134** of the ice maker **100**. That is, the cold-air duct **44** may connect the upper cold-air

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ejector **421** located at a center of the freezing compartment **4** in the horizontal direction and the ice maker **100** located at an upper end of the freezing compartment **4**, so that a portion of the cold-air discharged from the upper cold-air ejector **421** may be supplied directly into the ice maker **100**.

The cold-air duct **44** may be disposed at one end of the upper cold-air ejector **421** which extends in the horizontal direction. That is, the cold-air discharged from the upper cold-air ejector **421** is discharged to the freezing compartment **4**, and cold-air discharged from one side close to the cold-air duct **44** of the cold-air may be directed to the ice maker **100** through the cold-air duct **44**.

Thus, a rear end of the cold-air duct **44** may be recessed to receive one end of the upper cold-air ejector **421**. Further, an opened rear face of the cold-air duct **44** may be shaped in a shape corresponding to a shape of the grill pan **42**, and may be in contact with the grill pan **42** to prevent the cold-air from leaking. Further, a coupled portion **444** may be formed at a rear end of the cold-air duct **44**, and may be fixed to a front face of the grill pan **42** by a screw.

A cross-section of the cold-air duct **44** may decrease forwardly. Further, a duct outlet **446** on a front face of the cold-air duct **44** may be inserted into the cold-air hole **134** to concentrically supply the cold-air into the ice maker **100**.

In one example, the cold-air duct **44** may be constituted by an upper duct **443** forming an upper portion of the cold-air duct **44** and a lower duct **442** forming a lower portion of the cold-air duct **44**, and may define a whole cold-air passage by coupling of the upper duct **443** and the lower duct **442**. Further, the upper duct **443** and lower duct **442** may be coupled to each other by a connector **443**. The connector **443**, which has a hooking structure like a hook, may be formed on each of the upper duct **443** and the lower duct **442**.

FIG. 6 is a cross-sectional side view of a freezing compartment in a state in which a freezing compartment drawer and an ice bin are retracted therein, according to an embodiment of the present disclosure. Further, FIG. 7 is a partially-cut perspective view of a freezing compartment in a state in which a freezing compartment drawer and an ice bin are extended therefrom.

As shown in the drawings, the ice maker **100** may be mounted on the top face of the freezing compartment **4**. That is, the upper casing **120**, which forms an outer shape of the ice maker **100**, may be mounted on the mounting cover **43**.

In one example, the refrigerator **1** is installed to be tilted such that a front end of the cabinet **2** is slightly higher than a rear end thereof, so that the door **6** may be closed by a self weight after opening. Thus, the top face of the freezing compartment **4** may also be tilted relative to a ground on which the refrigerator **1** is installed, at the same slope as the cabinet **2**.

In this connection, when the ice maker **100** is mounted flush with the top face of the freezing compartment **4**, a water level of the water supplied inside the ice maker **100** may also be tilted, which may result in a problem of a difference in a size of ice cubes respectively made in the chambers. In particular, in a case of the ice maker **100** according to the present embodiment for making the spherical ice, when the water level is tilted, amounts of water received in the chambers are different from each other, so that a uniform spherical ice may not be made.

In order to avoid such problems, the ice maker **100** may be mounted to be tilted relative to the top face of the freezing compartment **4**, that is, based on top and bottom faces of the cabinet **2**. In detail, the ice maker **100** may be mounted to be in a state in which the top face of the upper casing **120** is

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pivoted counterclockwise (when viewed in FIG. 6) by a set angle α based on the top face of the freezing compartment **4** or the top face of the mounting cover **43**. In this connection, the set angle α may be equal to the slope of the cabinet **2**, and may be approximately 0.7° to 0.8° . Further, the front end of the upper casing **120** may be approximately 3 mm to 5 mm lower than the rear end thereof.

In a state of being mounted in the freezing compartment **4**, the ice maker **100** may be tilted by the set angle α , so that the ice maker **100** may be horizontal to the ground on which the refrigerator **1** is installed. Thus, the level of the water supplied into the ice maker **100** may become level with the ground, and the same amount of water may be received in the plurality of chambers to make ice of uniform size.

Further, in a state in which the ice maker **100** is mounted, the cold-air hole **134** at the rear end of the upper casing **120** may be connected to the upper cold-air ejector **421**. Thus, the cold-air for the ice-making may be concentrically supplied to an inner upper portion of the upper casing **120** to increase an ice-making efficiency.

In one example, the ice bin **102** may be mounted inside the freezing compartment drawer **41**. The ice bin **102** is positioned correctly below the ice maker **100** in a state in which the freezing compartment drawer **41** is retracted. To this end, the freezing compartment drawer **41** may have a bin mounting guide **411** which guides a mounting position of the ice bin **102**. The bin mounting guides **411** may respectively protrude upwardly from positions corresponding to four corners of the bottom face of the ice bin **102**, and may be arranged to enclose the four corners of the bottom face of the ice bin **102**. Thus, the ice bin **102** may remain in position in a state of being mounted in the freezing compartment drawer **41**, and may be positioned vertically below the ice maker **100** in a state in which the freezing compartment drawer **41** is retracted.

As shown in FIG. 6, a bottom of the ice maker **100** may be received inside the ice bin **102** in a state in which the freezing compartment drawer **41** is retracted. That is, the bottom of the ice maker **100** may be located inside the ice bin **102** and the freezing compartment drawer **41**. Thus, the ice removed from the ice maker **100** may fall and be stored in the ice bin **102**. Further, a volume loss inside the freezing compartment **4** due to arrangement of the ice maker **100** and the ice bin **102** may be minimized by minimizing the space between the ice maker **100** and the ice bin **102**. In another example, the bottom of the ice maker **100** and the bottom face of the ice bin **102** may be spaced apart each other by an appropriate distance to ensure a distance for storing an appropriate amount of ice.

In one example, in a state in which the ice maker **100** is mounted therein, the freezing compartment drawer **41** may be extended or retracted as shown in FIG. 7. Further, in this connection, at least a portion of rear faces of the ice bin **102** and the freezing compartment drawer **41** may be opened to prevent interference with the ice maker **100**.

In detail, a drawer opening **412** and a bin opening **102a** may be respectively defined in the rear faces of the freezing compartment drawer **41** and the ice bin **102** corresponding to the position of the ice maker **100**. The drawer opening **412** and the bin opening **102a** may be respectively defined at positions facing each other. Further, the drawer opening **412** and the bin opening **102a** may be respectively defined to open from the top of the freezing compartment drawer **41** and the top of the ice bin **102** to positions lower than the bottom of the ice maker **100**.

Thus, even when the freezing compartment drawer **41** is extended in a state in which the ice maker **100** is mounted

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therein, the ice maker **100** may be prevented from interfering with the ice bin **102** and the freezing compartment drawer **41**.

In particular, even in a state in which the ice maker **100** removes the ice and the lower assembly **200** is pivoted, or in a state in which an ice-full state detection lever **700** is pivoted to detect an ice-full state, the drawer opening **412** and the bin opening **102a** may be in a shape of being recessed further downward from the bottom of the ice maker **100** to prevent interference with the freezing compartment drawer **41** or the ice bin **102**.

A drawer opening guide **412a** extending rearward along a perimeter of the drawer opening **412** may be formed. The drawer opening guide **412a** may extend rearward to guide the cold-air flowing downward from the upper cold-air ejector **421** into the freezing compartment drawer **41**.

Further, a bin opening guide **102b** extending rearward along a perimeter of the bin opening **102a** may be included. The cold-air flowing downward from the upper cold-air ejector **421** may flow into the ice bin **102** through the bin opening guide **102b**.

In one example, a cover casing **130** in a plate shape may be disposed on a rear face of the upper casing **120** of the ice maker **100**. The cover plate **130** may be formed to cover at least a portion of the ice bin opening **102a** such that the ice inside the ice bin **102** does not fall downward through the bin opening **102a** and the drawer opening **412**.

The cover plate **130** extends downward from a rear face of the upper casing **120** of the ice maker **100** and may extend into the bin opening **102a**. As shown in FIG. 6, in a state in which the freezing compartment drawer **41** is retracted, the cover plate **130** is positioned inside the bin opening **102a** to cover at least a portion of the bin opening **102a**. Thus, even when the ice is moved backwards by inertia at the moment the freezing compartment drawer **41** is extended or retracted, the ice may be blocked by the cover plate **130**, and prevented from falling out of the ice bin **102**.

Further, the cover plate **130** may have a plurality of openings defined therein to allow the cold-air to pass therethrough. Thus, in a state in which the freezing compartment drawer **41** is closed as shown in FIG. 6, the cold-air may pass through the cover plate **130** and flow into the ice bin **102**.

The cover plate **130** may be formed to have a size for not interfering with the drawer opening **412** and the bin opening **102a**. Thus, the cover plate **130** may not interfere with the freezing compartment drawer **41** or the ice bin **102** when the freezing compartment drawer **41** is extended as shown in FIG. 7.

The cover plate **130** may be molded separately and joined to the upper casing **120** of the ice maker **100**. Alternatively, the rear face of the upper casing **120** may protrude further downward to form the cover plate **130**.

Hereinafter, the ice maker **100** will be described in detail with reference to the accompanying drawings.

FIG. 8 is a perspective view of an ice maker viewed from above. Further, FIG. 9 is a perspective view of a lower portion of an ice maker viewed from one side. Further, FIG. 10 is an exploded perspective view of an ice maker.

Referring to FIGS. 8 to 10, the ice maker **100** may include an upper assembly **110** and a lower assembly **200**.

The lower assembly **200** may be fixed to the upper assembly **110** such that one end thereof is pivotable. The pivoting may open and close an inner space defined by the lower assembly **200** and the upper assembly **110**.

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In detail, the lower assembly **200** may make the spherical ice together with the upper assembly **110** in a state in which the lower assembly **200** is in close contact with the upper assembly **110**.

That is, the upper assembly **110** and the lower assembly **200** define an ice chamber **111** for making the spherical ice. The ice chamber **111** is substantially a spherical chamber. The upper assembly **110** and the lower assembly **200** may define a plurality of divided ice chambers **111**. Hereinafter, an example in which three ice chambers **111** are defined by the upper assembly **110** and the lower assembly **200** will be described. Note that there is no limit to the number of ice chambers **111**.

In a state in which the upper assembly **110** and the lower assembly **200** define the ice chamber **111**, the water may be supplied to the ice chamber **111** via a water supply **190**. The water supply **190** is coupled to the upper assembly **110**, and direct the water supplied from the outside to the ice chamber **111**.

After the ice is made, the lower assembly **200** may pivot in a forward direction. Then, the spherical ice made in the space between the upper assembly **110** and the lower assembly **200** may be separated from the upper assembly **110** and the lower assembly **200**, and may fall to the ice bin **102**.

In one example, the ice maker **100** may further include a driver **180** such that the lower assembly **200** is pivotable relative to the upper assembly **110**.

The driver **180** may include a driving motor and a power transmission for transmitting power of the driving motor to the lower assembly **200**. The power transmission may include at least one gear, and may provide a suitable torque for the pivoting of the lower assembly **200** by a combination of the plurality of gears. Further, the ice-full state detection lever **700** may be connected to the driver **180**, and the ice-full state detection lever **700** may be pivoted by the power transmission.

The driving motor may be a bidirectionally rotatable motor. Thus, bidirectional pivoting of the lower assembly **200** and ice-full state detection lever **700** is achieved.

The ice maker **100** may further include an upper ejector **300** such that the ice may be separated from the upper assembly **110**. The upper ejector **300** may cause the ice in close contact with the upper assembly **110** to be separated from the upper assembly **110**.

The upper ejector **300** may include an ejector body **310** and at least one ejecting pin **320** extending in a direction intersecting the ejector body **310**. The ejecting pin **320** may include ejecting pins of the same number as the ice chamber **111**, and each ejecting pin may remove ice made in each ice chamber **111**.

The ejecting pin **320** may press the ice in the ice chamber **111** while passing through the upper assembly **110** and being inserted into the ice chamber **111**. The ice pressed by the ejecting pin **320** may be separated from the upper assembly **110**.

Further, the ice maker **100** may further include a lower ejector **400** such that the ice in close contact with the lower assembly **200** may be separated therefrom. The lower ejector **400** may press the lower assembly **200** such that the ice in close contact with the lower assembly **200** is separated from the lower assembly **200**.

An end of the lower ejector **400** may be located within a pivoting range of the lower assembly **200**, and may press an outer side of the ice chamber **111** to remove the ice in the pivoting process of the lower assembly **200**. The lower ejector **400** may be fixedly mounted to the upper casing **120**.

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In one example, a pivoting force of the lower assembly 200 may be transmitted to the upper ejector 300 in the pivoting process of the lower assembly 200 for ice-removal. To this end, the ice maker 100 may further include a connector 350 connecting the lower assembly 200 and the upper ejector 300 with each other. The connector 350 may include at least one link.

In one example, the connector 350 may include pivoting arms 351 and 352 and a link 356. The pivoting arms 351 and 352 may be connected to the driver 180 together with the lower support 270 and pivoted together. Further, ends of the pivoting arms 351 and 352 may be connected to the lower support 270 by an elastic member 360 to be in close contact with the upper assembly 110 in a state in which the lower assembly 200 is closed.

The link 356 connects the lower support 270 with the upper ejector 300, so that the pivoting force of the lower support 270 may be transmitted to the upper ejector 300 when the lower support 270 pivots. The upper ejector 300 may move vertically in association with the pivoting of the lower support 270 by the link 356.

In one example, when the lower assembly 200 pivots in the forward direction, the upper ejector 300 may descend by the connector 350, so that the ejecting pin 320 may press the ice. On the other hand, during when the lower assembly 200 pivots in a reverse direction, the upper ejector 300 may ascend by the connector 350 to return to an original position thereof.

Hereinafter, the upper assembly 110 and the lower assembly 200 will be described in more detail.

The upper assembly 110 may include an upper tray 150 that forms an upper portion of the ice chamber 111 for making the ice. Further, the upper assembly 110 may further include the upper casing 120 and an upper support 170 to fix the upper tray 150.

The upper tray 150 may be positioned below the upper casing 120, and the upper support 170 may be positioned below the upper tray 150. As such, the upper casing 120, the upper tray 150, and the upper support 170 may be arranged in the vertical direction one after the other, and may be fastened by a fastener and formed as a single assembly. That is, the upper tray 150 may be fixedly mounted between the upper casing 120 and the upper support 170 by the fastener. Thus, the upper tray 150 may be maintained at a fixed position, and may be prevented from being deformed or separated from the upper assembly 110.

In one example, the water supply 190 may be disposed at an upper portion of the upper casing 120. The water supply 190 is for supplying the water into the ice chamber 111, which may be disposed to face the ice chamber 111 from above the upper casing 120.

Further, the ice maker 100 may further include a temperature sensor 500 for sensing a temperature of the water or the ice in the ice chamber 111. The temperature sensor 500 may indirectly sense the temperature of the water or the ice in the ice chamber 111 by sensing a temperature of the upper tray 150.

The temperature sensor 500 may be mounted on the upper casing 120. Further, at least a portion of the temperature sensor 500 may be exposed through the opened side of the upper casing 120.

In one example, the lower assembly 200 may include a lower tray 250 that forms a lower portion of the ice chamber 111 for making the ice. Further, the lower assembly 200 may further include a lower support 270 supporting a lower portion of the lower tray 250 and a lower casing 210 covering an upper portion of the lower tray 250.

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The lower casing 210, lower tray 250, and the lower support 270 may be arranged in the vertical direction one after the other, and may be fastened by a fastener and formed as a single assembly.

In one example, the ice maker 100 may further include a switch 600 for turning the ice maker 100 on or off. The switch 600 may be disposed on a front face of the upper casing 120. Further, when the user manipulates the switch 600 to be turned on, the ice may be made by the ice maker 100. That is, when the switch 600 is turned on, operations of components, including the ice maker, for ice-making may be started. That is, when the switch 600 is turned on, the water is supplied to the ice maker 100, and an ice-making process in which the ice is made by the cold-air and an ice-removal process in which the lower assembly 200 is pivoted and the ice is removed may be repeatedly performed.

On the other hand, when the switch 600 is manipulated to be turned off, the components for the ice-making, including the ice maker 100, will remain inactive and will not be able to make the ice through the ice maker 100.

Further, the ice maker 100 may further include the ice-full state detection lever 700. The ice-full state detection lever 700 may detect whether the ice bin 102 is in the ice-full state while receiving the power of the driver 180 and pivoting.

One side of the ice-full state detection lever 700 may be connected to the driver 180 and the other side of the ice-full state detection lever 700 may be pivotably connected to the upper casing 120, so that the ice-full state detection lever 700 may pivot based on the operation of the driver 180.

The ice-full state detection lever 700 may be positioned below an axis of pivoting of the lower assembly 200, so that the ice-full state detection lever 700 does not interfere with the lower assembly 200 during the pivoting of the lower assembly 200. Further, both ends of the ice-full state detection lever 700 may be bent many times. The ice-full state detection lever 700 may be pivoted by the driver 180, and may detect whether a space below the lower assembly 200, that is, the space inside the ice bin 102 is in the ice-full state.

In one example, an internal structure of the driver 180 is not shown in detail, but will be briefly described for the operation of the ice-full state detection lever 700. The driver 180 may further include a cam rotated by the rotational power of the motor and a moving lever moving along a cam face. A magnet may be provided on the moving lever. The driver 180 may further include a hall sensor that may detect the magnet when the moving lever moves.

A first gear to which the ice-full state detection lever 720 is engaged among a plurality of gears of the driver 180 may be selectively engaged with or disengaged from a second gear that engages with the first gear. In one example, the first gear is elastically supported by the elastic member, so that the first gear may be engaged with the second gear when no external force is applied thereto.

On the other hand, when a resistance greater than an elastic force of the elastic member is applied to the first gear, the first gear may be spaced apart from the second gear.

A case in which the resistance greater than the elastic force of the elastic member is applied to the first gear is, for example, a case in which the ice-full state detection lever 700 is caught in the ice in the ice-removal process (in the case of the ice-full state). In this case, the first gear may be spaced apart from the second gear, so that breakage of the gears may be prevented.

The ice-full state detection lever 700 may be pivoted together in association with the lower assembly 200 by the

plurality of gears and the cam. In this connection, the cam may be connected to the second gear or may be linked to the second gear.

Depending on whether the hall sensor senses the magnet, the hall sensor may output first and second signals that are different outputs. One of the first signal and the second signal may be a high signal, and the other may be a low signal.

The ice-full state detection lever **700** may be pivoted from a standby position to an ice-full state detection position for the ice-full state detection. Further, the ice-full state detection lever **700** may identify whether the ice bin **102** is filled with the ice of equal to or greater than the predetermined amount while passing through an inner portion of the ice bin **102** in the pivoting process.

Hereinafter, the ice-full state detection lever **700** will be described in more detail with reference to FIG. **10**.

The ice-full state detection lever **700** may be a lever in a form of a wire. That is, the ice-full state detection lever **700** may be formed by bending a wire having a predetermined diameter a plurality of times.

The ice-full state detection lever **700** may include a detection body **710**. The detection body **710** may pass a position of a set vertical level inside the ice bin **102** in the pivoting process of the ice-full state detection lever **700**, and may be substantially the lowest portion of the ice-full state detection lever **700**.

Further, the ice-full state detection lever **700** may be positioned such that an entirety of the detection body **710** is located below the lower assembly **200** to prevent the interference between the lower assembly **220** and the detection body **710** in the pivoting process of the lower assembly **200**.

The detection body **710** may be in contact with the ice in the ice bin **102** in the ice-full state of the ice bin **102**. The ice-full state detection lever **700** may include the detection body **710**. The detection body **710** may extend in a direction parallel to a direction of extension of the connection shaft **370**. The detection body **710** may be positioned lower than a lowest point of the lower assembly **200** regardless of the position.

Further, the ice-full state detection lever **700** may include a pair of extensions **720** and **730** respectively extending upward from both ends of the detection body **710**. The pair of extensions **720** and **730** may extend substantially in parallel with each other.

A distance between the pair of extensions **720** and **730**, that is, a length of the detection body **710** may be larger than a horizontal length of the lower assembly **200**. Thus, in the pivoting process of the ice-full state detection lever **700** and the pivoting process of the lower assembly **200**, the pair of extensions **720** and **730** and the detection body **710** may be prevented from interfering with the lower assembly **200**.

The pair of extensions **720** and **730** may include a first extension **720** extending to a lever receiving portion **187** of the driver **180** and a second extension **710** extending to the lever receiving hole **120a** of the upper casing **120**. The pair of extensions **720** and **730** may be bent at least once, so that the ice-full state detection lever **700** is not deformed even after repeated contact with the ice and maintains a more reliable detection state.

For example, the extensions **720** and **730** may include a first bent portion **721** extending from each of both ends of the detection body **710** and a second bent portions **722** extending from each of ends of the first bent portions **721** to the driver **180**. Further, the first bent portion **721** and second bent portion **722** may be bent at a predetermined angle. The first bent portion **721** and the second bent portion **722** may

intersect with each other at an angle in a range approximately from 140° to 150° . Further, a length of the first bent portion **721** may be larger than a length of the second bent portion **722**. Due to such structure, the ice-full state detection lever **700** may reduce a radius of pivoting, and may detect the ice in the ice bin **102** while minimizing interference with other components.

Further, a pair of inserted portions **740** and **750**, which are respectively bent outwardly, may be formed at top of the pair of extensions **720** and **730**, respectively. The pair of inserted portions **740** and **750** may include a first inserted portion **740** that is bent at the end of the first extension **720** and inserted into the lever receiving portion **187** and a second inserted portion **750** that is bent at the end of the second extension **710** and inserted into the lever receiving hole **120a**. The first inserted portion **740** and second inserted portion **750** may be formed to be respectively coupled to and pivotably inserted into the lever receiving portion **187** and the lever receiving hole **120a**.

That is, the first inserted portion **740** may be coupled to the driver **180** and pivoted by the driver **180**, and the second inserted portion **750** may be pivotably coupled to the lever receiving hole **120a**. Thus, the ice-full state detection lever **700** may be pivoted based on the operation of the driver **180**, and may detect whether the ice bin **102** is in the ice-full state.

In one example, the ice maker **100** may be equipped with the cover plate **130**.

Hereinafter, a structure of the cover plate **130** will be described in detail with reference to the accompanying drawings.

FIG. **11** is an exploded perspective view showing a coupling structure of an ice maker and a cover plate.

Referring to FIGS. **6**, **7**, and **11**, the lever receiving hole **120a** may be defined in one face of the upper casing **120**, and a pair of bosses **120b** may respectively protrude from both left and right sides of the lever receiving hole **120a**. Further, a stepped plate seat **120c** may be formed above the pair of bosses **120b**. In this connection, one face of the upper casing **120** in which the lever receiving hole **120a** is defined and on which the plate seat **120c** is formed is a face adjacent to the rear face of the freezing compartment **4** as shown in FIGS. **6** and **7**. Further, the cover plate **130** may be coupled to said one face of the upper casing **120**.

The cover plate **130** may be formed in a rectangular plate shape, and may be formed to have a width corresponding to a width of the upper casing **120**. Further, the cover plate **130** extends further below the bottom of the upper casing **120**, and may extend to cover a large portion of the bin opening **102a** when the freezing compartment drawer **41** is closed.

A plate bent portion **130d** may be formed at a top of the cover plate **130**, and the plate bent portion **130d** may be seated on the plate seat **120c**. Further, the cover plate **130** may be formed with an exposing opening **130c** defined therein exposing the lever receiving hole **120a** and the second inserted portion **750**. The second inserted portion **750** is not interfered by the exposing opening **130c** when the ice-full state detection lever **700** is pivoted, thereby ensuring the operation of the ice-full state detection lever **700**.

Further, boss-receiving portions **130b** may protrude from left and right sides of the exposing opening **130c**, respectively. The boss-receiving portions **130b** are shaped to respectively accommodate the pair of the bosses **120b** protruding from the upper casing **120**. Further, the boss-receiving portion **130b** and the boss **120b** may be coupled with each other by a fastener such as the screw fastened to the boss-receiving portion **130b**, and the cover plate **130** may be fixed.

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In one example, a plurality of ventilation holes **130a** may be defined at a lower portion of the cover plate **130**. The ventilation holes **130a** may be defined in series, and the lower portion of the cover plate **130** may be shaped like a grill. The ventilation hole **130a** may extend vertically, and may extend from a bottom of the upper casing **120** to a bottom of the cover plate **130**. Therefore, the cold-air may be smoothly flowed into the ice bin **102** by the ventilation holes **130a**. Further, the cover plate **130** may be formed with a plate rib **130c**.

The plate rib **130e** is for reinforcing the cover plate **130**, which may be formed along the perimeter of the cover plate **130**. Further, the plate rib **130e** may be formed to cross the cover plate **130** and may be formed between the ventilation holes **130a**.

A sufficient strength of the cover plate **130** may be ensured by the plate rib **130c**. Thus, when the freezing compartment drawer **41** is extended and retracted to be opened and closed, the cover plate **130** may prevent the ice inside the ice bin **102** from rolling and passing through the bin opening **102a**. In this connection, the cover plate **130** may not be deformed or damaged from an impact of the ice.

The ice made in the present embodiment, which is substantially spherical or nearly spherical in shape, inevitably rolls or moves inside the ice bin **102**. Accordingly, such structure of the cover plate **130** may prevent the spherical ice from falling out of the ice bin **102**. Further, the cover plate **130** is formed so as not to block the flow of the cold-air into the ice bin **102**.

In one example, the cover plate **130** may be molded separately and mounted on the upper casing **120**. In another example, if necessary, one side of the upper casing **120** may be extended to have a shape corresponding to that of the cover plate **130**.

Hereinafter, a structure of the upper casing **120** constituting the ice maker **100** will be described in detail with reference to the accompanying drawings.

FIG. **12** is a perspective view of an upper casing according to an embodiment of the present disclosure viewed from above. Further, FIG. **13** is a perspective view of an upper casing viewed from below. Further, FIG. **14** is a side view of an upper casing.

Referring to FIGS. **12** to **14**, the upper casing **120** may be fixedly mounted to the top face of the freezing compartment **4** in a state in which the upper tray **150** is fixed.

The upper casing **120** may include an upper plate **121** for fixing the upper tray **150**. The upper tray **150** may be disposed on a bottom face of the upper plate **121**, and the upper tray **150** may be fixed to the upper plate **121**. The upper plate **121** may have a tray opening **123** defined therein through which a portion of the upper tray **150** passes. Further, a portion of a top face of the upper tray **150** may pass through the tray opening **123** and exposed. The tray opening **123** may be defined along an array of the plurality of ice chambers **111**.

The upper plate **121** may include a cavity **122** recessed downwardly from the upper plate **121**. A tray opening **123** may be defined in a bottom **122a** of the cavity **122**.

When the upper tray **150** is mounted on the upper plate **121**, a portion of the top face of the upper tray **150** may be located inside the space where the cavity **122** is defined, and may pass through the tray opening **123** and protrude upward.

A heater-mounted portion **124** in which an upper heater **148** for heating the upper tray **150** for ice-removal may be defined in the upper casing **120**. The heater-mounted portion may be defined in the bottom of the cavity **122**.

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Further, the upper casing **120** may further include a pair of sensor-fixing ribs **128** and **129** for mounting the temperature sensor **500**. The pair of sensor-fixing ribs **128** and **129** may be spaced apart from each other, and the temperature sensor **500** may be located between the pair of sensor-fixing ribs **128** and **129**. The pair of sensor-fixing ribs **128** and **129** may be provided on the upper plate **121**.

The upper plate **121** may have a plurality of slots **131** and a plurality of slots **132** defined therein for coupling with the upper tray **150**. Portions of the upper tray **150** may be inserted into the plurality of slots **131** and the plurality of slots **132**. The plurality of slots **131** and the plurality of slots **132** may include a first upper slot **131** and a second upper slot **132** positioned opposite to the first upper slot **131** around the tray opening **123**.

The first upper slot **131** and the second upper slot **132** may be arranged to face each other, and the tray opening **123** may be located between the first upper slot **131** and the second upper slot **132**.

The first upper slot **131** and the second upper slot **132** may be spaced apart from each other with the tray opening **123** therebetween. Further, each of the plurality of the first upper slots **131** and each of the plurality of second upper slots **132** may be spaced apart from each other along a direction in which the ice chambers **111** are successively arranged.

The first upper slot **131** and the second upper slot **132** may be defined in a curved shape. Thus, the first upper slot **131** and second upper slot **132** may be defined along a periphery of the ice chamber **111**. Such structure may allow the upper tray **150** to be more firmly fixed to the upper casing **120**. In particular, deformation of dropout of the upper tray **150** may be prevented by fixing the periphery of the ice chamber **111** of the upper tray **150**.

A distance from the first upper slot **131** to the tray opening **123** may differ from a distance from the second upper slot **132** to the tray opening **123**. In one example, the distance from the second upper slot **132** to the tray opening **123** may be shorter than the distance from the first upper slot **131** to the tray opening **123**.

The upper plate **121** may further include a sleeve **133** for inserting a coupling boss **175** of the upper support **170** to be described later therein. The sleeve **133** may be formed in a cylindrical shape, and may extend upward from the upper plate **121**.

In one example, a plurality of sleeves **133** may be arranged on the upper plate **121**. The plurality of sleeves **133** may be arranged successively in the extending direction of the tray opening, and may be spaced apart from each other at a regular interval.

Some of the plurality of sleeves **133** may be positioned between two adjacent first upper slots **131**. Some of the remaining sleeves **133** may be positioned between two adjacent second upper slots **132** or may be positioned to face a region between the two second upper slots **132**. Such structure may allow the coupling between the first upper slot **131** and the second upper slot **132** and the protrusions of the upper tray **150** to be very tight.

The upper casing **120** may further include a plurality of hinge supports **135** and **136** to allow the lower assembly **200** to pivot. Further, a first hinge hole **137** may be defined in each of the hinge supports **135** and **136**. The plurality of hinge supports **135** and **136** may be spaced apart from each other, so that both ends of the lower assembly **200** may be pivotably coupled to the plurality of hinge supports **135** and **136**.

The upper casing **120** may include through-openings **139b** and **139c** defined therein for a portion of the connector

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350 to pass therethrough. In one example, the links 356 located on both sides of the lower assembly 200 may pass through the through-openings 139b and 139c, respectively.

In one example, the upper casing 120 may be formed with a horizontal extension 142 and a vertical extension 140. The horizontal extension 142 may form the top face of the upper casing 120, and may be brought to be in contact with the top face of the freezing compartment 4, the inner casing 21. In another example, the horizontal extension 142 may be brought to be in contact with the mounting cover 43 rather than inner casing 21.

The horizontal extension 142 may be provided with a hook 138 and a threaded portion 142a for fixedly mounting the upper casing 120 to the inner casing 21 or the mounting cover 43.

The hook 138 may be formed on each of both rear ends of the horizontal extension 142, and may be configured to be fastened to the inner casing 21 or the mounting cover 43. In detail, the hook 138 may include a vertical hook 138b protruding upward from the horizontal extension 142 and a horizontal hook 138a extending rearward from an end of the vertical hook 138b. Thus, an entirety of the hook 138 may be formed in a hook shape. Further, one side of the inner casing 21 or the mounting cover 43 may be inserted and fastened into a space defined between the vertical hook 138b and the horizontal hook 138a to be locked to each other.

In one example, the hook 138 may protrude from an outer face of the vertical extension 140. That is, a side end of the hook 138 may be coupled to and integrally formed with the vertical extension 140. Thus, the hook 138 may satisfy a strength necessary to support the ice maker 100. Further, the hook 138 will not break during attachment and detachment process of the ice maker 100.

Further, an extended end of the horizontal hook 138a may be formed with an inclined portion 138d inclined upward, so that the hook 138 may be guided to a restraint position more easily when the ice maker 100 is mounted. Further, at least one protrusion 138c may be formed on a top face of the horizontal hook 138a. The protrusion 138c may be in contact with the inner casing 21 or the mounting cover 43, and therefore, vertical movement of the ice maker 100 may be prevented and the ice maker 100 may be more firmly mounted.

In one example, a threaded portion 142a may be formed at each of both front ends of the horizontal extension 142. The threaded portion 142a may protrude downward, and may be coupled with the inner casing 21 or the mounting cover 43 by the screw for fixing the upper casing 120.

Therefore, for the installation of the ice maker 100, after placing the module-shaped ice maker 100 inside the freezing compartment 4, the hook 138 is fastened to the inner casing 21 or the mounting cover 43, and then the ice maker 100 is pressed upward. In this connection, a coupling hook 140a on the vertical extension 140 may be coupled with the mounting cover 43, so that the ice maker 100 may be in an additional provisionally-fixed state. In this state, the screw may be fastened to the threaded portion 142a, so that the front end of the upper casing 120 may be coupled to the inner casing 21 or mounting cover 43, thereby completing the installation of the ice maker 100.

In other words, the ice maker 100 may be mounted by fastening the rear end of the ice maker 100 and fixing the front end thereof with the screw without any complicated structure or component for mounting the ice maker 100. The ice maker 100 may be easily detached in a reverse order.

In one example, an edge rib 120d may be formed along a perimeter of the horizontal extension 142. The edge rib 120d

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may protrude vertically upward from the horizontal extension 142, and may be formed along ends except for the rear end of the horizontal extension 142.

When the ice maker 100 is mounted, the edge rib 120d may be brought into close contact with the outer face of the inner casing 21 or the mounting cover 43, or may allow the ice maker 100 to be mounted horizontally with the ground on which the refrigerator 1 is installed.

To this end, a vertical level of the edge rib 120d may decrease from a front end thereof to a rear end thereof. In detail, a portion of the edge rib 120d formed along the front end of the horizontal extension 142 may be formed to have a highest vertical level and have a uniform vertical level. Further, a portion of the edge rib 120d, which is formed along each of both sides of the horizontal extension 142, may have a highest vertical level at a front end thereof, and a vertical level thereof may decrease rearwardly.

The vertical level of the front end, which has the highest vertical level in the edge rib 120d, may be approximately 3 to 5 mm. Thus, as shown in FIG. 6, the horizontal extension 142, which forms the top face of the ice maker 100, may be disposed to have an inclination of approximately 7 to 8° downwards relative to the outer face of the inner casing 21 or the mounting cover 43.

With such arrangement, even when the cabinet 2 is placed at an angle, the water level of the water supplied into the ice maker 100 may be horizontal, and the same amount of water may be received in the plurality of ice chambers 111, so that the spherical ice cubes having the same size may be made.

In one example, the vertical extension 140 may be formed inward of the horizontal extension 142 and may extend vertically upward along the perimeter of the upper plate 121. The vertical extension 140 may include at least one coupling hook 140a. The upper casing 120 may be hooked to the mounting cover 43 by the coupling hook 140a. Further, the water supply 190 may be coupled to the vertical extension 140.

The upper casing 120 may further include a side wall 143. The side wall 143 may extend downward from the horizontal extension 142. The side wall 143 may be disposed to surround at least a portion of the perimeter of the lower assembly 200. In other words, the side wall 143 prevents the lower assembly 200 from being exposed to the outside.

The side wall 143 may include a first side wall 143a in which a cold-air hole 134 is defined, and a second side wall 143b facing away from the first side wall 143a. When the ice maker 100 is mounted in the freezing compartment 4, the first side wall 143a may face a rear wall or one of both sidewalls of the freezing compartment 4.

The lower assembly 200 may be located between the first side wall 143a and the second side wall 143b. Further, since the ice-full state detection lever 700 pivots, an interference-prevention groove 148 may be defined in the side wall 143 such that interference is prevented in the pivoting operation of the ice-full state detection lever 700.

The through-openings 139b and 139c may include the first through-opening 139b positioned adjacent to the first side wall 143a and the second through-opening 139c positioned adjacent to the second side wall 143b. Further, the tray opening 123 may be defined between the through-openings 139b and 139c.

The cold-air hole 134 in the first side wall 143a may extend in the horizontal direction. The cold-air hole 134 may be defined in a corresponding size such that the front end of the cold-air duct 44 may be inserted therein. Therefore, an

entirety of the cold-air supplied through the cold-air duct **44** may flow into the upper casing **120** through the cold-air hole **134**.

The cold-air guide **145** may be formed between both ends of the cold-air hole **134**, and the cold-air flowing into the cold-air hole **134** may be guided toward the tray opening **123** by the cold-air guide **145**. Further, a portion of the upper tray **150** exposed through the tray opening **123** may be exposed to the cold-air and directly cooled.

In one example, in the ice-full state detection lever **700**, the first inserted portion **740** is connected to the driver **180** and the second inserted portion **750** is coupled to the first side wall **143a**.

The driver **180** is coupled to the second side wall **143a**. In the ice-removal process, the lower assembly **200** is pivoted by the driver **180**, and the lower tray **250** is pressed by the lower ejector **400**. In this connection, relative movement between the driver **180** and the lower assembly **200** may occur in the process in which the lower tray **250** is pressed by the lower ejector **400**.

A pressing force of the lower ejector **400** applied on the lower tray **250** may be transmitted to an entirety of the lower assembly **200** or to the driver **180**. In one example, a torsional force is applied on the driver **180**. The force acting on the driver **180** then acts on the second side wall **143b** too. When the second side wall **143b** is deformed by the force acting on the second side wall **143b**, a relative position between the driver **180** and the connector **350** installed on the second side wall **143b** may change. In this case, there is a possibility that the shaft of the driver **180** and the connector **350** are separated.

Therefore, a structure for minimizing the deformation of the second side wall **143b** may be further provided on the upper casing **120**. In one example, the upper casing **120** may further include at least one first rib **148a** connecting the upper plate **121** and the vertical extension **140** with each other, and a plurality of first ribs **148a** and **148b** may be spaced apart from each other.

An electrical-wire guide **148c** for guiding the electrical-wire connected to the upper heater **148** or the lower heater **296** may be disposed between two adjacent first ribs **148a** and **148b** among the plurality of first ribs **148a** and **148b**.

The upper plate **121** may include at least two portions in a stepped form. In one example, the upper plate **121** may include a first plate portion **121a** and a second plate portion **121b** positioned higher than the first plate portion **121a**.

In this case, the tray opening **123** may be defined in first plate portion **121a**.

The first plate portion **121a** and the second plate portion **121b** may be connected with each other by a connection wall **121c**. The upper plate **121** may further include at least one second rib **148d** connecting the first plate portion **121a**, the second plate portion **121b**, and the connection wall **121a** with each other.

The upper plate **121** may further include the electrical-wire guide hook **147** that guides the electrical wire to be connected with the upper heater **148** or lower heater **296**. In one example, the electrical-wire guide hook **147** may be provided in an elastically deformable form on the first plate portion **121a**.

Hereinafter, a cold-air guide structure of the upper casing **120** will be described in detail with reference to the accompanying drawings.

FIG. **15** is a partial plan view of an ice maker viewed from above. Further, FIG. **16** is an enlarged view of a portion A of FIG. **15**. Further, FIG. **17** shows flow of cold-air on a top

face of an ice maker. Further, FIG. **18** is a perspective view of FIG. **16** taken along a line **18-18'**.

As shown in FIGS. **15** and **18**, the cold-air hole **134** is not positioned in line with the ice chamber **111** and the tray opening **123**. Thus, the cold-air guide **145** may be formed to guide the cold-air flowed from the cold-air hole **134** toward the ice chamber **111** and the tray opening **123**.

When there is no cold-air guide on the upper casing **120**, the cold-air flowed through the cold-air hole **134** may not pass through the ice chamber **111** and the tray opening **123** or pass through only small portions thereof, which may reduce the cooling efficiency.

However, in the present embodiment, the cold-air introduced through the cold-air hole **134** may be led to sequentially pass upward of the ice chamber **111** and then through the tray opening **123** by the cold-air guide **145**. Thus, effective ice-making may be achieved in the ice chamber **111**, and ice-making speeds in the plurality of ice chambers **111** may be the same as or similar to each other.

The cold-air guide **145** may include a horizontal guide **145a** and a plurality of vertical guides **145b** and **145c** for guiding the cold-air passed through the cold-air hole **134**.

The horizontal guide **145a** may guide the cold-air to upward of the upper plate **121** in which the tray opening **123** is defined, at a position at or below the lowest point of the cold-air hole **134**. Further, the horizontal guide **145a** may connect the first side wall **143a** and the upper plate **121** with each other. The horizontal guide **145a** may substantially form a portion of the bottom face of the upper plate **121**.

The plurality of vertical guides **145b** and **145c** may be arranged to intersect or to be perpendicular to the horizontal guide **145a**. The plurality of vertical guides **145b** and **145c** may include a first vertical guide **145b** and a second vertical guide **145c** spaced apart from the first vertical guide **145b**.

Further, an end of each of the first vertical guide **145b** and the second vertical guide **145c** may extend toward an ice chamber **111** on one side closest to the cold-air hole **134** among the plurality of ice chambers **111**.

The plurality of ice chambers **111** may include a first ice chamber **111a**, a second ice chamber **111b**, and a third ice chamber **111c** that are sequentially arranged in a direction to be farther away from the cold-air hole **134**. That is, the first ice chamber **111a** may be located closest to the cold-air hole **134** and the third ice chamber **111c** may be located farthest from the cold-air hole **134**. The number of the ice chambers **111** may be three or more, and when the number of the ice chambers **111** is three or more, the number is not limited.

The first vertical guide **145b** may extend from one end of the cold-air hole **134** to ends of the first ice chamber **111a** and second ice chamber **111b**. In this connection, the first vertical guide **145b** may have a predetermined curvature or a bent shape, so that the cold-air flowed from the cold-air hole **134** may be directed to the first ice chamber **111a**.

Further, the extended end of the first vertical guide **145b** may be bent toward the second ice chamber **111b**. Thus, a portion of the cold-air discharged by the first vertical guide **145b** may be directed toward the second ice chamber **111b** after passing the end of the first ice chamber **111a**.

Further, the first vertical guide **145b** may be formed not to extend to the second ice chamber **111b** and formed in a bent or rounded shape, so that interference with electrical-wires provided on the upper plate **121** may not occur.

The second vertical guide **145c** may extend toward the first ice chamber **111a** from the other end of the cold-air hole **134**, which is facing away from the end where the first vertical guide **145b** extends.

The second vertical guide **145c** may be spaced apart from the extended end of the first vertical guide **145b**, and the first ice chamber **111a** may be positioned between the ends of the first vertical guide **145b** and the second vertical guide **145c**, so that the discharged cold-air may be directed toward the first ice chamber **111a** by the cold-air guide **145**.

In one example, the second vertical guide **145c** forms a portion of a perimeter of the first through-opening **139b**. This prevents the cold-air flowing along the cold-air guide **145** from entering the first through-opening **139b** directly.

The cold-air guided by the cold-air guide **145** may be directed towards the first ice chamber **111a**. Further, the discharged cold-air may pass the plurality of ice chambers **111** sequentially, and finally, pass through the second through-opening **139c** defined next to the third ice chamber **111c**.

Thus, as shown in FIG. 17, the cold-air passed through the cold-air hole **134** may be concentrated above the upper plate **121** by the cold-air guide **145**. Further, the cold-air that passed the upper plate **121** passes through the first and second through-openings **139b** and **139c**.

Further, the supplied cold-air may be supplied to pass the plurality of ice chambers **111** sequentially along a direction of arrangement of the plurality of ice chambers **111** by the cold-air guide **145**. Further, the cold-air may be evenly supplied to all of the ice chambers **111**, so that the ice-making may be performed more effectively. Further, the ice-making speeds in the plurality of ice chambers **111** may be uniform.

In one example, it may be seen that the supplied cold-air is concentrated in the first ice chamber **111a** by the cold-air guide **145** due to the arrangement of the ice chambers **111** as shown in FIG. 17. Therefore, it will be apparent that an ice formation speed in the first ice chamber **111a**, where the cold-air is concentratedly supplied, will be high in an early state of the ice-making.

In detail, the ice inside the ice chamber **111** may be made in an indirect cooling scheme. In particular, the supply of the cold-air is concentrated on the upper tray **150** side, and the lower tray **250** is naturally cooled by the cold-air in the refrigerator. In particular, in the present embodiment, in order to make the transparent spherical ice, the lower tray **250** is periodically heated by the lower heater **296** disposed in the lower tray **250**, so that the ice formation starts from the top of the ice chamber **111** and gradually proceeds downward. Thus, bubbles generated during the ice formation inside the ice chamber **111** may be concentrated in a lower portion of the lower tray **250**, so that ice transparent except for a bottom thereof where the bubbles are concentrated may be made.

Due to the nature of such cooling scheme, the ice formation occurs first in the upper tray **150**. The cold-air is concentrated in the first ice chamber **111a**, so that the ice formation may occur quickly in the first ice chamber **111a**. Further, due to the sequential flow of the cold-air, the ice formation begins sequentially in upper portions of the second ice chamber **111b** and the third ice chamber **111c**.

Water expands in a process of being phase-changed into ice. When an ice making speed is high in the first ice chamber **111a**, an expansion force of the water is applied to the second ice chamber **111b** and the third ice chamber **111c**. Then, the water in the first ice chamber **111a** passes between the upper tray **150** and the lower tray **250** and flows toward the second ice chamber **111b**, and then the water in the second ice chamber **111b** may sequentially flows toward the third ice chamber **111c**. As a result, water of an amount greater than the set amount may be supplied into the third ice

chamber **111c**. Thus, ice made in the third ice chamber **111c** may not have a relatively complete spherical shape, and may have a size different from that of ice cubes made in other ice chambers **111a** and **111b**.

In order to prevent such a problem, the ice formation in the first ice chamber **111a** should be prevented from being performed relatively faster, and preferably, the ice formation speed should be uniform in the ice chambers **111**. Further, the ice formation may occur in the second ice chamber **111b** first rather than in the first ice chamber **111a** to prevent water from concentrating into one ice chamber **111**.

To this end, a shield **125** may be formed in the tray opening **123** corresponding to the first ice chamber **111a**, and may minimize an area of exposure of the upper tray **150** corresponding to the first ice chamber **111a**.

In detail, the shield **125** may be formed in the cavity **122** corresponding to the first ice chamber **111a**, and a bottom of the cavity **122**, which defines the tray opening **123**, may extend toward a center portion thereof to form the shield **125**. That is, a portion of the tray opening **123** corresponding to the first ice chamber **111a** has an area which is significantly small, and portions of the tray opening **123** respectively corresponding to the remaining second ice chamber **111b** and third ice chamber **111c** have larger areas.

Thus, as in a state in which the upper tray **150** is coupled to the upper casing **120** shown in FIG. 15, the top face of the upper tray **150** where the first ice chamber **111a** is formed may be further shielded by the shield **125**.

The shield **125** may be rounded or inclined in a shape corresponding to an upper portion of an outer face of a portion corresponding to the first ice chamber **111a** of the upper tray **150**. The shield **125** may extend centerward from the bottom of the cavity **122**, and may extend upward in a rounded or inclined manner. Further, an extended end of the shield **125** may define a shield opening **125a**. The shield opening **125a** may have a size to be correspond to the ejector-receiving opening **154** in communication with the first ice chamber **111a**. Accordingly, in a state in which the upper casing **120** and the upper tray **150** are coupled with each other, only the ejector-receiving opening **154** may be exposed through the portion of the tray opening **123** corresponding to the first ice chamber **111a**.

Due to such structure, even when the cold-air supplied to pass the upper plate **121** is concentratedly supplied into the first ice chamber **111a** by the cold-air guide **145**, the shield **125** may reduce the cold-air transmission into the first ice chamber **111a**. In other words, an adiabatic effect by the shield **125** may reduce the transmission of the cold-air into the first ice chamber **111a**. As a result, the ice formation in the first ice chamber **111a** may be delayed, and the ice formation may not proceed in the first ice chamber **111a** faster than in other ice chambers **111b** and **111c**.

Further, the shield opening **125a** may have a radially recessed rib groove **125c** defined therein. The rib groove **125c** may receive a portion of the first connection rib **155a** radially disposed in the ejector-receiving opening **154**. To this end, the rib groove **125c** may be recessed from a circumference of the shield opening **125a** at a position corresponding to the first connection rib **155a**. A portion of the top of the first connection rib **155a** is accommodated in the rib groove **125c**, so that the top face of the upper tray **150** that is rounded may be effectively surrounded.

Further, the portion of the top of the first connection rib **155a** is accommodated in the rib groove **125c**, so that the top of the upper tray **150** may remain in place without leaving the shield **125**. Further, the deformation of the upper tray **150** may be prevented and the upper tray **150** may be

maintained in a fixed shape, so that the ice made in the first ice chamber **111a** may be ensured to have the spherical shape always.

In one example, a shield cut **125b** may be defined in one side of the shield **125**. The shield cut **125b** may be defined by being cut at a position corresponding to the second connection rib **162** to be described below, and may be defined to receive the second connection rib **162** therein.

The shield **125** may be cut in a direction toward the second ice chamber **111b**, and may shield the remaining portion except for a portion where the second connection rib **162** is formed and the ejector-receiving opening **154** in communication with the first ice chamber **111a**.

The shield **125** may not be completely in contact with the top face of the upper tray **150** and may be spaced from the top face of the upper tray **150** by a predetermined distance. Due to such structure, an air layer may be formed between the shield **125** and the upper tray **150**. Therefore, heat insulation between the first ice chamber **111a** and the corresponding portion may be further improved.

In one example, the first through-opening **139b** and the second through-opening **139c** may be defined in both sides of the tray opening **123**. Unit guides **181** and **182** to be described below and the first link **356** moving vertically along the unit guides **181** and **182** may pass through the first through-opening **139b** and the second through-opening **139c**.

In particular, a stopper in contact with each of the unit guides **181** and **182** may protrude upward from each of the first through-opening **139b** and the second through-opening **139c** to restrain a horizontal movement of each of the unit guides **181** and **182**.

In detail, a first stopper **139ba** and a second stopper **189bb** may protrude from the first through-opening **139b**. The first stopper **139ba** and the second stopper **189bb** may be separated from each other to support the first unit guide **181** from both sides. In this connection, the second stopper **189bb** may be formed by bending the end of the second vertical guide **145c**.

Further, a third stopper **189ca** and a fourth stopper **189cb** may protrude from the second through-opening **139c**. The third stopper **189ca** and fourth stopper **189cb** may be spaced apart from each other to support the second unit guide **182** from both sides.

Because of such structure, the horizontal movement of the unit guides **181** and **182** may be prevented fundamentally. Therefore, the movement of the upper ejector **300** along the unit guides **181** and **182** may also be prevented. In the vertical movement, the upper ejector **300** may press the upper tray **150** to deform or detach the upper tray **150**, so that the upper ejector **300** should be vertically moved at a fixed position. Thus, the upper ejector **300** is not interfered with the upper tray **150** by the stopper during the vertical movement process.

In one example, the fourth stopper **189cb** among the stoppers may have a height slightly smaller than that of the other stoppers **139ba**, **139bb**, and **139ca**. This is to allow the cold-air flowing along the upper tray **150** to pass the fourth stopper **189cb** and be discharged smoothly through the second through-opening **139c**.

Hereinafter, the upper tray **150** will be described in more detail with reference to the accompanying drawings.

FIG. **19** is a perspective view of an upper tray according to an embodiment of the present disclosure viewed from above. Further, FIG. **20** is a perspective view of an upper tray viewed from below. Further, FIG. **21** is a side view of an upper tray.

Referring to FIGS. **19** to **21**, the upper tray **150** may be made of a flexible or soft material that may be returned to its original shape after being deformed by an external force.

In one example, the upper tray **150** may be made of a silicone material. When the upper tray **150** is made of the silicone material as in the present embodiment, in the ice-removal process, even when the upper tray **150** is deformed by the external force, the upper tray **150** returns to its original shape, so that the spherical ice may be made despite the repetitive ice generation.

Further, when the upper tray **150** is made of the silicone material, the upper tray **150** may be prevented from melting or being thermally deformed by heat provided from the upper heater **148** to be described later.

The upper tray **150** may include the upper tray body **151** forming the upper chamber **152** that is a portion of the ice chamber **111**. A plurality of upper chambers **152** may be sequentially formed on the upper tray body **151**. The plurality of upper chambers **152** may include a first upper chamber **152a**, a second upper chamber **152b**, and a third upper chamber **152c**, which may be sequentially arranged in series on the upper tray **151**.

The upper tray body **151** may include three chamber walls **153** that form three independent upper chambers **152a**, **152b**, and **152c**, and the three chamber walls **153** may be integrally formed and connected to each other.

The upper chamber **152** may be formed in a hemispherical shape. That is, an upper portion of the spherical ice may be formed by the upper chamber **152**.

An ejector-receiving opening **154** through which the upper ejector **300** may enter or exit for the ice-removal may be defined in an upper portion of the upper tray body **151**. The ejector-receiving opening **154** may be defined in a top of each of the upper chambers **152**. Therefore, each upper ejector **300** may independently push the ice cubes in each of the ice chambers **111** to remove the ice cubes. In another example, the ejector-receiving opening **154** has a diameter sufficient for the upper ejector **300** to enter and exit, which allows the cold-air flowing along the upper plate **121** to enter and exit.

In one example, in order to minimize the deformation of the portion of the upper tray **150** near the ejector-receiving opening **154** in a process in which the upper ejector **300** is inserted through the ejector-receiving opening **154**, an opening-defining wall **155** may be formed on the upper tray **150**. The opening-defining wall **155** may be disposed along the circumference of the ejector-receiving opening **154**, and may extend upward from the upper tray body **151**.

The opening-defining wall **155** may be formed in a cylindrical shape. Thus, the upper ejector **300** may pass through an internal space of the opening-defining wall **155** and pass through the ejector-receiving opening **154**.

The opening-defining wall may act as a guide for movement of the upper ejector **300**, and at the same time, may define extra space to prevent the water contained in the ice chamber **111** from overflowing. Therefore, the internal space of the opening-defining wall **155**, that is, the space in which the ejector-receiving opening **154** is defined, may be referred to as a buffer.

Since the buffer is formed, even when the water of the amount equal to or greater than the predefined amount is flowed into the ice chamber **111**, the water will not overflow. When the water inside the ice chamber **111** overflows, ice cubes respectively contained in adjacent ice chambers **111** may be connected with each other, so that the ice may not be easily separated from the upper tray **150**. Further, when the water inside the ice chamber may overflow from the

upper tray 150, serious problems, such as induction of attachment of the ice cubes in the ice chambers may occur.

In the present embodiment, the buffer is formed by the opening-defining wall 155 to prevent the water inside the ice chamber 111 from overflowing. When a height of the opening-defining wall 155 becomes excessively large to form the buffer, the buffer may interfere with the movement of the cold-air of passing the upper plate 121 and inhibit smooth movement of the cold-air. On the contrary, when the height of the opening-defining wall 155 becomes excessively small, a role of the buffer may not be expected and it may be difficult to guide the movement of the upper ejector 300.

In one example, a preferred height of the buffer may be a height corresponding to the horizontal extension 142 of the upper tray 150. Further, a capacity of the buffer may be set based on an inflow amount of ice debris that may be attached along a circumference of the upper tray body 151. Therefore, it is preferable that an internal volume of the buffer is defined to have a capacity of 2 to 4% of a volume of the ice chamber 111.

When an inner diameter of the buffer is too large, the top of the completed ice may have an excessively wide flat shape, and thus, an image of the spherical ice may not be provided to the user. Therefore, the buffer should be formed to have a proper inner diameter.

The inner diameter of the buffer may be larger than a diameter of the upper ejector 300 to facilitate entry and exit of the upper ejector 300, and may be determined to satisfy the water capacity and height of the buffer.

In one example, the first connection rib 155a for connecting the side of the opening-defining wall 155 and the top face of the upper tray body 151 with each other may be formed on the circumference of the opening-defining wall 155. A plurality of the first connection ribs 155a may be formed at regular intervals along the circumference of the opening-defining wall 155. Thus, the opening-defining wall 155 may be supported by the first connection rib 155a such that the opening-defining wall 155 is not deformed easily. Even when the upper ejector 300 is in contact with the opening-defining wall 155 in a process of being inserted into the ejector-receiving opening 154, the opening-defining wall 155 may maintain its shape and position without being deformed.

The first connection rib 155a may be formed on each of all the first upper chamber 152a and second upper chamber 152b and third upper chamber 152c.

In one example, two opening-defining walls 155 respectively corresponding to the second upper chamber 152b and the third upper chamber 152c may be connected with each other by a second connection rib 162. The second connection rib 162 may connect the second upper chamber 152b and the third upper chamber 152c with each other to further prevent the deformation of the opening-defining wall 155, and at the same time, to prevent deformation of top faces of the second upper chamber 152b and the third upper chamber 152c.

In one example, the second connection rib 162 may also be disposed between the first upper chamber 152a and the second upper chamber 152b to connect the first upper chamber 152a and the second upper chamber 152b with each other, but the second connection rib 162 may be omitted since the second receiving space 161 in which the temperature sensor 500 is disposed is defined between the first upper chamber 152a and the second upper chamber 152b.

The water-supply guide 156 may be formed on the opening-defining wall 155 corresponding to one of the three upper chambers 152a, 152b, and 152c.

Although not limited, the water-supply guide 156 may be formed on the opening-defining wall 155 corresponding to the second upper chamber 152b. The water-supply guide 156 may be inclined upward from the opening-defining wall 155 in a direction farther away from the second upper chamber 152b. Even when only one water-supply guide is formed on the upper chamber 152, the upper tray 150 and the lower tray 250 may not be closed during the water-supply, so that water may be evenly filled in all the ice chambers 111.

The upper tray 150 may further include a first receiving space 160. The first receiving space 160 may accommodate the cavity 122 of the upper casing 120 therein. The cavity 122 includes a heater-mounted portion 124, and the heater-mounted portion 124 includes the upper heater 148, so that it may be understood that the upper heater 148 is accommodated in the first receiving space 160.

The first receiving space 160 may be defined in a form surrounding the upper chambers 152a, 152b, and 152c. The first receiving space 160 may be defined as the top face of the upper tray body 151 is recessed downward.

The temperature sensor 500 may be accommodated in the second receiving space 161, and the temperature sensor 500 may be in contact with an outer face of the upper tray body 151 while the temperature sensor 500 is mounted.

The chamber wall 153 of the upper tray body 151 may include a vertical wall 153a and a curved wall 153b.

The curved wall 153b may be upwardly rounded in a direction farther away from the upper chamber 152. In this connection, a curvature of the curved wall 153b may be the same as a curvature of a curved wall 260b of the lower tray 250 to be described below. Thus, when the lower tray 250 pivots, the upper tray 150 and the lower tray 250 do not interfere with each other.

The upper tray 150 may further include a horizontal extension 164 extending in a horizontal direction from a perimeter of the upper tray body 151. The horizontal extension 164 may, for example, extend along a perimeter of a top edge of the upper tray body 151.

The horizontal extension 164 may be in contact with the upper casing 120 and the upper support 170. A bottom face 164b of the horizontal extension 164 may be in contact with the upper support 170, and a top face 164a of the horizontal extension 164 may be in contact with the upper casing 120. Thus, at least a portion of the horizontal extension 164 may be fixedly mounted between the upper casing 120 and the upper support 170.

The horizontal extension 164 may include a plurality of upper protrusions 165 respectively inserted into the plurality of upper slots 131 and a plurality of upper protrusions 166 respectively inserted into the plurality of upper slots 132.

The plurality of upper protrusions 165 and 166 may include a plurality of first upper protrusions 165 and a plurality of second upper protrusions 166 positioned opposite to the first upper protrusions 165 around the ejector-receiving opening 154.

The first upper protrusion 165 may be formed in a shape corresponding to the first upper slot 131 to be inserted into the first upper slot 131, and the second upper protrusion 166 may be formed in a shape corresponding to the second upper slot 132 to be inserted into the second upper slot 132. Further, the first upper protrusion 165 and the second upper protrusion 166 may protrude from the top face 164a of the horizontal extension 164.

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The first upper protrusion **165** may be, for example, formed in a curved shape. Further, the second upper protrusion **166** may be, for example, formed in a curved shape. Further, the first upper protrusion **165** and the second upper protrusion **166** may be arranged to face away from each other around the ice chamber **111**, so that the perimeter of the ice chamber **111** may be maintained in a firmly coupled state, in particular.

The horizontal extension **164** may further include a plurality of lower protrusions **167** and a plurality of lower protrusions **168**. Each of the plurality of lower protrusions **167** and each of the plurality of lower protrusions **168** may be respectively inserted into lower slots **176** and **177** of the upper support **170** to be described later.

The plurality of lower protrusions **167** and **168** may include a first lower protrusion **167** and a second lower protrusion **168** positioned opposite to the first lower protrusion **167** around the upper chamber **152**.

The first lower protrusion **167** and the second lower protrusion **168** may protrude downward from the bottom face **164b** of the horizontal extension **164**. The first lower protrusion **167** and the second lower protrusion **168** may be formed in the same shape as the first upper protrusion **165** and the second upper protrusion **166**, and may be formed to protrude in a direction opposite to a protruding direction of the first upper protrusion **165** and the second upper protrusion **166**.

Thus, because of the upper protrusions **165** and **166** and the lower protrusions **167** and **168**, not only the upper tray **150** is coupled between the upper casing **120** and the upper support, but also deformation of the ice chamber **111** or the horizontal extension **264** adjacent to the ice chamber **111** is prevented in the ice-making or ice-removal process.

The horizontal extension **164** may have a through-hole **169** defined therein to be penetrated by a coupling boss of the upper support **170** to be described later. Some of a plurality of through-holes **169** may be located between two adjacent first upper protrusions **165** or two adjacent first lower protrusions **167**. Some of the remaining through-holes **169** may be located between two adjacent second lower protrusions **168** or may be defined to face a region between the two second lower protrusions **168**.

In one example, an upper rib **153d** may be formed on the bottom face **153c** of the upper tray body **151**. The upper rib **153d** is for hermetic sealing between the upper tray **150** and the lower tray **250**, which may be formed along the perimeter of each of the ice chambers **111**.

In a structure in which the ice chamber **111** is formed by the coupling of the upper tray **150** and the lower tray **250**, even when the upper tray **150** and the lower tray **250** remain in close contact with each other at first, a gap is defined between the upper tray **150** and the lower tray **250** due to a volume expansion occurring in a process in which the water is phase-changed into the ice. When the ice formation occurs in a state in which the upper tray **150** and the lower tray **250** are separated from each other, a burr that protrudes in a shape of an ice strip is generated along a circumference of the completed spherical ice. Such burr generation causes a poor shape of the spherical ice itself. In particular, when the ice is connected to ice debris formed in a circumferential space between the upper tray **150** and the lower tray **250**, the shape of the spherical ice becomes worse.

In order to solve such problem, in the present embodiment, the upper rib **153d** may be formed at the bottom of the upper tray **150**. The upper rib **153d** may shield between the upper tray **150** and the lower tray **250** even when the volume expansion of the water due to the phase-change occurs. Thus

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the burr may be prevented from being formed along the circumference of the completed spherical ice.

In detail, the upper rib **153d** may be formed along the perimeter of each of the upper chambers **152**, and may protrude downward in a thin rib shape. Therefore, in a situation where the upper tray **150** and the lower tray **250** are completely closed, deformation of the upper rib **153d** will not interfere with the sealing of the upper tray **150** between the lower tray **250**.

Therefore, the upper rib **153d** may not be formed excessively long. Further, it is preferable that the upper rib **153d** is formed to have a height sufficient to cover the gap between the upper tray **150** and the lower tray **250**. In one example, the upper tray **150** and the lower tray **250** may be separated from each other by about 0.5 mm to 1 mm when the ice is formed, and correspondingly the upper rib **153d** may be formed with a height **h1** of about 0.8 mm.

In one example, the lower tray **250** may be pivoted in a state in which a pivoting shaft thereof is positioned outward (rightward in FIG. 21) of the curved wall **153b**. In such structure, when the lower tray **250** is closed by pivoting, a portion thereof close to the pivoting shaft is brought to be in contact with the upper tray **150** first, and then a portion thereof far away from the pivoting shaft is sequentially brought to be in contact with the upper tray **150** as the upper tray **150** and the lower tray **250** are compressed.

Thus, when the upper rib **153d** is formed along an entirety of the perimeter of the bottom of the upper chamber **152**, interference of the upper rib **153d** may occur at a position near the pivoting shaft, which may cause the upper tray **150** and the lower tray **250** not to be closed completely. In particular, there is a problem that the upper tray **150** and the lower tray **250** are not closed at a position far away from the pivoting shaft.

In order to prevent such problem, the upper rib **153d** may be formed to be inclined along the perimeter of the upper chamber **152**. The upper rib **153d** may be formed such that a height thereof increases toward the vertical wall **153a** and decreases toward the curved wall **153b**. One end of the upper rib **153d** close to the vertical wall **153b** may have a maximum height **h1**, the other end of the upper rib **153d** close to the curved wall **153b** may have a minimum height, and the minimum height may be zero.

Further, the upper rib **153d** may not be formed on the entirety of the upper chamber **152**, but may be formed on the remaining portion of the upper chamber **152** except for a portion thereof near the curved wall **153b**. In one example, as shown in FIG. 21, based on a length **L** of an entire width of the bottom of the upper tray **150**, the upper rib **153d** may start to protrude from a position away from an end at which the curved wall **153b** is formed by $\frac{1}{5}$ length **L1** and extend to an end at which the vertical wall **153b** is formed. Therefore, a width of the upper rib **153d** may be $\frac{4}{5}$ length **L2** based on the length **L** of the entire width of the bottom of the upper tray **150**. In one example, when the width of the bottom of the upper tray **150** is 50 mm, the upper rib **153d** extends downwards from a position 10 mm away from the end of the curved wall **153b**, and may extend to the end adjacent to the vertical wall **153a**. In this connection, the width of the upper rib **153d** may be 40 mm.

In another example, there may be some differences, but the point where the upper rib **153d** starts to protrude may be a point away from the curved wall **153b** such that the interference may be minimized when the lower tray **250** is closed, and at the same time, the gap between the upper tray **150** and the lower tray **250** may be covered.

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Further, the height of the upper rib **153d** may increase from the curved wall **153b** side to the vertical wall **153a** side. Thus, when the lower tray **250** is opened by the freezing, the gap between the upper tray **150** and the lower tray **250** having varying height may be effectively covered.

Hereinafter, the upper support **170** will be described in more detail with reference to the accompanying drawings.

FIG. **22** is a perspective view of an upper support according to an embodiment of the present disclosure viewed from above. Further, FIG. **23** is a perspective view of an upper support viewed from below. Further, FIG. **24** is a cross-sectional view showing a coupling structure of an upper assembly according to an embodiment of the present disclosure.

Referring to FIGS. **22** to **24**, the upper support **170** may include a plate shaped support plate **171** that supports the upper tray **150** from below. Further, a top face of the support plate **171** may be in contact with the bottom face **164b** of the horizontal extension **164** of the upper tray **150**.

The support plate **171** may have a plate opening **172** defined therein to be penetrated by the upper tray body **151**. A side wall **174**, which is bent upward, may be formed along an edge of the support plate **171**. The side wall **174** may be in contact with a perimeter of the side of the horizontal extension **164** to restrain the upper tray **150**.

The support plate **171** may include a plurality of lower slots **176** and a plurality of lower slots **177**. The plurality of lower slots **176** and the plurality of lower slots **177** may include a plurality of first lower slots **176** into which the first lower protrusions **167** are inserted respectively and a plurality of second lower slots **177** into which the second lower protrusions **168** are inserted respectively.

The plurality of first lower slots **176** and the plurality of second lower slots **177** may be formed to be inserted into each other in a shape corresponding to a position corresponding to the first lower protrusion **167** and the second lower protrusion **168**, respectively.

The first lower slot **176** may be defined to have a shape corresponding to the first lower protrusion **167** at a position corresponding to the first lower protrusion **167** such that the first lower protrusion **167** may be inserted into the first lower slot **176**. Further, the second lower slot **177** may be defined to have a shape corresponding to the second lower protrusion **168** at a position corresponding to the second lower protrusion **168** such that the second lower protrusion **168** may be inserted into the second lower slot **177**.

The support plate **171** may further include a plurality of coupling bosses **175**. The plurality of coupling bosses **175** may protrude upward from the top face of the support plate **171**. Each coupling boss **175** may be inserted into the sleeve **133** of the upper casing **120** by passing through the through-hole **169** of the horizontal extension **164**.

In a state in which the coupling boss **175** is inserted into the sleeve **133**, a top face of the coupling boss **175** may be located at the same vertical level or below the top face of the sleeve **133**. The fastener such as a bolt may be fastened to the coupling boss **175**, so that the assembly of the upper assembly **110** may be completed, and the upper casing **120**, the upper tray **150**, and upper support **170** may be rigidly coupled to each other.

The upper support **170** may further include a plurality of unit guides **181** and **182** for guiding the connector **350** connected to the upper ejector **300**. The plurality of unit guides **181** and **182** may be respectively formed at both ends of the upper plate **170** to be spaced apart each other, and may be respectively formed at positions facing away from each other.

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The unit guides **181** and **182** may respectively extend upwards from the both ends of the support plate **171**. Further, a guide slot **183** extending in the vertical direction may be defined in each of the unit guides **181** and **182**.

In a state in which each of both ends of the ejector body **310** of the upper ejector **300** penetrates the guide slot **183**, the connector **350** is connected to the ejector body **310**. Thus, in the pivoting process of the lower assembly **200**, when the pivoting force is transmitted to the ejector body **310** by the connector **350**, the ejector body **310** may vertically move along the guide slot **183**.

In one example, a plate electrical-wire guide **178** extending downward may be formed at one side of the support plate **171**. The plate electrical-wire guide **178** is for guiding the electrical wire connected to the lower heater **296**, which may be formed in a hook shape extending downward. The plate electrical-wire guide **178** is formed on an edge of the support plate **171** to minimize interference of the electrical-wire with other components.

Further, an electrical-wire opening **178a** may be defined in the support plate **171** to correspond to the plate electrical-wire guide **178**. The electrical-wire opening **178a** may direct the electrical-wire guided by the plate electrical-wire guide **178** to pass through the support plate **171** and toward the upper casing **120**.

In one example, as shown in FIGS. **13** and **24**, the heater-mounted portion **124** may be formed in the upper casing **120**. The heater-mounted portion **124** may be formed on the bottom of the cavity **122** defined along the tray opening **123**, and may include a heater-receiving groove **124a** defined therein for accommodating the upper heater **148** therein.

The upper heater **148** may be a wire type heater. Thus, the upper heater **148** may be inserted into the heater-receiving groove **124a**, and may be disposed along a perimeter of the tray opening **123** of the curved shape. The upper heater **148** is brought to be in contact with the upper tray **150** by the assembling the upper assembly **110**, so that the heat transfer to the upper tray **150** may be achieved.

Further, the upper heater **148** may be a DC powered DC heater. When the upper heater **148** is operated for the ice-removal, heat from the upper heater **148** may be transferred to the upper tray **150**, so that the ice may be separated from a surface (inner face) of the upper tray **150**.

When the upper tray **150** is made of the metal material and as the heat from the upper heater **148** is strong, after the upper heater **148** is turned off, a portion of the ice heated by the upper heater **148** adheres again to the surface of the upper tray **150**, so that the ice becomes opaque.

In other words, an opaque strip of a shape corresponding to the upper heater is formed along a circumference of the ice.

However, in the present embodiment, the DC heater having a low output is used, and the upper tray **150** is made of silicone, so that an amount of the heat transferred to the upper tray **150** is reduced and a thermal conductivity of the upper tray **150** itself is lowered.

Therefore, since the heat is not concentrated in a local portion of the ice, and a small amount of the heat is gradually applied to the ice, the formation of the opaque strip along the circumference of the ice may be prevented while the ice is effectively separated from the upper tray **150**.

The upper heater **148** may be disposed to surround the perimeter of each of the plurality of upper chambers **152** such that the heat from the upper heater **148** may be evenly transferred to the plurality of upper chambers **152** of the upper tray **150**.

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In one example, as shown in FIG. 24, in a state in which the upper heater 148 is coupled to the heater-mounted portion 124 of the upper casing 120, the upper assembly may be assembled by coupling the upper casing 120, the upper tray 150, and upper support 170 with each other.

In this connection, the first upper protrusion 165 of the upper tray 150 may be inserted into the first upper slot 131 of the upper casing 120, and the second upper protrusion 166 of the upper tray 150 may be inserted into the second upper slot 132 of the upper casing 120.

Further, the first lower protrusion 167 of the upper tray 150 may be inserted into the first lower slot 176 of the upper support 170, and the second lower protrusion 168 of the upper tray may be inserted into the second lower slot 177 of the upper support 170.

Then, the coupling boss 175 of the upper support 170 passes through the through-hole 169 of the upper tray 150 and is received within the sleeve 133 of the upper casing 120. In this state, the fastener such as the bolt may be fastened to the coupling boss 175 from upward of the coupling boss 175.

When the upper assembly 110 is assembled, the heater-mounted portion 124 in combination with the upper heater 148 is received in the first receiving space 160 of the upper tray 150. In a state in which the heater-mounted portion 124 is received in the first receiving space 160, the upper heater 148 is in contact with the bottom face 160a of the first receiving space 160.

As in the present embodiment, when the upper heater 148 is accommodated in the heater-mounted portion 124 in the recessed form and in contact with the upper tray body 151, the transferring of the heat from the upper heater 148 to other components other than the upper tray body 151 may be minimized.

In one example, the present disclosure may also include another example of another ice maker. In another embodiment of the present disclosure, there are differences only in a structure of the upper tray 150 and a structure of the shield 125 of the upper casing 120, and other components will be identical. The same component will not be described in detail and will be described using the same reference numerals.

Hereinafter, structures of the upper tray and the shield according to another embodiment of the present disclosure will be described with reference to the drawings.

FIG. 25 is a perspective view of an upper tray according to another embodiment of the present disclosure viewed from above. Further, FIG. 26 is a cross-sectional view of FIG. 25 taken along a line 26-26'. Further, FIG. 27 is a cross-sectional view of FIG. 25 taken along a line 27-27'. Further, FIG. 28 is a partially-cut perspective view showing a structure of a shield of an upper casing according to another embodiment of the present disclosure.

As shown in FIGS. 25 to 28, an upper tray 150' according to another embodiment of the present disclosure differs only in structures of the opening-defining wall 155 and the top face of the upper chamber 152 connected with the opening-defining wall 155, but other components thereof are the same as in the above-described embodiment.

The upper tray 150' includes the horizontal extension 142 formed thereon. Further, the horizontal extension 142 may include the first upper protrusion 165, the second upper protrusion 166, the first lower protrusion 167, and the second lower protrusion 168 formed thereon. Further, the through-hole 169 may be defined in the horizontal extension 142.

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Further, the upper chamber 152 may be formed in the upper tray body 151 extending downward from the horizontal extension 142. The upper chamber 152 may include the first upper chamber 152a, the second upper chamber 152b, and the third upper chamber 152c arranged successively from a side close to the cold-air guide 145.

The opening-defining wall 155 that defines the ejector-receiving opening 154 may be formed on each of the upper chambers 152. Further, the water-supply guide 156 may be formed on the opening-defining wall 155 of the second upper chamber 152b. In one example, a plurality of ribs that connect the outer face of the opening-defining wall 155 and the top face of the upper chamber 152 may be arranged on the opening-defining wall 155 of each the upper chambers 152.

In detail, the plurality of radially arranged first connection ribs 155a may be formed on the first upper chamber 152a and the second upper chamber 152b. The first connection rib 155a may prevent the deformation of the opening-defining wall 155. Further, the first upper chamber 152a and the second upper chamber 152b may be connected with each other by a second connection rib 162, and the deformation of the first upper chamber 152a, the second upper chamber 152b, and the opening-defining wall 155 may be further prevented.

Further, the third upper chamber 152c may be spaced apart for mounting the temperature sensor 500. Thus, a plurality of third connection ribs 155c may be formed to prevent deformation of the opening-defining wall 155 formed upward of the third upper chamber 152c. The plurality of third connection ribs 155c may be formed in the same shape as the first connection rib 155a, and may be arranged at an interval narrower than in the first upper chamber 152a or the second upper chamber 152b. That is, the third upper chamber 152c will have more ribs than the other chambers 152a and 152b. Thus, even when the third upper chamber 152c is placed separately, a shape the third upper chamber 152c may be maintained, and the third upper chamber 152c may be prevented from deforming easily.

In one example, a thermally-insulating portion 152e may be formed on the top face of the first upper chamber 152a. The thermally-insulating portion 152e is for further blocking the cold-air passing through the upper tray 150' and upper casing 120, which further protrudes along the perimeter of the first upper chamber 152a. The thermally-insulating portion 152e is a face exposed through the top face of the first upper chamber 152a, that is, exposed upwardly of the upper tray 150', which is formed along the perimeter of the bottom of the opening-defining wall 155.

In detail, as shown in FIGS. 26 and 27, a thickness D1 of the upper face of the first upper chamber 152a may be larger than a thickness D2 of the upper faces of the second upper chamber 152b and of the third upper chamber 152c by the thermally-insulating portion 152e.

When the thickness of the first upper chamber 152a is larger by the thermally-insulating portion 152e, even in a state in which the supplied cold-air is concentrated on the first upper chamber 152a side by the cold-air guide 145, the amount of the cold-air transferred to the first upper chamber 152a may be reduced. As a result, the thermally-insulating portion 152e may reduce the ice formation speed in the first upper chamber 152a. Thus, the ice formation may occur first in the second upper chamber 152b or the ice formation may occur at a uniform speed in the upper chambers 152.

In one example, the shield 126 that extends from the cavity 122 of the upper casing 120 may be formed upward of the first upper chamber 152a. The shield 126 protrudes

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upward to cover the top face of the first upper chamber **152a**, and may be formed round or inclined.

A shield opening **126a** is defined at a top of the shield **126**, and the shield opening **126a** is in contact with the top of the ejector-receiving opening **154**. Therefore, when the upper tray **150'** is viewed from above, the remaining portion of the first upper chamber **152a** except for the ejector-receiving opening **154** is covered by the shield **126**. That is, a region of the thermally-insulating portion **152e** is covered by the shield **126**.

Further, a rib groove **126c** to be inserted into the top of the first connection rib **155a** may be defined along a circumference of the shield opening **126a**, so that positions of the top of the first upper chamber **152a** and the opening-defining wall **155** may be maintained in place.

With such structure, the first upper chamber **152a** may be thermally-insulated further, and the ice formation speed in the first upper chamber **152a** may be reduced despite the cold-air concentratedly supplied by the cold-air guide **145**.

In one example, a cut **126e** may be defined in the shield **126** corresponding to the second connection rib **162**. The cut **126e** is formed by cutting a portion of the shield **125**, which may be opened to allow the second connection rib **162** to pass therethrough completely.

When the cut **126e** is too narrow, in a process in which the upper tray **150'** is deformed during the ice-removal process by the upper ejector **300**, the second connection rib **162** may be deviated from the cut **126e** and jammed. In this case, the second connection rib **162** is unable to return to its original position after the ice-removal, causing defects during the ice-making. On the contrary, when the cut **126e** is too wide, the thermal insulation effect may be significantly reduced due to the inflow of the cold-air.

Thus, in the present embodiment, a width of the cut **126e** may decrease upwardly. That is, both ends **126b** of the cut **126e** may be formed in an inclined or rounded shape, so that a width of a bottom of the cut **126e** may be the widest and a width of a top of the cut **126e** may be the narrowest. Further, the width of the top of the cut **126e** may correspond to or be somewhat larger than the thickness of the second connection rib **162**.

Therefore, when the upper tray **150'** is deformed and then restored during the ice-removal by the upper ejector **300**, the second connection rib **162** may be easily inserted into the cut **126e** and moved along both ends of the cut **126e**, so that the upper tray **150'** may be restored at a correct position.

In one example, when the opening of the bottom of the cut **126e** becomes large, the cold-air may be introduced through the bottom of the cut **126e**. In order to prevent this, fourth connection ribs **155b** may be formed along the perimeter of the first upper chamber **152a**.

Like the first connection rib **155a**, the fourth connection rib **155b** may be formed to connect the outer face of the opening-defining wall **155** and the upper face of the first upper chamber **152a** with each other, and an outer end thereof may be inclined. Further, a height of the fourth connection rib **155b** may be smaller than that of the first connection rib **155a**, so that the fourth connection rib **155b** may be in contact with the bottom face of the shield without interfering with the top of the shield **126**.

The fourth connection ribs **155b** may be respectively located at both left and right sides around the second connection rib **162**. Further, the fourth connection ribs **155b** may be respectively located at positions corresponding to the both ends of the cut **126e** or slightly outward of the both ends of the cut **126e**. The fourth connection ribs **155b** may be in close contact with the inner face of the shield **126**.

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Thus, a space between the shield **126** and the top face of the first upper chamber **152a** may be shielded to prevent the cold-air from entering through the cut **126c**.

The shield **126** and the top face of the first upper chamber **152a** may be somewhat spaced apart from each other, and an air layer may be formed therebetween. The inflow of the cold-air from the air layer may be blocked by the fourth connection rib **155b**. Therefore, the top face of the first upper chamber **152a** may be further thermally insulated to further reduce the ice formation speed in the first upper chamber **152a**.

Hereinafter, the lower assembly **200** will be described in more detail with reference to the accompanying drawings.

FIG. **29** is a perspective view of a lower assembly according to an embodiment of the present disclosure. Further, FIG. **30** is an exploded perspective view of a lower assembly viewed from above. Further, FIG. **31** is an exploded perspective view of a lower assembly viewed from below.

As shown in FIGS. **29** to **31**, the lower assembly **200** may include a lower tray **250**, a lower support **270** and a lower casing **210**.

The lower casing **210** may surround a portion of a perimeter of the lower tray **250**, and the lower support **270** may support the lower tray **250**. Further, the connector **350** may be coupled to both sides of the lower support **270**.

The lower casing **210** may include a lower plate **211** for fixing the lower tray **250**. A portion of the lower tray **250** may be fixed in contact with a bottom face of the lower plate **211**. The lower plate **211** may be provided with an opening **212** defined therein through which a portion of the lower tray **250** penetrates.

In one example, when the lower tray **250** is fixed to the lower plate **211** in a state of being positioned below the lower plate **211**, a portion of the lower tray **250** may protrude upward of the lower plate **211** through the opening **212**.

The lower casing **210** may further include a side wall **214** surrounding the portion of the lower tray **250** passed through the lower plate **211**. The side wall **214** may include a vertical portion **214a** and a curved portion **215**.

The vertical portion **214a** is a wall extending vertically upward from the lower plate **211**. The curved portion **215** is a wall that is rounded upwardly in a direction farther away from the opening **212** upwards from the lower plate **211**.

The vertical portion **214a** may include a first coupling slit **214b** defined therein to be coupled with the lower tray **250**. The first coupling slit **214b** may be defined as a top of the vertical portion **214a** is recessed downward.

The curved portion **215** may include a second coupling slit **215a** defined therein to be coupled with the lower tray **250**. The second coupling slit **215a** may be defined as a top of the curved portion **215** is recessed downward. The second coupling slit **215a** may restrain a lower portion of the second coupling protrusion **261** protruding from the lower tray **250**.

Further, a protruding confiner **213** protruding upward may be formed on a rear face of the curved portion **215**. The protruding confiner **213** may be formed at a position corresponding to the second coupling slit **215a**, and may protrude outward from a face in which the second coupling slit **215a** is defined to restrain an upper portion of the second coupling protrusion **261**.

That is, both top and bottom of the second coupling protrusion **261** may be restrained by the second coupling slit **215a** and the protruding confiner **213**, respectively. Thus, the lower tray **250** may be firmly fixed to the lower casing **210**.

Structure of the second coupling protrusion **261**, the second coupling slit **215a**, and the protruding confiner **213** will be described in more detail below.

In one example, the lower casing **210** may further include a first coupling boss **216** and a second coupling boss **217**. The first coupling boss **216** may protrude downward from the bottom face of the lower plate **211**. In one example, a plurality of first coupling bosses **216** may protrude downward from the lower plate **211**.

The second coupling boss **217** may protrude downward from the bottom face of the lower plate **211**. In one example, a plurality of second coupling bosses **217** may protrude from the lower plate **211**.

In the present embodiment, a length of the first coupling boss **216** and a length of the second coupling boss **217** may be different. In one example, the length of the second coupling boss **217** may be larger than the length of the first coupling boss **216**.

A first fastener may be fastened to the first coupling boss **216** from upward of the first coupling boss **216**. On the other hand, a second fastener may be fastened to the second coupling boss **217** from below of the second coupling boss **217**.

A groove **215b** for a movement of the fastener may be defined in the curved portion **215** such that the first fastener does not interfere with the curved portion **215** in a process in which the first fastener is fastened to the first coupling boss **216**.

The lower casing **210** may further include a slot **218** for coupling with the lower tray **250** defined therein. A portion of the lower tray **250** may be inserted into the slot **218**. The slot **218** may be located adjacent to the vertical portion **214a**.

The lower casing **210** may further include a receiving groove **218a** defined therein for insertion of a portion of the lower tray **250**. The receiving groove **218a** may be defined as a portion of the lower plate **211** is recessed toward the curved portion **215**.

The lower casing **210** may further include an extension wall **219** in contact with a portion of a perimeter of a side of the lower plate **212** in a state in which the lower casing **210** is coupled with the lower tray **250**.

In one example, the lower tray **250** may be made of a flexible material or a flexible material such that the lower tray **250** may be deformed by an external force and then returned to its original form.

In one example, the lower tray **250** may be made of a silicone material. When the lower tray **250** is made of the silicone material as in the present embodiment, even when the external force is applied to the lower tray **250** and the shape of the lower tray **250** is deformed in the ice-removal process, the lower tray **250** may be returned to its original shape. Thus, the spherical ice may be generated despite the repeated ice generation.

Further, when the lower tray **250** is made of the silicone material, the lower tray **250** may be prevented from being melted or thermally deformed by heat provided from a lower heater to be described later.

In one example, the lower tray **250** may be made of the same material as the upper tray **150**, or may be made of a material softer than the material of the upper tray **150**. That is, when the lower tray **250** and the upper tray **150** come into contact with each other for the ice-making, since the lower tray **250** has a lower hardness, while the top of the lower tray **250** is deformed, the upper tray **150** and the lower tray **250** may be pressed and sealed with each.

Further, since the lower tray **250** has a structure that is repeatedly deformed by direct contact with the lower ejector

400, the lower tray **250** may be made of a material having a low hardness to facilitate the deformation.

However, when the hardness of the lower tray **250** is too low, another portion of the lower chamber **252** may be deformed too. Thus, it is preferable that the lower tray **250** is formed to have an appropriate hardness to maintain the shape.

The lower tray **250** may include a lower tray body **251** that forms a lower chamber **252** that is a portion of the ice chamber **111**. The lower tray body **251** may form a plurality of lower chambers **252**.

In one example, the plurality of lower chambers **252** may include a first lower chamber **252a**, a second lower chamber **252b**, and a third lower chamber **252c**.

The lower tray body **251** may include three chamber walls **252d** forming the three independent lower chambers **252a**, **252b**, and **252c**. The three chamber walls **252d** may be formed integrally to form the lower tray body **251**. Further, the first lower chamber **252a**, the second lower chamber **252b**, and the third lower chamber **252c** may be arranged in series.

The lower chamber **252** may be formed in a hemispherical form or a form similar to the hemisphere. That is, a lower portion of the spherical ice may be formed by the lower chamber **252**. Herein, the form similar to the hemisphere means a form that is not a complete hemisphere but is almost close to the hemisphere.

The lower tray **250** may further include a lower tray mounting face **253** extending horizontally from a top edge of the lower tray body **251**. The lower tray mounting face **253** may be formed sequentially along a circumference of the top of the lower tray body **251**. Further, in coupling with the upper tray **150**, the lower tray mounting face **253** may be in close contact with the top face **153c** of the upper tray **150**.

The lower tray **250** may further include a side wall **260** extending upwardly from an outer end of the lower tray mounting face **253**. Further, the side wall **260** may surround the upper tray body **151** seated on the top face of the lower tray body **251** in a state in which the upper tray **150** and the lower tray **250** are coupled together.

The side wall **260** may include a first wall **260a** surrounding the vertical wall **153a** of the upper tray body **151** and a second wall **260b** surrounding the curved wall **153b** of the upper tray body **151**.

The first wall **260a** is a vertical wall extending vertically from the top face of the lower tray mounting face **253**. The second wall **260b** is a curved wall formed in a shape corresponding to the upper tray body **151**. That is, the second wall **260b** may be rounded upwardly from the lower tray mounting face **253** in a direction farther away from the lower chamber **252**. Further, the second wall **260b** is formed to have a curvature corresponding to the curved wall **153b** of the upper tray body **151**, so that the lower assembly **200** may maintain a predetermined distance from the upper assembly **110** and may not interfere with the upper assembly **110** in a process of being pivoted.

The lower tray **250** may further include a tray horizontal extension **254** extending in the horizontal direction from the side wall **260**. The tray horizontal extension **254** may be positioned higher than the lower tray mounting face **253**. Thus, the lower tray mounting face **253** and the tray horizontal extension **254** form a step.

The tray horizontal extension **254** may include a first upper protrusion **255** formed thereon to be inserted into the slot **218** of the lower casing **210**. The first upper protrusion **255** may be spaced apart from the side wall **260** in the horizontal direction.

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In one example, the first upper protrusion **255** may protrude upward from the top face of the tray horizontal extension **254** at a location adjacent to the first wall **260a**. The plurality of first upper protrusions **255** may be spaced apart from each other. The first upper protrusion **255** may extend, for example, in a curved form.

The tray horizontal extension **254** may further include a first lower protrusion **257** formed thereon to be inserted into a protrusion groove of the lower support **270** to be described later. The first lower protrusion **257** may protrude downward from a bottom face of the tray horizontal extension **254**. A plurality of first lower protrusions **257** may be spaced apart from each other.

The first upper protrusion **255** and the first lower protrusion **257** may be located on opposite sides of the tray horizontal extension **254** in the vertical direction. At least a portion of the first upper protrusion **255** may overlap the second lower protrusion **257** in the vertical direction.

In one example, the tray horizontal extension **254** may include a plurality of through-holes **256** defined therein. The plurality of through-holes **256** may include a first through-hole **256a** through which the first coupling boss **216** of the lower casing **210** penetrates, and a second through-hole **256b** through which the second coupling boss **217** of the lower casing **210** penetrates.

A plurality of first through-holes **256a** and a plurality of second through-holes **256b** may be located opposite to each other around the lower chamber **252**. Some of the plurality of second through-holes **256b** may be located between two adjacent first upper protrusions **255**. Further, some of the remaining second through-holes **256b** may be located between two adjacent first lower protrusions **257**.

The tray horizontal extension **254** may further include a second upper protrusion **258**. The second upper protrusion **258** may be located opposite to the first upper protrusion **255** around the lower chamber **252**.

The second upper protrusion **258** may be spaced apart from the side wall **260** in the horizontal direction. In one example, the second upper protrusion **258** may protrude upward from the top face of the tray horizontal extension **254** at a location adjacent to the second wall **260b**.

The second upper protrusion **258** may be received in the receiving groove **218a** of the lower casing **210**. The second upper protrusion **258** may be in contact with the curved portion **215** of the lower casing **210** in a state in which the second upper protrusion **258** is received in the receiving groove **218a**.

The side wall **260** of the lower tray **250** may include a first coupling protrusion **262** for coupling with the lower casing **210** formed thereon.

The first coupling protrusion **262** may protrude in the horizontal direction from the first wall **260a** of the side wall **260**. The first coupling protrusion **262** may be located on an upper portion of a side of the first wall **260a**.

The first coupling protrusion **262** may include neck portion **262a** which is reduced in diameter compared to other portions. The neck portion **262a** may be inserted into the first coupling slit **214b** which is defined in the side wall **214** of the lower casing **210**.

The side wall **260** of the lower tray **250** may further include a second coupling protrusion **261**. The second coupling protrusion **261** may be coupled with the lower casing **210**.

The second coupling protrusion **261** may protrude from the second wall **260b** of the side wall **260** and may be formed in a direction opposite to the first coupling protrusion **262**. Further, the first coupling protrusion **262** and the second

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coupling protrusion **261** may be arranged to face away from each other around a center of the lower chamber **252**. Thus, the lower tray **250** may be firmly fixed to the lower casing **210**, and in particular, deviation and deformation of the lower chamber **252** may be prevented.

The tray horizontal extension **254** may further include a second lower protrusion **266**. The second lower protrusion **266** may be positioned opposite the second lower protrusion **257** around the lower chamber **252**.

The second lower protrusion **266** may protrude downward from the bottom face of the tray horizontal extension **254**. The second lower protrusion **266** may extend, for example, in a straight line form. Some of the plurality of first through-holes **256a** may be located between the second lower protrusion **266** and the lower chamber **252**. The second lower protrusion **266** may be received in a guide groove defined in the lower support **270** to be described later.

The tray horizontal extension **254** may further include a lateral stopper **264**. The lateral stopper **264** restricts a horizontal movement of the lower tray **250** in a state in which the lower casing **210** and the lower support **270** are coupled with each other.

The lateral stopper **264** protrudes laterally from the side of the tray horizontal extension **254**, and a vertical length of the lateral stopper **264** is larger than a thickness of the tray horizontal extension **254**. In one example, a portion of the lateral stopper **264** is positioned higher than the top face of the tray horizontal extension **254**, and another portion thereof is positioned lower than the bottom face of the tray horizontal extension **254**.

Thus, a portion of the lateral stopper **264** may be in contact with a side of the lower casing **210** and another portion thereof may be in contact with a side of the lower support **270**. The lower tray body **251** may further include a convex portion **251b** having an upwardly convex lower portion. That is, the convex portion **251b** may be disposed to be convex inwardly of the ice chamber **111**.

In one example, the lower support **270** may include a support body **271** for supporting the lower tray **250**.

The support body **271** may include three chamber-receiving portions **272** defined therein for respectively accommodating the three chamber walls **252d** of the lower tray **250** therein. The chamber-receiving portion **272** may be defined in a hemispherical shape.

The support body **271** may include a lower opening **274** defined therein to be penetrated by the lower ejector **400** in the ice-removal process. In one example, three lower openings **274** may be defined in the support body **271** to respectively correspond to the three chamber-receiving portions **272**. A reinforcing rib **275** for strength reinforcement may be formed along a circumference of the lower opening **274**.

A lower support step **271a** for supporting the lower tray mounting face **253** may be formed on a top of the support body **271**. Further, the lower support step **271a** may be formed to be stepped downward from a lower support top face **286**. Further, the lower support step **271a** may be formed in a shape corresponding to the lower tray mounting face **253**, and may be formed along a circumference of a top of the chamber-receiving portion **272**.

The lower tray mounting face **253** of the lower tray **250** may be seated in the lower support step **271a** of the support body **271**, and the lower support top face **286** may surround the side of the lower tray mounting face **253** of the lower tray **250**. In this connection, a face connecting the lower support

top face **286** with the lower support step **271a** may be in contact with the side of the lower tray mounting face **253** of the lower tray **250**.

The lower support **270** may further include a protrusion groove **287** defined therein for accommodating the first lower protrusion **257** of the lower tray **250**. The protrusion groove **287** may extend in a curved shape. The protrusion groove **287** may be formed, for example, in the lower support top face **286**.

The lower support **270** may further include a first fastener groove **286a** into which a first fastener **B1** passed through the first coupling boss **216** of the upper casing **210** is fastened. The first fastener groove **286a** may be defined, for example, in the lower support top face **286**. Some of a plurality of first fastener grooves **286a** may be located between two adjacent protrusion grooves **287a**.

The lower support **270** may further include an outer wall **280** disposed to surround the lower tray body **251** while being spaced apart from the outer face of the lower tray body **251**. The outer wall **280** may, for example, extend downwardly along an edge of the lower support top face **286**.

The lower support **270** may further include a plurality of hinge bodies **281** and **282** to be respectively connected to hinge supports **135** and **136** of the upper casing **210**. The plurality of hinge bodies **281** and **282** may be spaced apart from each other. Since the hinge bodies **281** and **282** differ only in mounting positions thereof, and structures and shapes thereof are identical, only a hinge body **292** at one side will be described.

Each of the hinge bodies **281** and **282** may further include a second hinge hole **282a** defined therein. The second hinge hole **282a** may be penetrated by a shaft connector **352b** of the pivoting arms **351** and **352**. The connection shaft **370** may be connected to the shaft connector **352b**.

Further, each of the hinge bodies **281** and **282** may include a pair of hinge ribs **282b** protruding along a circumference of each of the hinge bodies **281** and **282**. The hinge rib **282b** may reinforce the hinge bodies **281** and **282** and prevent the hinge bodies **281** and **282** from breaking.

The lower support **270** may further include a coupling shaft **283** to which the link **356** is pivotably connected. A pair of coupling shafts **383** may be provided on both faces of the outer wall **280**, respectively.

Further, the lower support **270** may further include an elastic member receiving portion **284** to which the elastic member **360** is coupled. The elastic member receiving portion **284** may define a space **284a** in which a portion of the elastic member **360** may be accommodated. As the elastic member **360** is received in the elastic member receiving portion **284**, the elastic member **360** may be prevented from interfering with a surrounding structure.

Further, the elastic member receiving portion **284** may include a stopper **284a** to which a bottom of the elastic member **370** is hooked. Further, the elastic member receiving portion **284** may include an elastic member shield **284c** that covers the elastic member **360** to prevent insertion of a foreign material or fall of the elastic member **360**.

In one example, a link shaft **288** to which one end of the link **356** is pivotably coupled may protrude at a position between the elastic member receiving portion **284** and each of the hinge bodies **281** and **282**. The link shaft **288** may be provided forward and downward from a center of pivoting of each of the hinge bodies **281** and **282**. With such arrangement, a vertical stroke of the upper ejector **300** may be secured, and the link **356** may be prevented from interfering with other components.

Hereinafter, the coupling structure of the lower tray **250** and the lower casing **210** will be described in more detail with reference to the accompanying drawings.

FIG. **32** is a partial perspective view illustrating a protruding confiner of a lower casing according to an embodiment of the present disclosure. Further, FIG. **33** is a partial perspective view illustrating a coupling protrusion of a lower tray according to an embodiment of the present disclosure. Further, FIG. **34** is a cross-sectional view of a lower assembly. Further, FIG. **35** is a cross-sectional view of FIG. **27** taken along a line **35-35'**.

As shown in FIGS. **32** to **35**, a protruding confiner **213** may protrude from the curved wall **215** of the upper casing **120**. The protruding confiner **213** may be formed at a location corresponding to the second coupling slit **215a** and the second coupling protrusion **261**.

In detail, the protruding confiner **213** may include a pair of lateral portions **213b** and a connector **213c** connecting tops of the lateral portions **213b** with each other. The pair of lateral portions **213b** may be located on both sides around the second coupling slit **215a**. Thus, the second coupling slit **215a** may be located in an insertion space **213a** defined by the pair of lateral portions **213b** and the connector **213c**. Further, the second coupling protrusion **261** may be inserted into the insertion space **213a**. Thus, the lower portion of the second coupling protrusion **261** may be press-fitted into the second coupling slit **215a**.

The pair of lateral portions **213b** may extend to a vertical level corresponding to the top of the second coupling protrusion **261**. Further, a confining rib **213d** extending downwards may be formed inside the connector **213c**.

The confining rib **213d** may be inserted into the protrusion groove **261d** defined in the top of the second coupling protrusion **261**, and may restrain the second coupling protrusion **261** from falling. As such, both the upper and lower portions of the second coupling protrusion **261** may be fixed, and the lower tray **250** may be firmly fixed to the lower casing **210**.

The second coupling protrusion **261** may protrude outwardly of the second wall **260b**, and a thickness thereof may increase upwardly. That is, due to a self-load of the second coupling protrusion **261**, the second wall **260b** does not roll inward or deform, and the top of the second wall **260b** is pulled outward.

Thus, in a process in which the lower tray **250** pivots in a reverse direction, the second coupling protrusion **261** prevents an end of the second wall **260b** of the lower tray **250** from deforming in contact with the upper tray **150**.

When the end of the second wall **260b** of the lower tray **250** is deformed in contact with the upper tray **150**, the lower tray **250** may be moved to a water-supply position while being inserted into the upper chamber **152** of the upper tray **150**. In this state, when the ice-making is completed after the water supply is performed, the ice is not produced in the spherical form.

Thus, when the second coupling protrusion **261** protrudes from the second wall **260a**, the deformation of the second wall **260a** may be prevented. Thus, the second coupling protrusion **261** may be referred to as a deformation preventing protrusion.

The second coupling protrusion **261** may protrude in the horizontal direction from the second wall **260a**. The second coupling protrusion may extend upward from a lower portion of the outer face of the second wall **260b**, and a top of the second coupling protrusion **261** may extend to the same vertical level as the top of the second wall **260a**.

Further, the second coupling protrusion **261** may include a protrusion lower portion **261a** forming a lower portion thereof and a protrusion upper portion **261b** forming an upper portion thereof.

The protrusion lower portion **261a** may be formed to have a corresponding width to be inserted into the second coupling slit **215a**. Thus, when the second coupling protrusion **261** is inserted into the insertion space of the protruding confiner **213**, the protrusion lower portion **261a** may be press-fitted into the second coupling slit **215a**.

The protrusion upper portion **261b** extends upward from a top of the protrusion lower portion **261a**. The protrusion upper portion **261b** may extend upward from a top of the second coupling slit **215a**, and may extend to the connector **213c**. In this connection, the protrusion upper portion **261b** may protrude further rearward than the protrusion lower portion **261a**, and may have a width larger than that of the protrusion lower portion **261a**. Thus, the second wall **260b** may be directed further outwards by a self-load of the protrusion upper portion **261b**. That is, the protrusion upper portion **261b** may pull the top of the second wall **260b** outward to maintain the outer face of the second wall **260b** and the curved wall **153b** to be in close contact with each other.

Further, a protrusion groove **261d** may be defined in a top face of the protrusion upper portion **261b**, that is, a top face of the second coupling protrusion **261**. The protrusion groove **261d** is defined such that the confining rib **213d** extending downward from the connector **213c** may be inserted therein.

Thus, a bottom of the second coupling protrusion **261** may be pressed into the second coupling slit **215a** and a top thereof may be restrained by the connector **213c** and the confining rib **213d** in a state of being received inside the insertion space **213a**. Thus, the second coupling protrusion **261** may be in a state of being completely in close contact with and fixed to the lower casing **210** so as not to be in contact with the upper tray **150** during the pivoting process of the lower tray **250**.

A round face **260e** may be formed on the top of the second coupling protrusion **261** to prevent the second coupling protrusion **261** from interfering with the upper tray **150** in the pivoting process of the lower tray **250**.

A lower portion **260d** of the second coupling protrusion **261** may be spaced apart from the tray horizontal extension **254** of the lower tray **250** such that the lower portion **260d** of the second coupling protrusion **261** may be inserted into the second coupling slit **215a**.

In one example, as shown in FIG. 35, the lower support **270** may further include a boss through-hole **286b** to be penetrated by the second coupling boss **217** of the upper casing **210**. The boss through-hole **286b** may be, for example, defined in the lower support top face **286**. The lower support top face **286** may include a sleeve **286c** surrounding the second coupling boss **217** passed through the boss through-hole **286b**. The sleeve **286c** may be formed in a cylindrical shape with an open bottom.

The first fastener **B1** may be fastened into the first fastener groove **286a** after passing through the first coupling boss **216** from upward of the lower casing **210**. Further, the second fastener **B2** may be fastened to the second coupling boss **217** from downward of the lower support **270**.

A bottom of the sleeve **286c** may be positioned flush with the bottom of the second coupling boss **217** or lower than the bottom of the second coupling boss **217**.

Thus, in the fastening process of the second fastener **B2**, a head of the second fastener **B2** may be in contact with the

second coupling boss **217** and a bottom face of the sleeve **286c** or in contact with the bottom face of the sleeve **286c**.

The lower casing **210** and the lower support **270** may be firmly coupled to each other by the fastening of the first fastener **B1** and the second fastener **B2**. Further, the lower tray **250** may be fixed between the lower casing **210** and the lower support **270**.

In one example, the lower tray **250** comes into contact with the upper tray **150** by the pivoting, and the upper tray **150** and the lower tray may always be sealed with each other during the ice-making. Hereinafter, a sealing structure based on the pivoting of the lower tray **250** will be described in detail with reference to the accompanying drawings.

FIG. 36 is a plan view of a lower tray. Further, FIG. 37 is a perspective view of a lower tray according to another embodiment of the present disclosure. Further, FIG. 38 is a cross-sectional view that sequentially illustrates a pivoting state of a lower tray. Further, FIG. 39 is a cross-sectional view showing states of an upper tray and a lower tray immediately before or during ice-making. Further, FIG. 40 shows states of upper and lower trays upon completion of ice-making.

Referring to FIGS. 36 to 40, the lower chamber **252** opened upwards may be defined in the lower tray **250**. Further, the lower chamber **252** may include the first lower chamber **252a**, the second lower chamber **252b**, and the third lower chamber **252c** arranged in series. Further, the side wall **260** may extend upward along the perimeter of the lower chamber **252**.

In one example, the lower tray mounting face **253** may be formed along a perimeter of top of the lower chamber **252**. The lower tray mounting face **253** forms a face that is in contact with the bottom face **153c** of the upper tray **150** when the lower tray **250** is pivoted and closed.

The lower tray mounting face **253** may be formed in a planar shape, and may be formed to connect the tops of the lower chambers **252** with each other. Further, the side wall **260** may extend upwardly along the outer end of the lower tray mounting face **253**.

A lower rib **253a** may be formed on the lower tray mounting face **253**. The lower rib **253a** is for sealing between the upper tray **150** and the lower tray **250**, which may extend upward along the perimeter of the lower chamber **252**.

The lower rib **253a** may be formed along the circumference of each of the lower chambers **252**. Further, the lower rib **253a** may be formed at a position to face away from the upper rib **153d** in the vertical direction.

Further, the lower rib **253a** may be formed in a shape corresponding to the upper rib **153d**. That is, the lower rib **253a** may extend starting from a position separated by a predetermined distance from one end of the lower chamber **252**, which is close to the pivoting shaft of the lower tray **250**. Further, a height of the lower tray **250** may increase in a direction farther away from the pivoting shaft of the lower tray **250**.

The lower rib **253a** may be in close contact with the inner face of the upper tray **150** in a state in which the lower tray **250** is completely closed. For this purpose, the lower rib **253a** protrudes upwards from the top of the lower chamber **252**, and may be flush with the inner face of the lower chamber **252**. Thus, in a state in which the lower tray **250** closed, as shown in FIG. 39, an outer face of the lower rib **253a** may come into contact with an inner face of the upper rib **153d**, and the upper tray **150** and the lower tray **250** may be completely sealed with each other.

In this connection, due to the driving of the driver **180**, the first pivoting arm **351** and the second pivoting arm **352** may be further pivoted, and the elastic member **360** may be tensioned to press the lower tray **250** toward the upper tray **150**.

When the upper tray **150** and the lower tray **250** are further closed by the pressurization of the elastic member **360**, the upper rib **153d** and the lower rib **253a** may be bent inward to allow the upper tray **150** and the lower tray **250** to be further sealed with each other.

In one example, before the ice-making, when the lower tray **250** is filled with water, and when the lower tray **250** is closed as shown in FIG. **39**, the upper rib **153d** and the lower rib **253a** may overlap and sealed. In this connection, the top of the lower rib **253a** may come into contact with an inner face of the bottom of the upper chamber **152** of the upper tray **150**. Therefore, a step of a coupling portion inside the ice chamber **111** may be minimized to generate the ice.

In order to fill the water in all of the plurality of ice chambers **111**, the water is supplied in a state in which the lower tray **250** is slightly open. Then, when the water supply is complete, the lower tray **250** is pivoted and closed as shown in FIG. **39**. Accordingly, the water may flow into spaces **G1** and **G2** defined between the side wall **260** and the chamber wall **153** and be filled to a water level the same as that in the ice chamber **111**. Further, the water in the spaces **G1** and **G2** between the side wall **260** and the chamber wall **153** may be frozen during the ice-making operation.

However, the ice chamber **111** and the spaces **G1** and **G2** may be completely separated from each other by the upper rib **153d** and the lower rib **253a**, and may maintain the separated state by the upper rib **153d** and the lower rib **253a** even when the ice-making is completed. Therefore, the ice strip may not be formed on the ice made in the ice chamber **111**, and the ice may be removed in a state of being completely separated from ice debris in the spaces **G1** and **G2**.

When viewing a state in which the ice-making is completed in the ice chamber **111** through FIG. **40**, due to the expansion of the water resulted from the phase-change, the lower tray **250** is inevitably opened at a certain angle. However, the upper rib **153d** and lower rib **253a** may remain in contact with each other, and thus, the ice inside the ice chamber **111** will not be exposed into the space. That is, even when the lower tray **250** is slowly opened during the ice-making process, the upper tray **150** and the lower tray **250** may be maintained to be shielded by the upper rib **153d** and the lower rib **253a**, thereby forming the spherical ice.

In one example, as shown in FIG. **40**, when the ice-making is completed and the lower tray **250** is opened at the maximum angle, the upper tray **150** and the lower tray **250** may be separated from each other by approximately 0.5 to 1 mm. Therefore, a length of the lower rib **253a** is preferably approximately 0.3 mm. In another example, a height of the lower rib **253a** is only an example, and the lengths of the upper rib **153d** and the lower rib **253a** may be appropriately selected depending on the distance between the upper tray **150** and the lower tray **250**.

Further, when an area of the lower tray mounting face **253** is large enough, a pair of lower ribs **253a** and **253b** may be formed on the lower tray mounting face **253**. The pair of lower ribs **253a** and **253b** may be formed in the same shape as the lower rib **253a**, but may be composed of an inner rib **253b** disposed close to the lower chamber **252** and an outer rib **253a** outward of the inner rib **253b**. The inner rib **253b** and the outer rib **253a** are spaced apart from each other to define a groove therebetween. Therefore, when the lower

tray **250** is pivoted and closed, the upper rib **153d** may be inserted into the groove between the inner rib **253b** and the outer rib **253a**.

Due to such double-rib structure, the upper rib **153d** and the lower ribs **253a** and **253b** may be more sealed with each other. However, such a structure may be applicable when the lower tray mounting face **253** is provided with sufficient space for the inner rib **253b** and outer rib **253a** to be formed.

In one example, the lower tray **250** may be pivoted about the hinge bodies **281** and **282**, and may be pivoted by an angle of about 140° such that the ice-removal may be achieved even when the ice is placed in the lower chamber **252**. The lower tray **250** may be pivoted as shown in FIG. **38**. Even during such pivoting, the side wall **260** and chamber wall **153** should not interfere with each other.

More specifically, the water supply is inevitably performed in a state in which the lower tray **250** is slightly open for supplying the water into the plurality of the lower chambers **252**. In this situation, the side wall **260** of the lower tray **250** may extend upwards above a water-supply level in the ice chamber **111** to prevent water leakage.

Further, since the lower tray **250** opens and closes the ice chamber **111** by the pivoting, the spaces **G1** and **G2** are inevitably defined between the side wall **260** and the chamber wall **153**. When the spaces **G1** and **G2** between the side wall **260** and the chamber wall **153** are too narrow, interference with the upper tray **150** may occur during the pivoting process of the lower tray **250**. Further, when the spaces **G1** and **G2** between the side wall **260** and the chamber wall **153** are too wide, during the water supplying into the lower chamber **252**, an excessive amount of water is flowed into the spaces **G1** and **G2** and lost, and thus, an excessive amount of ice debris is generated. Therefore, widths of the spaces **G1** and **G2** between the side wall **260** and the chamber wall **153** may be equal to or less than about 0.5 mm.

In one example, the curved wall **153b** of the upper tray **150** and the curved wall **260b** of the lower tray **250** of the side wall **260** and the chamber wall **153** may be formed to have the same curvature. Thus, as shown in FIG. **38**, the curved wall **153b** of the upper tray **150** and the curved wall **260b** of the lower tray **250** do not interfere with each other in an entire region where the lower tray **250** is pivoted.

In this connection, a radius **R2** of the curved wall **153b** of the upper tray **150** is slightly larger than a radius **R1** of the curved wall **260b** of the lower tray **250**, so that the upper tray **150** and lower tray **250** may have a water-supplyable structure without interfering with each other during the pivoting.

In one example, a center of pivoting **C** of the hinge bodies **281** and **282**, which is the axis of pivoting of the lower tray **250**, may be located somewhat lower than the top face **286** of the upper lower support **270** or the lower tray mounting face **253**. The bottom face **153c** of the upper tray **150** and the lower tray mounting face **253** are in contact with each other when the lower tray **250** is pivoted and closed.

The lower tray **250** may have a structure to be in close contact with the upper tray **150** in the closing process. Therefore, when the lower tray **250** is pivoted and closed, a portion of the upper tray **150** and a portion of the lower tray **250** may be engaged with each other at a position close to the pivoting shaft of the lower tray **250**. In such a situation, even when the lower tray **250** is pivoted to be closed completely, ends of the upper tray **150** and the lower tray **250** at points far from the pivoting shaft may be separated from each other due to the interference in the engaged portion.

To solve such problem, the center of pivoting CI of the hinge bodies **281** and **282**, which is the pivoting shaft of the lower tray **250**, is moved somewhat downward. For example, the center of pivoting CI of the hinge bodies **281** and **282** may be located 0.3 mm below the top face of the lower support **270**.

Thus, when the lower tray **250** is closed, the ends of the upper tray **150** and the lower tray **250** close to the pivoting shaft may not be engaged with each other first, but the lower tray mounting face **253** and the entirety of the bottom face **153c** of the upper tray **150** may be in close contact with each other.

In particular, since the upper tray **150** and the lower tray **250** are made of an elastic material, tolerances may occur during the assembly, or coupling may be loosened or micro deformation may occur during the use. However, such structure may solve the problem of the ends of the upper tray **150** and the lower tray **250** engaging with each other first.

In one example, the pivoting shaft of the lower tray **250** may be substantially the same as the pivoting shaft of the lower support **270**, and the hinge bodies **281** and **282** may also be formed on the lower support **270**.

Hereinafter, the upper ejector **300** and the connector **350** connected to the upper ejector **300** will be described with reference to the drawings.

FIG. **41** is a perspective view showing a state in which an upper assembly and a lower assembly are closed, according to an embodiment of the present disclosure. Further, FIG. **42** is an exploded perspective view showing a coupling structure of a connector according to an embodiment of the present disclosure. Further, FIG. **43** is a side view showing a disposition of a connector. Further, FIG. **44** is a cross-sectional view of FIG. **41** taken along a line **44-44**.

As shown in FIGS. **41** and **44**, the upper ejector **300** is positioned at a topmost position when the lower assembly **200** and the upper assembly **110** are fully closed. Further, the connector **350** will remain stationary.

The connector **350** may be pivoted by the driver **180**, and the connector **350** may be connected to the upper ejector **300** mounted on the upper support **170** and the lower support **270**.

Therefore, when the lower assembly **200** is opened in the pivoting, the upper ejector **300** may be moved downward by the connector **350** and may remove the ice in the upper chamber **152**.

The connector **350** may include a pivoting arm **352** for pivoting the lower support **270** under the power of the driver **180** and a link **356** connected to the lower support **270** to transfer a pivoting force of the lower support **270** to the upper ejector **300** when the lower support **270** pivots.

In detail, a pair of pivoting arms **351** and **352** may be disposed at both sides of the lower support **270**, respectively. A second pivoting arm **352** of the pair of pivoting arms **351** and **352** may be connected to the driver **180**, and a first pivoting arm **351** may be disposed opposite to the second pivoting arm **352**. Further, the first pivoting arm **351** and the second pivoting arm **352** may be respectively connected to both ends of the connection shaft **370**, which pass through the hinge bodies **281** and **282** at both sides, respectively. Therefore, the first pivoting arm **351** and the second pivoting arm **352** may be pivoted together when the driver **180** is operated.

To this end, the shaft connector **352b** may protrude inwardly of each of the first pivoting arm **351** and the second pivoting arm **352**. Further, the shaft connector **352b** may be coupled to second hinge holes **282a** of the hinge body **282** in both sides. The second hinge hole **282a** and the shaft

connector **352b** may be formed in structures to be coupled with each other to allow the transmission of the power.

In one example, the second hinge hole **282a** and the shaft connector **352b** may have shapes corresponding to each other, but may be formed to have a predetermined play (FIG. **44**) in the direction of pivoting. Thus, when the lower assembly **200** is closed in pivoting, the driver **180** may be rotated further by a set angle while the lower tray **250** is in contact with the upper tray **150**, thereby further pivoting the pivoting arms **351** and **352**. The lower tray **250** may be further pressed toward the upper tray **150** by an elastic force of the elastic member **360** generated at this time.

In one example, a power connector **352ac** that is coupled to a rotation shaft of the driver **180** may be formed on an outer face of the second pivoting arm **352**. The power connector **352a** may be formed in a polygonal hole, and the rotation shaft of the driver **180** formed in the corresponding shape may be inserted into the power connector **352a** to allow the transmission of the power.

In one example, the first pivoting arm **351** and second pivoting arm **352** may extend above the elastic member receiving portion **284**. Further, the elastic member connectors **351c** and **352c** may be formed at the extended ends of the first pivoting arm **351** and the second pivoting arm **352**, respectively. One end of the elastic member **360** may be connected to each of the elastic member connectors **351c** and **352c**. The elastic member **360** may be, for example, a coil spring.

The elastic member **360** may be located inside the elastic member receiving portion **284**, and the other end of the elastic member **360** may be fixed to a locking portion **284a** of the lower support **270**. The elastic member **360** provides an elastic force to the lower support **270** to keep the upper tray **150** and the lower tray **250** in contact with each other in a pressed state.

The elastic member **360** may provide an elastic force that allows the lower assembly **200** to be in a close contact with the upper assembly **200** in a closed state. That is, when the lower assembly **200** pivots to close, the first pivoting arm **351** and the second pivoting arm **352** are also pivoted together until the lower assembly **200** is closed, as shown in FIG. **41**.

Further, in a state in which the lower assembly **200** is pivoted to a set angle and in contact with the upper assembly **200**, the first pivoting arm **351** and the second pivoting arm **352** may be further pivoted by the rotation of the driver **180**. The pivoting of the first pivoting arm **351** and second pivoting arm **352** may cause the elastic member **360** to be tensioned. Further, the lower assembly **200** may be further pivoted in the closing direction by the elastic force provided by the elastic member **360**.

When the elastic member **360** is not provided and the lower assembly **200** is further pivoted by the driver **180** to press the lower assembly to the upper assembly **110**, an excessive load may be concentrated on the driver **180**. Further, when the water is phase-changed and expands and the lower tray **250** pivots in the open direction, a reverse force is applied to the gear of the driver **180**, so that the driver **180** may be damaged. Further, when the driver **180** is turned off, the lower tray **250** sags due to a play of the gears. However, all of these problems may be solved when the lower assembly **200** is pulled to be closed contacted by the elastic force provided by the elastic member **360**.

That is, the lower assembly **200** may be provided with the elastic force through the elastic member **360** in a tensioned

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state without additional power from the driver **180**, and may allow the lower assembly **200** to be closer to the upper assembly **110**.

Further, even when the lower tray **250** is stopped by the driver **180** before being fully pressed against the upper tray **150**, an elastic restoring force of the elastic member **360** allows the lower tray **250** to be pivoted further to be completely in contact with the upper tray **150**. In particular, an entirety of the lower tray **250** may be in close contact with the upper tray **150** without a gap by the elastic members **360** arranged on both sides.

The elastic member **360** will sequentially provide the elastic force to the lower assembly **200**. Therefore, even when the ice is produced in the ice chamber **111** and expands, the elastic force is applied to the lower assembly **200**, so that the lower assembly **200** may not be excessively opened.

In one example, the link **356** may link the lower tray **250** and the upper ejector **300** with each other. The link **356** is formed in a bent shape, so that the link **356** does not interfere with each of the hinge bodies **281** and **282** during the pivoting process of the lower tray **250**.

A tray connector **356a** may be formed at a bottom of the link **356**, and the link shaft **288** may pass through the tray connector **356a**. Thus, a bottom of the link **356** may be pivotably connected to the lower support **270**, and may pivot together upon the pivoting of the lower support **270**.

The link shaft **288** may be located between each of the hinge bodies **281** and **282** and the elastic member receiving portion **284**. Further, the link shaft **288** may be located further below a center of pivoting of each of the hinge bodies **281** and **282**. Therefore, the link shaft **288** may be positioned close to a vertical movement path of the upper ejector **300**, so that the upper ejector **300** may be effectively moved vertically. Further, the upper face **300** may descend to a required position, and at the same time, the upper ejector **300** may not be moved to an excessively high position when the upper ejector **300** moves upward. Therefore, heights of the upper ejector **300** and the unit guides **181** and **182** that are exposed upwardly of the ice maker **100** may be further lowered, so that an upper space lost when the ice maker **100** is installed in the freezing compartment **4** may be minimized.

The link shaft **288** protrudes vertically outward from an outer face of the lower support **270**. In this connection, the link shaft **288** may extend to pass through the tray connector **356a**, but may be covered by the pivoting arms **351** and **352**. Each of the pivoting arms **351** and **352** becomes very close to the link and the link shaft **288**. Thus, the link **356** may be prevented from being separated from the link shaft **288** by each of the pivoting arms **351** and **352**. Each of the pivoting arms **351** and **352** may shield the link shaft **288** at any point in the path of pivoting. Thus, the pivoting arms **351** and **352** may be formed to have a width enough to cover the link shaft **288**.

An ejector connector **356b** through which an end of the ejector body **310**, that is, the stopper protrusion **312** passes may be formed on the top of the link **356**. The ejector connector **356b** may also be pivotably mounted with the end of the ejector body **310**. Therefore, when the lower support **270** is pivoted, the upper ejector **300** may be moved together in the vertical direction.

Hereinafter, states of the upper ejector **300** and the connector **350** based on the operation of the lower assembly **200** will be described with reference to the drawings.

FIG. **45** is a cross-sectional view of FIG. **41** taken along a line **45-45'**. Further, FIG. **46** is a perspective view showing

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a state in which upper and lower assemblies are open. Further, FIG. **47** is a cross-sectional view of FIG. **46** taken along a line **47-47'**.

As shown in FIGS. **41** and **45**, during the ice-making of the ice maker **100**, the lower assembly **200** may be closed.

In this state, the upper ejector **300** is located at the topmost position, and the ejecting pin **320** may be located outward of the ice chamber **111**. Further, the upper tray **150** and the lower tray **250** may be completely in close contact with each other and sealed by the pivoting arms **351** and **352** and the elastic member **360**.

In such state, the ice formation may proceed in the ice chamber **111**.

During the ice-making operation, the upper heater **148** and the lower heater **296** are operated periodically, so that the ice formation proceeds from the upper portion of the ice chamber **111**, thereby producing the transparent spherical ice. Further, when the ice formation is completed inside the ice chamber **111**, the driver **180** is operated to pivot the lower assembly **200**.

As shown in FIGS. **46** and **47**, during the ice-removal of the ice maker **100**, the lower assembly **200** may be open. The lower assembly **200** may be fully opened by the operation of the driver **180**.

When the lower assembly **200** opens in the open direction, the bottom of the link **356** pivots with the lower tray **250**. Further, the top of the link **356** moves downward. The top of the link **356** may be connected to the ejector body **310** to move the upper ejector **300** downward, and may be moved downward without being guided by the unit guides **181** and **182**.

When the lower assembly **200** is fully pivoted, the ejecting pin **320** of the upper ejector **300** may pass through the ejector-receiving opening **154** and move to the bottom of the upper chamber **152** or a position adjacent thereto to remove the ice from the upper chamber **152**. In this connection, the link **356** is also pivoted to the maximum angle, but the link **356** has a bent shape, and at the same time, the link shaft **288** may be located forwards and downwards of each of the hinge bodies **281** and **282**, so that interference of the link **356** with other components may be prevented.

In one example, the lower assembly **200** may partially sag while in a closed state. In detail, in the present embodiment, the driver **180** has a structure of being connected to the second pivoting arm **352** among the pivoting arms **351** and **352** on both sides, and the second pivoting arm **352** has a structure of being connected to the first pivoting arm **351** by the connection shaft **370**. Therefore, the rotational force is transmitted to the first pivoting arm **351** through the connection shaft **370**, so that the first pivoting arm **351** and the second pivoting arm **352** may pivot simultaneously.

However, the first pivoting arm **351** has a structure of being connected to the connection shaft **370**. Further, for the connection, a tolerance inevitably occurs at a connected portion. Such tolerance may cause slippage during the pivoting of the connection shaft **370**.

In addition, since the lower assembly **200** extends in the direction of power transmission, a portion of the first pivoting arm **351** positioned at a relatively far may sag, and a torque may not be 100% transmitted thereto.

Because of such structure, when the first pivoting arm **351** pivots less than the second pivoting arm **352**, the upper tray **150** and the lower tray **250** are not completely in contact with each other and sealed, and there is a region partially open between the upper tray **150** and the lower tray **250** at a side close to the first pivoting arm **351**. Therefore, when the lower tray **250** sags or tilts, and thus, a water surface

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inside the ice chamber **111** is tilted, the spherical ice of a uniform size and shape may not be generated. Further, when water leaks through open portion, more serious problems may be caused.

To avoid such problem, a vertical level of the extended top of the first pivoting arm **351** may be different from that of the extended top of the second pivoting arm **352**.

Referring to FIGS. **48**, **49**, and **50**, a vertical level **h2** from the bottom face of the lower assembly **200** to the elastic member connector **351c** of the first pivoting arm **351** may be higher than a vertical level **h3** from the bottom face of the lower assembly **200** to the elastic member connector **352c** of the second pivoting arm **352**.

Thus, when the lower assembly **200** pivots to be closed, the first pivoting arm **351** and second pivoting arm **352** pivot together. Further, because the vertical level of the first pivoting arm is high, when the lower tray **250** and the upper tray **150** begin to be in contact with each other, the elastic member **360** connected to the first pivoting arm **351** is further tensioned.

That is, in a state in which the lower tray **250** is completely in contact with the upper tray **150**, the elastic force of the elastic member **360** of the first pivoting arm **351** becomes greater. This compensates for the sagging of the lower tray **250** at the first pivoting arm **351**. Thus, the entirety of the top face of the lower tray **250** may be in close contact and sealed with the bottom face of the upper tray **150**.

In particular, in a structure where the driver **180** is located on one side of the lower tray **250** and is directly connected only to the second pivoting arm **352**, due to the tolerance occurred in the assembly of the connection shaft **370**, the first pivoting arm **351** may be less pivoted. However, as in the embodiment of the present disclosure, the first pivoting arm **351** pivots the lower tray **250** with a force greater than that of the second pivoting arm **352**, so that the lower tray **250** is prevented from sagging or less pivoting.

In another example, the first pivoting arm **351** and second pivoting arm **352** may be pivotably coupled both ends of the connection shaft **370** respectively to be alternated with each other by a set angle with respect to the connection shaft **370**. Thus, the top of the first pivoting arm **351** may be positioned higher than the top of the second pivoting arm **352**.

Further, in another example, shapes of the first pivoting arm **351** and the second pivoting arm **352** may be different from each other such that the first pivoting arm **351** extends longer than the second pivoting arm **352**, and thus, a point where the first pivoting arm **351** is connected to the elastic member **360** becomes higher than a point where the second pivoting arm **352** is connected to the elastic member **360**.

Further, in another example, an elastic modulus of the elastic member **360** connected to the first pivoting arm **351** may be made larger than an elastic modulus of the elastic member **360** connected to the second pivoting arm **352**.

When the lower assembly **200** is completely closed, as shown in FIG. **50**, the top of the lower casing **210** and the bottom of the upper support **170** may be spaced apart from each other by a predetermined distance **h4**. Further, a portion of the upper tray **150** may be exposed through the gap. In this connection, the space is defined between the upper casing **210** and the upper support **170**, but the upper tray **150** and the lower tray **250** remain in close contact with each other.

In other words, even when the upper tray **150** and the lower tray **250** are completely in contact and sealed with

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each other, the top of the lower casing **210** and the bottom of the upper support **170** may be spaced apart from each other.

When the top of the lower casing **210** and the bottom of the upper support **170**, which are injection-molded structures, are in contact with each other, an impact may strain and damage the driver **180**.

Further, when the top of the lower casing **210** and the bottom of the upper support **170** are spaced apart from each other, a space where the upper tray **150** and the lower tray **250** may be pressed and deformed may be defined. Therefore, in order to ensure close contact between the upper tray **150** and the lower tray **250** in various situations, such as the assembly tolerance and the deformation on use, the top of the lower casing **210** and the bottom of the upper support **170** must be spaced apart from each other. To this end, the side wall **260** of the lower tray **250** may extend higher than the top of the upper casing **120**.

Hereinafter, a structure of an upper ejector **300** will be described with reference to the drawings.

FIG. **50** is a front view of an ice maker. Further, FIG. **51** is a partial cross-sectional view showing a coupling structure of an upper ejector.

As shown in FIGS. **50** and **51**, the ejector body **310** has passing-through portions **311** at both ends thereof, and the passing-through portion **311** may pass through the guide slot **183** and the ejector connector **356b**. Further, a pair of stopper protrusions **312** may protrude in opposite directions from both ends of the ejector body **310**, that is, from respective ends of the passing-through portions **311**, respectively. Thus, each of the both ends of the ejector body **310** may be prevented from being separated from the ejector connector **356b**. Further, the stopper protrusion **312** abuts an outer face of the link **356** and extends vertically to prevent generation of the play between the stopper protrusion **312** and the link **356**.

Further, a body protrusion **313** may be further formed on the ejector body **310**. The body protrusion **313** may protrude downwardly at a position spaced apart from the stopper protrusion **312** and may extend to be in contact with an inner face of the link **356**. The body protrusion **313** may be inserted into the guide slot **183**, and may protrude by a predetermined length to be in contact with the inner face of the link **356**.

In this connection, the stopper protrusion **312** and the body protrusion **313** may respectively abut both faces of the link **356**, and may be arranged to face each other. Thus, the both face of the link may be supported by the stopper protrusion **312** and the body protrusion **313**, thereby effectively preventing the link **356** from moving.

When the ejector body **310** moves in a horizontal direction, the position of the ejecting pin **320** may be moved in the horizontal direction. Thus, the ejecting pin **320** may press the upper tray **150** in a process of passing through the ejector-receiving opening **154**, so that the upper tray **150** may be deformed or detached. Further, the ejecting pin **320** may get caught in the upper tray **150** and may not move.

Thus, in order to ensure that the ejecting pin **320** exactly passes through a center of the ejector-receiving opening **154** without moving, the stopper protrusion **312** and the body protrusion **313** may prevent the link **356** from moving, so that the ejecting pin **320** may move vertically a set position.

In addition, as shown in FIG. **15**, a first stopper **139ba** and a second stopper **139bb** may be provided at the first through-opening **139b** of the upper casing **120** through which the pair of the unit guides **181** and **182** are passed, and a third stopper **139ca** and a fourth stopper **139cb** are provided at the second

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through-opening 139c, so that the movement of the unit guides 181 and 182 that guide the vertical movement of the ejector body 310 may also be prevented.

Therefore, the present embodiment has a structure that prevents the movements of not only the ejector body 310 but also of the unit guides 181 and 182, and the ejecting pin 320, which moves a relatively long distance in the vertical direction, does not move and enters the ejector-receiving opening 154 along a set path, so that contact or interference with the upper tray 150 may be completely prevented.

Hereinafter, a mounting structure of the driver 180 will be described with reference to the drawings.

FIG. 52 is an exploded perspective view of a driver according to an embodiment of the present disclosure. Further, FIG. 53 is a partial perspective view showing a driver being moved for provisional fixing of a driver. Further, FIG. 54 is a partial perspective view of a driver, which has been provisionally-fixed. Further, FIG. 55 is a partial perspective view for showing restraint and coupling of a driver.

As shown in FIGS. 52 to 55, the driver 180 may be mounted on an inner face of the upper casing 120. The driver 180 may be disposed adjacent to a side wall 143 far away from the cold-air hole 134, that is, the second side wall.

In one example, the driver 180 may have a pair of fixed protrusions 185a protruding from the top face. The fixed protrusion 185a may be formed in a plate shape. The fixed protrusion 185a may extend in a direction from the top face of the driver casing 185 to the cold-air hole 134.

Further, the rotation shaft 186 of the driver 180 may protrude in the protruding direction of the fixed protrusion 185a. Further, a lever connector 187 to which the ice-full state detection lever 700 is mounted may be formed on one side away from the rotation shaft 186. The top face of the driver casing 185 may further include a screw-receiving portion 185b formed thereon a through which a screw B3 for fixing the driver 180 penetrates.

An opening 149c may be defined in a bottom face of the upper plate 121 of the upper casing 120 in which the driver 180 is mounted. The opening 149c is defined such that the screw-receiving portion 185b may be passed therethrough. Further, a screw groove 149d may be defined at one side of the opening 149c.

Further, a driver mounted portion 149a on which the driver 180 is seated may be formed on the bottom face of the upper plate 121. The driver mounted portion 149a may be located closer to the cold-air hole 134 than the opening 149c, and the driver mounted portion 149a may further include an electrical-wire receiving hole 149e defined therein through which the electrical-wire connected to the driver 180 enters.

Further, the bottom face of the upper plate 121 may be formed with a fixed protruding confiner 149b into which the fixed protrusion 185a is inserted. The fixed protruding confiner 149b is positioned closer to the cold-air hole 134 than the driver mounted portion 149a. Further, the fixed protruding confiner 149b may have an insertion hole opening defined therein in a corresponding shape such that the fixed protrusion 185a may be inserted therein.

Hereinafter, a mounting process of the driver 180 having the structure as described above will be described.

As shown in the FIG. 52, the operator directs the top face of the driver 180 to the inner side of the upper casing 120, and insert the driver 180 into a mounting position of the driver 180.

Next, as shown in the FIG. 53, the operator moves the driver 180 horizontally toward the cold-air hole 134 in a state in which the fixed protrusion 185a is in close contact

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with the driver mounted portion 149a. The fixed protrusion 185a is inserted into the fixed protruding confiner 149b through such moving operation.

When the fixed protrusion 185a is fully inserted, as shown in FIG. 54, the fixed protrusion 185a is fixed inside the fixed protruding confiner 149b. Further, the top face of the driver casing 185 may be seated on the driver mounted portion 149a.

In this state, as shown in FIG. 55, the screw-receiving portion 185b may protrude upward and be exposed through the opening 149c. Further, the screw B3 is inserted and fastened into the screw-receiving portion 185b through the screw groove 149d. The driver 180 may be fixed to the upper casing 120 by the fastening of the screw B3.

In one example, the screw groove 149d may be defined at the end of the upper plate 121 corresponding to the screw-receiving portion 185b, thereby facilitating fastening and separating of the screw B3 to and from the screw-receiving portion 185b.

Hereinafter, the ice-full state detection lever 700 will be described with reference to the drawings.

FIG. 56 is a side view of an ice-full state detection lever positioned at a topmost position, which is an initial position, according to an embodiment of the present disclosure. Further, FIG. 57 is a side view of an ice-full state detection lever positioned at a bottommost position, which is a detection position.

As shown in FIG. 56 and FIG. 57, the ice-full state detection lever 700 may be connected to the driver 180 and may be pivoted by the driver 180. Further, the ice-full state detection lever 700 may pivot together when the lower assembly 200 pivots for the ice-removal to detect whether the ice bin 102 is in the ice-full state. In another example, the ice-full state detection lever 700 may be operated independently of the lower assembly 200 if necessary.

The ice-full state detection lever 700 has a shape bent in one direction (toward the left side of FIG. 56) due to the first bent portion 721 and the second bent portion 722. Therefore, even when the ice-full state detection lever 700 pivots as shown in FIG. 57 to detect the ice-full state, the ice-full state detection lever 700 may effectively detect whether the ice stored in the ice bin 102 has reached the predefined vertical level without interfering with other components. The lower assembly 200 and the ice-full state detection lever 700 may pivot counterclockwise at a degree greater than a degree as shown FIG. 57. In one example, the lower assembly 200 and the ice-full state detection lever 700 may pivot by about 140° for effective ice-removal.

A length L1 of the ice-full state detection lever 700 may be defined as the vertical distance from the pivoting shaft of the ice-full state detection lever 700 to the detection body 710. Further, the length of the ice-full state detection lever 700 may be larger than the distance L2 of the bottom branch of the lower assembly 200. If the length L1 of the ice-full state detection lever 700 is smaller than the distance L2 of the end branch of the lower assembly 200, the ice-full state detection lever 700 and the lower assembly 200 may interfere with each other in the process in which the ice-full state detection lever 700 and the lower assembly 200 pivot.

To the contrary, if the ice-full state detection lever 700 is too long and when the lever 799 extends to the location of the ice I placed at the bottom of the ice bin 102, there is a high probability of false detection. The ice made in this embodiment may be spherical and thus may roll and move inside the ice bin. Therefore, if the length of the ice-full state detection lever 700 is long enough to detect ice at the bottom of the ice bin 102, there is a possibility of misdetection of the

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ice-full state due to the detection of the rolling ice even though the ice bin is not in an actual ice-full state.

Therefore, the ice-full state detection lever **700** may extend to a position higher by the diameter of the ice so that the lever may not detect the ice laid in one layer on the bottom of the ice bin **102**. In one example, the ice-full state detection lever **700** may extend to reach a position higher than the height **L5** by the diameter of the ice **I** from the bottom of the ice bin **102** upon the ice-full state detection.

That is, the ice may be stored at the bottom face of the ice bin **102**. Before the ice **I** entirely fills the first layer, the ice-full state detection lever **700** will not detect the ice-full state even when the lever pivots. When the refrigerator continues the ice-making and ice-removal processes, the ice spreads widely on the bottom face of the ice bin **102** instead of accumulating on the bottom of the ice bin **102** due to the characteristics of the spherical ice that is removed into the ice bin and thus sequentially forms an ice stack of multiple layers on the bottom face of the ice bin. Further, during the pivoting process of the lower assembly **200** or the movement process of the freezing compartment drawer **41**, the first layer ice **I** inside the ice bin **102** rolls to fill an empty space therein.

Once the first layer on the bottom of the ice bin **102** is fully filled with the ice, the removed ice may be stacked on top of the ice **I** of the first layer. In this connection, the vertical dimension of the ice in the second layer is not twice the diameter of the ice, but may be a sum of the diameter of an single ice and about $\frac{1}{2}$ to $\frac{3}{4}$ of the diameter of the ice. This is because the ice of the second layer is settled into a valley formed between the ices of the first layer.

In one example, when the ice-full state detection lever **700** detects the ice portion just above the height **L5** of the ice **I** of the first layer, the detection may be erroneous when the ice height of the first layer is increased due to ice debris, etc. Thus, it would be desirable for the lever **700** to detect the ice portion higher than the height **L5** of the ice **I** of the first layer by a predefined distance.

Thus, the ice-full state detection lever **700** may be formed to extend to any point which is higher than the height **L5** by the diameter of the ice and is lower than the height **L6** which is a sum of the $\frac{1}{2}$ to $\frac{4}{3}$ of the diameter of the single ice and the diameter of the single ice.

In one example, the ice-full state detection lever **700** is short as possible as long as it does not interfere with the lower tray **250**, thereby to secure the ice making amount. To prevent the erroneous detection due to the height difference caused by residual debris ices, the ice-full state detection lever **700** may have a length such that it extends to the top of the distance range **L6**. The top level of the vertical dimension **L6** may be equal to a sum of the $\frac{1}{2}$ to $\frac{4}{3}$ of the diameter of the single ice and the diameter of the single ice.

In this embodiment, an example in which the lever **799** detects the ice of the second layer is described. In a refrigerator having the ice bin **102** being a large vertical dimension and having an large amounts of spherical ices stored in the ice bin **102**, the lever **700** may detect the ice of the third layer or the ice of a higher layer. In this case, the ice-full state detection lever **700** may extend to a vertical level equal to a sum of the $\frac{1}{2}$ to $\frac{4}{3}$ of the diameter of the single ice and the diameters of the n ices from the bottom of the ice bin.

Hereinafter, the lower ejector **400** will be described with reference to the drawings.

FIG. **58** is an exploded perspective view showing a coupling structure of an upper casing and a lower ejector according to an embodiment of the present disclosure.

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Further, FIG. **59** is a partial perspective view showing a detailed structure of a lower ejector. Further, FIG. **60** shows a deformed state of a lower tray when the lower assembly fully pivots. Further, FIG. **61** shows a state just before a lower ejector passes through a lower tray.

As shown in FIG. **58** to FIG. **61**, the lower ejector **400** may be mounted onto the side wall **143**. An ejector mounted portion **441** may be formed at the bottom of the side wall **143**. The ejector mounted portion **441** may be positioned to face the lower assembly **200** when the lower assembly **200** pivots. The ejector mounted portion **441** may be recessed into a shape corresponding to the shape of the lower ejector **400**.

A pair of body fixing portions **443** may protrude from the top face of the ejector mounted portion **441**. The body fixing portion **443** may have a hole **443a** into which the screw is fastened. Further, the lateral portion **442** may be formed on each of both sides of the ejector mounted portion **441**. The lateral portion **442** may have a groove defined therein for receiving each of both ends of the lower ejector **400** so that the lower ejector **400** may be inserted in a slidable manner.

The lower ejector **400** may include a lower ejector body **410** fixed to the ejector mounted portion **441**, and a lower ejecting pin **420** protruding from the lower ejector body **410**. The lower ejector body **410** may be formed into a shape corresponding to a shape of the ejector mounted portion **441**. The face defined by the lower ejecting pin **420** may be inclined so that the lower ejecting pin **420** faces toward the lower opening **274** when the lower assembly **200** pivots.

The top face of the lower ejector body **410** may have a body groove **413** defined therein for receiving the body fixing portion **443**. In the body groove **413**, a hole **412** to which the screw is fastened may be defined. Further, an inclined groove **411** may be recessed in the inclined face of the lower ejector body **410** corresponding to the hole **412** to facilitate the fastening and detachment of the screw.

Further, a guide rib **414** may protrude on each of the both sides of the lower ejector body **410**. The guide rib **414** may be inserted into the lateral portion **442** of the ejector mounted portion **441** upon mounting of the lower ejector **400**.

In one example, the lower ejecting pin **420** may be formed on the inclined face of the ejector body **310**. The number of the lower ejecting pins **420** may be equal to the number of the lower chambers **252**. The lower ejecting pins **420** may push the lower chambers **252** respectively for ice removal.

The lower ejecting pin **420** may include a rod **421** and a head **422**. The rod **421** may support the head **422**. Further, the rod **421** may be formed to have a predetermined length and slope or roundness such that the lower ejecting pin **420** extends to the lower opening **274**. The head **422** is formed at the extended end of the rod **421** and pushes the curved outer surface of the lower chamber **252** for the ice-removal.

In detail, the rod **421** may be formed to have a predetermined length. In one example, the rod **421** may extend such that the end of the head **422** meets an extension **L4** of the top of the lower chamber **252** when the lower assembly **200** fully pivots for the ice-removal. That is, the rod **421** may extend to a sufficient length so that when the head **422** pushes the lower tray **250** for the removal of the ice from the lower chamber **252**, the ice is pushed by the head **422** until the ice may deviate from at least the hemisphere area so that ice may be separated from the lower chamber **252**.

If the rod **421** is further longer, interference may occur between the lower opening **274** and the rod **421** when the

lower assembly 200 pivots. If the rod 421 is too short, the removal of the ice from the lower tray 250 may not be carried out smoothly.

The rod 421 protrudes from the inclined surface of the lower ejector body 410 and has a predetermined inclination or roundness. The rod 421 may be configured to naturally pass through the lower opening 274 when the lower assembly 200 pivots. That is, the rod 421 may extend along the pivoting path of the lower opening 274.

In one example, the head 422 may protrude from the end of the rod 421. The head 422 may have a hollow 425 formed therein. Thus, the area of contact thereof with the ice surface may be increased such that the head 422 may push the ice effectively.

The head 422 may include an upper head 423 and a lower head 424 formed along the perimeter of the head 422. The upper head 423 may protrude more than the lower head 424. Therefore, the head 422 may effectively push the curved surface of the lower chamber 252 where the ice is accommodated, that is, push the convex portion 251b. When the head 422 pushes the convex portion 251b, both the upper head 423 and the lower head 424 are in contact with the curved face, thereby to push more reliably the ice for the ice-removal.

Thus, the spherical ice may be removed more effectively from the lower tray 250. In one example, when the upper head 423 of the head 422 protrudes more than the lower head 424, the lower opening 274 and the end of the upper head 423 may interfere with each other in the pivoting process of the lower assembly 200.

In order to prevent the interference, the protruding length of the upper head 423 may be maintained, but the top face of the upper head 423 may be formed in an obliquely cut off shape. That is, the upper head 423 may have the top face as inclined. In this connection, the inclination of the upper head 423 may be configured such that the vertical level may gradually be lower toward the extended end of the upper head 423. In order to form the cutoff portion of the upper head 423, the top face portion of the upper head 423 may be partially cut off by an area where interference thereof with the lower opening occurs, that is, by approximately C.

Thus, as shown in FIG. 61, the upper head 423 may extend to a sufficient length to effectively contact the curved surface, but may not interfere with the perimeter of the lower opening 274 due to the presence of the cut off portion. That is, the rod 421 may have a sufficient length while the head 422 may be constructed to improve the contact ability with the curved surface and at the same time prevent the interference with the lower opening 274, so that the ice-removal from the lower chamber 252 may be facilitated efficiently.

Hereinafter, the operation of the ice maker 100 will be described with reference to the drawings.

FIG. 62 is a cutaway view taken along a line 62-62' of FIG. 8. FIG. 63 is a view showing a state in which the ice generation is completed in FIG. 62.

Referring to FIG. 62 and FIG. 63, the lower support 270 may be equipped with a lower heater 296.

The lower heater 296 applies heat to the ice chamber 111 in the ice-making process, causing a top portion of water in the ice chamber 111 to be first frozen. Further, as the lower heater 296 periodically turns on and off in the ice-making process to generate heat. Thus, in the ice-making process, bubbles in the ice chamber 111 are moved downward. Thus, when the ice-making process is completed, a portion of the spherical ice except for the lowest portion may become transparent. That is, according to this embodiment, a substantially transparent spherical ice may be produced. In the

present embodiment, the substantially transparent sphere shaped ice is not perfectly transparent but has a degree of transparency at which the ice may be commonly referred to as transparent ice. The substantially sphere shape is not a perfect sphere, but means a roughly spherically shape.

In one example, the lower heater 296 may be a wire type heater. The lower heater 296 may be a DC heater, like the upper heater 148. The lower heater 296 may be configured to have a lower output than that of the upper heater 148. In one example, the upper heater 148 may have a heat capacity of 9.5 W, while the lower heater 296 may have a 6.0 W heat capacity. Thus, the upper heater 148 and lower heater 296 may maintain the condition at which the transparent ice is made by heating the upper tray 150 and the lower tray 250 periodically at low heat capacity.

The lower heater 296 may contact the lower tray 250 to apply heat to the lower chamber 252. In one example, the lower heater 296 may be in contact with the lower tray body 251.

In one example, the ice chamber 111 is defined as the upper tray 150 and the lower tray 250 are arranged vertically and contact each other. Further, a top face 251e of the lower tray body 251 is in contact with a bottom face 151a of the upper tray body 151.

In this connection, while the top face of the lower tray body 251 and the bottom face of the upper tray body 151 are in contact with each other, the elastic force of the elastic member 360 is exerted to the lower support 270. The elastic force of the elastic member 360 is then applied to the lower tray 250 via the lower support 270 such that the top face 251e of the lower tray body 251 presses the bottom face 151a of the upper tray body 151. Thus, while the top face of the lower tray body 251 is in contact with the bottom face of the upper tray body 151, the both faces are pressed against each other, thereby improving adhesion therebetween.

Thus, when the adhesion between the top face of the lower tray body 251 and the bottom face of the upper tray body 151 is increased, there may be no gap between the two faces to prevent formation of a thin strip shaped burr around the spherical ice after the completion of the ice-making process. Further, as in FIGS. 39 and 40, the upper rib 153d and the lower rib 253a may prevent the gap formation until the ice-making process is completed.

The lower tray body 251 may further include the convex portion 251b in which the lower portion of the body 251 is convex upward. That is, the convex portion 251b may be configured to be convex toward the inside of the ice chamber 111.

A convex shaped recess 251c may be formed below and in a corresponding manner to the convex portion 251b such that a thickness of the convex portion 251b is substantially equal to a thickness of the remaining portion of the lower tray body 251.

As used herein, the phrase "substantially equal" may mean being exactly equal to each other or being equal to each other within a tolerable difference.

The convex portion 251b may be configured to face the lower opening 274 of the lower support 270 in the vertical direction.

Further, the lower opening 274 may be located vertically below the lower chamber 252. That is, the lower opening 274 may be located vertically below the convex portion 251b.

As shown in FIG. 62, a diameter D3 of the convex portion 251b may be smaller than a diameter D4 of the lower opening 274.

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When cold-air is supplied to the ice chamber **111** while water has been supplied to the ice chamber **111**, the liquid water changes to solid ice. In this connection, the water expands in a process in which the water changes to the ice, such that a water expansion force is applied to each of the upper tray body **151** and the lower tray body **25**.

In this embodiment, while a portion (hereinafter, referred to as a corresponding portion) corresponding to the lower opening **274** of the support body **271** is not surrounded by the support body **271**, a remaining portion of the lower tray body **251** is surrounded by the support body **271**.

When the lower tray body **251** is formed in a perfect hemispherical shape, and when the expansion force of the water is applied to the corresponding portion of the lower tray body **251** corresponding to the lower opening **274**, the corresponding portion of the lower tray body **251** is deformed toward the lower opening **274**.

In this case, before the ice is produced, the water supplied to the ice chamber **111** is in a form of a sphere. However, after the ice has been produced, the deformation of the corresponding portion of the lower tray body **251** may allow an additional ice portion in a form of a protrusion to be formed to occupy a space created by the deformation of the corresponding portion.

Therefore, in this embodiment, the convex portion **251b** may be formed in the lower tray body **251** in consideration of the deformation of the lower tray body **251** such that the shape of the finally created ice is identical as possible as with the perfect sphere.

In this embodiment, the water supplied to the ice chamber **111** does not have a spherical shape until the ice is formed. However, after the ice generation is completed, the convex portion **251b** of the lower tray body **251** is deformed toward the lower opening **274** such that the spherical ice may be generated.

In the present embodiment, since the diameter D1 of the convex portion **251b** is smaller than the diameter D2 of the lower opening **274**, the convex portion **251b** may be deformed and invade inside the lower opening **274**.

Hereinafter, an ice manufacturing process by an ice maker according to an embodiment of the present disclosure will be described. FIG. **64** is a cross-sectional view taken along a line **62-62'** of FIG. **8** in a water-supplied state. Further, FIG. **65** is a cross-sectional view taken along a line **62-62'** of FIG. **8** in an ice-making process. Further, FIG. **66** is a cross-sectional view taken along a line **62-62'** of FIG. **8** in a state in which the ice-making process is completed. Further, FIG. **67** is a cross-sectional view taken along a line **62-62'** of FIG. **8** at an initial ice-removal state. Further, FIG. **68** is a cross-sectional view taken along a line **62-62'** of FIG. **8** in a state in which an ice-removal process is completed.

Referring to FIG. **64** to FIG. **68**, first, the lower assembly **200** is moved to the water-supplied position.

In the water-supplied position of the lower assembly **200**, the top face **251e** of the lower tray **250** is spaced apart from at least a portion of the bottom face **151e** of the upper tray **150**. In the present embodiment, a direction in which the lower assembly **200** pivots for the ice-removal is referred to as a forward direction (a counterclockwise direction in the drawing), while a direction opposite to the forward direction is referred to as a reverse direction (a clockwise direction in the drawing).

In one example, an angle between the top face **251e** of the lower tray **250** and the bottom face **151e** of the upper tray **150** in the water-supplied position of the lower assembly **200** may be approximately 8°. However, the present disclosure may not be limited thereto.

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In the water-supply position of the lower assembly **200**, the detection body **710** is located below the lower assembly **200**.

In this state, water is supplied by the water supply **190** to the ice chamber **111**. In this connection, water is supplied to the ice chamber **111** through one ejector-receiving opening of the plurality of ejector-receiving openings **154** of the upper tray **150**.

When the water supply is completed, a portion of the water as supplied may fill an entirety of the lower chamber **252**, while a remaining portion of the water as supplied may fill a space between the upper tray **150** and the lower tray **250**.

In one example, a volume of the upper chamber **151** and a volume of the space between the upper tray **150** and the lower tray **250** may be equal to each other. Then, water between the upper tray **150** and the lower tray **250** may fill an entirety of the upper tray **150**. Alternatively, the volume of the space between the upper tray **150** and the lower tray **250** may be smaller than the volume of the upper chamber **151**. In this case, the water may be present in the upper chamber **151**.

In the present embodiment, there is no channel for mutual communication between the three lower chambers **252** in the lower tray **250**.

Even when there is no channel for water movement in the lower tray **250**, a following result may be achieved because the lower tray **250** and the upper tray **150** are spaced apart from each other in the water-supply step as shown in FIG. **64**: in the water-supply process, when a specific lower chamber **252** is fully filled with water, the water may move to neighboring lower chambers **252** to fill all of the lower chambers **252**. Thus, each of the plurality of lower chambers **252** of the lower tray **250** may be fully filled with water.

Further, in this embodiment, since there is no channel for communication between the lower chambers **252** in the lower tray **250**, the presence of the additional ice portion in the form of the protrusion around the ice after the ice has been created may be suppressed.

When the water-supply is completed, the lower assembly **200** pivots in the reverse direction as shown in FIG. **30**. When the lower assembly **200** pivots in the reverse direction, the top face **251e** of the lower tray **250** is brought to be close to the bottom face **151e** of the upper tray **150**.

Then, water between the top face **251e** of the lower tray **250** and the bottom face **151e** of the upper tray **150** is divided into portions which in turn are distributed into the plurality of upper chambers **152** respectively. Further, when the top face **251e** of the lower tray **250** and the bottom face **151e** of the upper tray **150** come into a close contact state with each other, the upper chambers **152** may be filled with water.

In one example, when the lower assembly is in a closed state such that the upper tray **150** and lower tray **250** are in close contact with each other, the chamber wall **153** of the upper tray body **151** may be accommodated in the interior space of the side wall **260** of the lower tray **250**.

In this connection, the vertical wall **153a** of the upper tray **150** may face the vertical wall **260a** of the lower tray **250**, while the curved wall **153b** of the upper tray **150** may face the curved wall **260b** of the lower tray **250**.

The outer face of the chamber wall **153** of the upper tray body **151** is spaced apart from the inner face of the side wall **260** of the lower tray **250**. That is, a space (G2 in FIG. **39**) is formed between the outer face of the chamber wall **153** of the upper tray body **151** and the inner face of the side wall **260** of the lower tray **250**.

The water supplied from the water supply **180** may be supplied while the lower assembly **200** pivots at a predefined angle to be open such that the water fill the entire ice chamber **111**. Thus, the water as supplied will fill the lower chamber **252** and fill an entirety of the inner space defined with the side wall **260**, thereby to fill the neighboring lower chambers **252**. In this state, when the water supply to the predefined level is completed, the lower assembly **200** pivots to be closed so that the water level in the ice chamber **111** becomes the predefined level. In this connection, the space (G1, G2) between the inner faces of the side wall **260** of the lower tray **250** is inevitably filled with water.

In one example, when more than a predefined amount of water in the water-supply process or ice-making process is supplied to the ice chamber **111**, the water from the ice chamber **111** may flow into the ejector-receiving opening **154**, that is, into the buffer. Thus, even when more than the predefined amount of water is present in the ice chamber **111**, the water may be prevented from overflowing the ice maker **100**.

For this reason, while the top face of the lower tray body **251** contacts the bottom face of the upper tray body **151** such that the lower assembly is in a closed state, the top of the side wall **260** may be positioned at a higher level than the bottom of the ejector-receiving opening **154** of the upper tray **150** or the top of the upper chamber **152**.

The position of the lower assembly **200** while the top face **251e** of the lower tray **250** and the bottom face **151e** of the upper tray **150** contact each other may be referred to as the ice-making position. In the ice-making position of the lower assembly **200**, the detection body **710** is positioned below the lower assembly **200**.

Then, the ice-making process begins while the lower assembly **200** has moved to the ice-making position.

During the ice-making process, the pressure of the water is lower than the force for deforming the convex portion **251b** of the lower tray **250**, so that the convex portion **251b** remains undeformed.

When the ice-making process begins, the lower heater **296** may be turned on. When the lower heater **296** is turned on, heat from the lower heater **296** is transferred to the lower tray **250**.

Thus, when the ice-making is performed while the lower heater **296** is turned on, a top portion of the water the ice chamber **111** is first frozen.

In this embodiment, a mass or volume the water in the ice chamber **111** may vary or may not vary along a height of the ice chamber depending on the shape of the ice chamber **111**.

For example, when the ice chamber **111** has a cuboid shape, the mass or volume of the water in the ice chamber **111** may not vary along the height thereof.

To the contrary, when the ice chamber **111** has a sphere, an inverted triangle or a crescent shape, the mass or volume may vary along the height thereof.

When the temperature of the cold-air and the amount of the cold-air supplied to the freezing compartment **4** are constant, and when the output of the lower heater **296** is constant, a rate at which the ice is produced may vary along the height when the ice chamber **111** has a sphere, an inverted triangle or a crescent shape such that the mass or volume may vary along the height thereof.

For example, when the mass per unit height of water is small, ice formation rate is high, whereas when the mass per unit height of water is large, ice formation rate is low.

As a result, the rate at which ice is generated along the height of the ice chamber is not constant, such that the transparency of the ice may vary along the height. In

particular, when ice is generated at a high rate, bubbles may not move from the ice to the water, such that ice may contain bubbles, thereby lowering the ice transparency.

Therefore, in this embodiment, the output of the lower heater **296** may be controlled based on the mass per unit height of water of the ice chamber **111**.

When the ice chamber **111** is formed into a spherical shape, as shown in this embodiment, the mass per unit height of water in the ice chamber **111** increases in a range from a top to a middle level and then decreases in a range from the middle level to the bottom.

Thus, after the lower heater **296** turns on, the output of the lower heater **430** decreases gradually and then the output is minimal at the middle level of the chamber. Then, the output of the lower heater **296** may increase gradually from the middle level to the top of the chamber.

Thus, since the top portion of the water in the ice chamber **111** is first frozen, bubbles in the ice chamber **111** move downwards. In the process where ice is generated in a downward direction in the ice chamber **111**, the ice comes into contact with the top face of the convex portion **251b** of the lower tray **250**.

When the ice is sequentially generated in this state, the convex portion **251b** is deformed by the ice pressing the convex portion as shown in FIG. **31**. When the ice-making process is completed, the spherical ice may be generated.

A controller (not shown) may determine whether the ice-making is completed based on the temperature detected by the temperature sensor **500**.

The lower heater **296** may be turned off when the ice-making is completed or before ice-making is completed.

When the ice-making process is completed, the upper heater **148** may first be turned on for ice-removal of the ice. When the upper heater **148** is turned on, the heat from the upper heater **148** is transferred to the upper tray **150**, thereby to cause the ice to be separated from the inner face of the upper tray **150**.

After the upper heater **148** is activated for a predefined time, the upper heater **148** is turned off. Then, the driver **180** may be activated to pivot the lower assembly **200** in the forward direction.

As the lower assembly **200** pivot in a forward direction, as shown in FIG. **66**, the lower tray **250** is spaced apart from the upper tray **150**.

Further, the pivoting force of the lower assembly **200** is transmitted to the upper ejector **300** via the connector **350**. Then, the upper ejector **300** is lowered by the unit guides **181** and **182**, such that the ejecting pin **320** is inserted into the upper chamber **152** through the ejector-receiving opening **154**.

In the ice-removal process, the ice may be removed from the upper tray **250** before the ejecting pin **320** presses the ice. That is, the ice may be separated from the surface of the upper tray **150** due to the heat of the upper heater **148**.

In this case, the ice may be moved together with the lower assembly **200** while the ice is supported by the lower tray **250**.

Alternatively, the ice does not separate from the surface of the upper tray **150** even though the heat of the upper heater **148** is applied to the upper tray **150**.

Thus, when the lower assembly **200** pivots in a forward direction, the ice may be separated from the lower tray **250** while the ice is in close contact with the upper tray **150**.

In this state, in the pivoting process of the lower assembly **200**, the ice may be released from the upper tray **150** when the ejecting pin **320** passes through the ejector-receiving opening **154** and then presses the ice as is in close contact

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to the upper tray 150. The ice removed from the upper tray 150 may again be supported by the lower tray 250.

When the ice moves together with the lower assembly 200 while the ice is supported by the lower tray 250, the ice may be separated from the lower tray 250 by its own weight even when no external force is applied to the lower tray 250.

In the forward pivoting process of the lower assembly 200, the ice-full state detection lever 700 may move to the ice-full state detection position, as shown in FIG. 67. In this connection, when the ice bin 102 is in the ice-full state, the ice-full state detection lever 700 may move to the ice-full state detection position.

While the ice-full state detection lever 700 has moved to the ice-full state detection position, the detection body 700 is located below the lower assembly 200.

When, in the pivoting process of the lower assembly 200, the ice is not separated, via the weight thereof, from the lower tray 250, the ice may be removed from the lower tray 250 when the lower tray 250 is pressed by the lower ejector 400 as shown in FIG. 68.

Specifically, in the process in which the lower assembly 200 pivots, the lower tray 250 comes into contact with the lower ejecting pin 420.

Further, as the lower assembly 200 continues to pivot in the forward direction, the lower ejecting pin 420 will pressurize the lower tray 250, thereby deforming the lower tray 250. Thus, the pressing force of the lower ejecting pin 420 may be transferred to the ice, thereby causing the ice to be separated from the surface of the lower tray 250. Then, the ice separated from the surface of the lower tray 250 may fall downward and be stored in the ice bin 102.

After the ice is removed from the lower tray 250, the lower assembly 200 may pivot in the reverse direction by the driver 180.

When the lower ejecting pin 420 is spaced apart from the lower tray 250 in the process in which the lower assembly 200 pivots in the reverse direction, the deformed lower tray may be restored to its original form.

Further, in the reverse pivoting process of the lower assembly 200, the pivoting force is transmitted to the upper ejector 300 via the connector 350, thereby causing the upper ejector 300 to rise up. Then, the ejecting pin 320 is released from the upper chamber 152.

Further, the driver 180 will stop when the lower assembly 200 reaches the water-supplied position, and then the water supply begins again.

In addition to the above-described embodiments, other various embodiments of the present disclosure may be available. Another embodiment of the present disclosure is characterized in that a surface-treatment portion is formed on the inner face of the ice chamber. Thus, another embodiment of the present disclosure differs only in a structure of the inner face of the ice chamber, and the remaining components are exactly the same as the above-described embodiments. A description of other components except for the surface-treatment portion will be omitted, and the same components as those of the above-described embodiment will be indicated using the reference numerals of the above-described embodiment.

FIG. 69 is a cut perspective view illustrating an interior of an ice chamber according to another embodiment of the present disclosure. Further, FIG. 70 is a perspective view illustrating an open state of a lower assembly according to another embodiment of the present disclosure.

Referring to the drawing, the ice maker 100 may include the upper assembly 110 and the lower assembly 200. Further, the upper assembly 110 and the lower assembly 200

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define the ice chamber 111 for making the spherical ice. The ice chamber 111 is substantially the spherical chamber. The upper assembly 110 and the lower assembly 200 may define the plurality of divided ice chambers 111.

The water supply 190 is coupled to the upper assembly 110, and direct the water supplied from the outside to the ice chamber 111. The ice chamber 111 may include the plurality of ice chambers. In one example, three ice chambers, the first ice chamber 111a, the second ice chamber 111b, and the third ice chamber 111c, are sequentially arranged. The water supplied to the water supply 90 may flow to the plurality of ice chambers 111, and then the plurality of ice chambers 111 may be filled with the water of uniform amount.

The upper assembly 110 may include the upper tray 150 that forms the upper portion of the ice chamber 111 for making the ice. Further, the upper assembly 110 may further include the upper casing 120 and the upper support 170 to fix the upper tray 150.

The upper tray 150 may include the hemispherical upper chamber 152 opened downward. The upper chamber 152 may include the plurality of upper chambers arranged sequentially, and the plurality of upper chambers 152 may be arranged independently and not be connected to each other. For example, the upper chamber 152 may include the first upper chamber 152a, the second upper chamber 152b, and the third upper chamber 152c.

Further, the ejector-receiving opening 154 through which an end of the upper ejector 300 may enter or exit may be defined in each of the first upper chamber 152a, the second upper chamber 152b, and the third upper chamber 152c.

Further, a surface-treatment portion 112 may be formed on an entirety of inner face of the first upper chamber 152a of the upper chambers 152. The surface-treatment portion 112, which is a portion that in contact with the ice to be removed from the ice chamber 111, may be processed in a set roughness.

For example, the surface-treatment portion 112 may be formed to have a fine pattern or bump by etching, and may have fine protrusions 112a when the upper tray 150 is injection-molded.

In the drawings, the protrusion 112a is illustrated to be slightly larger in size to help understand, and is illustrated as having a predetermined pattern. However, when the size of the protrusion 112a is large, not only a surface of the removed spherical ice may be bumpy, but also a transparency may be significantly reduced. Thus, the protrusion 112a may be formed in a fine protrusion shape much smaller than that shown. For example, the protrusion 112a may be formed to have a size of several to several tens of micrometers, so that it is preferable that the protrusion 112a is generally invisible.

The surface-treatment portion 112 may be formed at a degree of roughness such that friction with the surface of the ice may occur. Therefore, the water flows to the inner face of the upper chamber 152, that is, between the surface-treatment portion 112 and the outer face of the ice, thereby preventing the water from flowing upward.

When the surface-treatment portion 112 is not formed, during the ice-removal process, the water phase-changes to the ice, which inevitably expands in volume. In this connection, as a gap is generated between the upper tray 150 and the lower tray 250, water in the neighboring ice chamber 111 or water outside the ice chamber may be introduced. Further, such water may flow upwards along the inner face of the upper chamber 152 and the surface of the ice, and then may

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be eventually frozen at the ejector-receiving opening 154 side, so that a protrusion may be formed on the outer face of the ice.

Attachment of the ice and the inner face of the upper chamber 152 and flowing of the water toward the ejector-receiving opening 154 may be prevented by the surface-treatment portion 112.

The surface-treatment portion 112 may be formed only on the inner face of the first ice chamber 111a. This is because the first ice chamber 111a is located at the position closest to the cold-air hole 134 through which the cold-air is introduced, which may be frozen first. Therefore, when the surface-treatment portion 112 is formed on the inner surface of the first ice chamber 111a, during the ice-removal, the ice making is first started in the first ice chamber 111a. Further, when the ice making is started in the first ice chamber 111a, the water may be prevented from entering the second ice chamber 111b. Therefore, the formation of the spherical ice of a set size may be ensured in the first ice chamber 111a, and the ice-removal may be performed sequentially in the second ice chamber 111b and the third ice chamber 111c.

Although not shown, the surface-treatment portion 112 may be formed only in the first upper chamber 152a of the first ice chamber. The ejector-receiving opening 154 is an only portion in which the ice piece protruding on the outer face of the ice, which is being removed, may grow. Accordingly, the surface-treatment portion is formed only in the first upper chamber 152a to prevent the water from flowing toward the ejector-receiving opening 154 during the ice-removal.

In another example, the surface-treatment portion 112 may be formed in all the first ice chamber 111a, the second ice chamber 111b, and the third ice chamber 111c. Further, the surface-treatment portion 112 may be formed only in the first upper chamber 152a, the second upper chamber 152b, and the third upper chamber 152c.

Further, the lower assembly 200 may include the lower tray 250 forming the lower portion of the ice chamber 111 for the ice formation. The lower tray 250 may form the hemispherical lower chamber 252 opened upward. The lower chamber 252 may include the plurality lower chambers arranged sequentially. For example, the lower chamber 252 may include the first lower chamber 252a, the second lower chamber 252b, and the third lower chamber 252c.

When the upper tray 150 and the lower tray 250 are in contact with each other and closed, the upper chamber 152 and the lower chamber 252 come into contact with each other to form the ice chamber 111.

Further, the surface-treatment portion 112 may also be formed on the inner face of the lower chamber 252. The surface-treatment portion 112 may be formed on the entirety of the inner face of the first lower chamber 252a of the lower chambers 252. The surface-treatment portion 112 formed in the lower chamber 252 has the same structure and function as the surface-treatment portion 112 formed in the upper chamber 152, and thus a detailed description thereof will be omitted.

FIG. 71 is a cross-sectional view illustrating states of an upper tray and a lower tray when an ice volume is increased in an ice-removal state of another embodiment of the present disclosure.

As shown, when the ice-removal begins after the water supply is complete, in the process in which the water phase-changes into the ice in the ice chamber 111, the ice inside the ice chamber 111 is increased in volume. Accord-

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ingly, the lower tray 250 is pivoted in the reverse direction, so that a gap between the upper tray 150 and the lower tray 250 may increase.

Further, when a relatively large amount of the cold-air is supplied to the first ice chamber 111a, which is close to the cold-air hole 134, among the plurality of ice chambers 111, the ice-making may be performed first in the ice chamber 111a.

The water inside the first ice chamber 111a becomes ice, but the ice-making may not be performed in the second ice chamber 111b and the third ice chamber 111c yet. In addition, the upper tray 150 and the lower tray 250 may be partially spaced apart from each other due to the increase in the volume of the ice inside the first ice chamber 111a. In this connection, the water of the neighboring second ice chamber 111b may flow into the first ice chamber 111a.

However, the ice inside the first ice chamber 111a is brought into close contact with the inner face of the first ice chamber 111a by the surface-treatment portion 112. Thus, a space into which the water in the second chamber 111b or the water outside the ice chamber 111a flows is not provided. Therefore, there is no additional water inflow and flow into the first ice chamber 111a, and the ice inside the first ice chamber 111a may be made into the spherical ice having the set size.

Further, since the water in the second ice chamber 111b and the third ice chamber 111c was not substantially introduced into the first ice chamber 111a, the ice cubes inside the second ice chamber 111b and the third ice chamber 111c may also have the same size and shape as the ice made in the first ice chamber 111a.

In addition to the above-described embodiment, other various embodiments of the present disclosure may be available. Another embodiment of the present disclosure differs only in the structure of the surface-treatment portion, and the remaining components are exactly the same as the above-described embodiment. Thus, a description of other components except for the surface-treatment portion will be omitted, and the same components as those of the above-described embodiment will be indicated using the reference numerals of the above-described embodiment.

FIG. 72 is a cut perspective view illustrating an interior of an ice chamber in another embodiment of the present disclosure.

As shown, the ice maker 100 according to another embodiment of the present disclosure forms the ice chamber 111 in a state in which the upper tray 150 and the lower tray 250 are closed. That is, the upper chamber 152 of the upper tray 150 and the lower chamber 252 of the lower tray 250 may be formed in a hemispherical shape. Further, the lower tray 250 pivots to form the spherical ice chamber 111 in a state in which the lower assembly 200 is closed.

In addition, the surface-treatment portion 113 may be formed on the inner face of the ice chamber 111. The surface-treatment portion 113 may remain attached to the surface of the ice made inside the ice chamber 111 by the friction during the ice-removal process.

The surface-treatment portion 113 may be a hydrophilic coating or film. That is, the surface-treatment portion 113 may be formed by performing hydrophilic coating or applying a hydrophilic paint on the inner face of the ice chamber 111, or may be formed by attaching the hydrophile film to the inner face of the ice chamber 111.

Thus, the surface-treatment portion 113 is in contact with the surface of the ice made in the ice chamber 111, so that no space is generated between the surface-treatment portion 113 and the outer surface of the ice. Therefore, even when

the lower tray is partially opened during the ice-removal process, the water inflow is prevented and the water flow toward the ejector-receiving opening 154 is prevented.

The surface-treatment portion 113 may be formed in the entire upper chamber 152 and the lower chamber 252, or may be formed only in the upper chamber 152. In addition, the surface-treatment portion 113 may be formed only in the ice chamber of one side, for example, the first ice chamber 111a, close to the cold-air hole 134 among the plurality of ice chambers 111.

In one example, an ice maker according to the present embodiment includes an upper tray made of an elastic material, wherein a plurality of hemispherical upper chambers opened downward are defined in the upper tray, a lower tray made of an elastic material, and having a plurality of lower chambers defined therein, which are rotatably arranged below the plurality of upper assembly, and in contact with the plurality of upper chambers by pivoting to define a plurality of spherical ice chambers, respectively, a driver for pivoting the lower tray, a water supply disposed above the ejector-receiving opening opened in a top face of the upper tray for supplying water for ice-making into the ice chamber, an upper ejector disposed above the ejector-receiving opening in a vertically movable manner, wherein the upper ejector passes through the ejector-receiving opening to remove each ice from each ice chamber, and a lower ejector disposed on a pivoting radius of the lower support to press an outer face of the lower chamber to remove ice from the lower chamber upon pivoting of the lower support, wherein a surface-treatment portion to be attached to a surface of the ice inside the ice chamber during the ice-removal may be formed in at least a portion of the ice chamber.

The surface-treatment portion may have a bump or a pattern formed thereon for increasing a roughness of a surface of the ice chamber.

The surface-treatment portion may be formed by etching.

The surface-treatment portion may be formed as a separate layer having hydrophilicity.

The surface-treatment portion may be formed by a hydrophilic coating, application of a hydrophilic paint, or adhesion of a hydrophilic film.

The surface-treatment portion may be formed only in the upper chamber.

A roughness of the inner face of the upper chamber may be higher than that of the lower chamber.

The surface-treatment portion may be formed inside an ice chamber among the plurality of ice chambers at a position closest to the cold-air hole through which the cold-air is introduced into the ice maker.

An opening-defining wall having a set height is formed along the circumference of the ejector-receiving opening. Further, an opening-defining wall rib for connecting the outer face of the opening-defining wall and the outer face of the upper chamber with each other may be radially disposed along a circumference of the opening-defining wall.

A chamber wall extending downward along the perimeter of the plurality of upper chambers may be formed on the bottom face of the upper tray. Further, a side wall extending upward along the perimeter of the plurality of lower chambers may be formed on the top face of the lower tray.

The chamber wall may be inserted into the side wall and portions of the upper tray and the lower tray may be spaced apart from each other, so that water supply for ice-removal may be achieved in a state in which the plurality of ice chambers are in communication with each other.

The side wall may extend to a vertical level corresponding to the top of the ice chamber.

As described above, the present disclosure is described with reference to the drawings. However, the present disclosure is not limited by the embodiments and drawings disclosed in the present specification. It will be apparent that various modifications may be made thereto by those skilled in the art within the scope of the present disclosure. Furthermore, although the effect resulting from the features of the present disclosure has not been explicitly described in the description of the embodiments of the present disclosure, it is obvious that a predictable effect resulting from the features of the present disclosure should be recognized.

What is claimed is:

1. An ice maker comprising:

a first tray that is made of a flexible material and defines a first chamber; and

a second tray that is made of a flexible material and defines a second chamber, the second chamber being configured to move between (i) a closed position at which the first tray and the second tray are in contact with each other for ice making and (ii) an open position at which the first tray and the second tray are spaced apart from each other for ice separating,

wherein the first chamber and the second chamber are configured to define an ice chamber based on the second tray moving to the closed position,

wherein the first tray and the second tray comprise a chamber wall that defines the ice chamber, the chamber wall having an opening that is defined at an upper portion of the chamber wall and that is fluid communication with the ice chamber, and

wherein the chamber wall comprises a surface-treatment portion that is provided at an inner surface of the upper portion of the chamber wall and that is configured to increase attaching force between the inner surface of the chamber wall and a surface of ice in the ice chamber.

2. The ice maker of claim 1, wherein the surface-treatment portion comprises a bump or a pattern that is provided at the inner surface of the chamber wall to increase a roughness of the inner surface of the chamber wall.

3. The ice maker of claim 2, wherein the surface-treatment portion is provided by etching.

4. The ice maker of claim 2, wherein the second tray is configured to be disposed below the first tray based on the second tray moving to the closed position,

wherein the opening is defined at an upper side of the first tray, and

wherein a roughness of an inner surface of the first tray is greater than a roughness of an inner surface of the second tray.

5. The ice maker of claim 1, wherein the surface-treatment portion comprises a hydrophilic coating or a hydrophilic film.

6. The ice maker of claim 1, wherein the surface-treatment portion comprises (i) a hydrophilic coating that is provided by applying a hydrophilic paint on the inner surface of the chamber wall or (ii) a hydrophilic film that is attached to the inner surface of the chamber wall.

7. The ice maker of claim 1, wherein the second tray is configured to be disposed below the first tray based on the second tray moving to the closed position,

wherein the opening is defined at an upper side of the first tray, and

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wherein the surface-treatment portion is provided only on an inner surface of the first tray and is not provided on an inner surface of the second tray.

8. The ice maker of claim 1, wherein the first and second trays are configured to, based on the second tray moving to the closed position, define a plurality of ice chambers that are arranged in a direction away from a cold-air hole configured to provide cold air into the ice maker, and

wherein the surface-treatment portion is provided only on an inner face of a chamber wall that defines one of the plurality of ice chambers disposed closest to the cold-air hole among the plurality of ice chambers.

9. The ice maker of claim 1, further comprising; an opening-defining wall that has a predetermined height and is disposed around the opening; and a connection rib that connects an outer surface of the opening-defining wall to an outer surface of the chamber wall.

10. The ice maker of claim 1, wherein the chamber wall comprises (i) a first chamber wall that is disposed at the first tray and defines the first chamber and (ii) a second chamber wall that is disposed at the second tray and defines the second chamber,

wherein the second tray further comprises a side wall configured to surround at least a portion of the first chamber wall based on the second tray moving to the closed position, and

wherein the surface-treatment portion is provided only at an inner face of the first chamber wall and is not provided on an inner surface of the second chamber wall.

11. A refrigerator comprising:

a cabinet that defines a storage space; and an ice maker provided in the storage space, wherein the ice maker comprises:

a first tray that is made of a flexible material and defines a first chamber, and

a second tray that is made of a flexible material and defines a second chamber, the second chamber being configured to move between (i) a closed position at which the first tray and the second tray are in contact with each other for ice making and (ii) an open position at which the first tray and the second tray are spaced apart from each other for ice separating,

wherein the first chamber and the second chamber are configured to define an ice chamber based on the second tray moving to the closed position,

wherein the first tray and the second tray comprise a chamber wall that defines the ice chamber, the chamber wall having an opening that is defined at an upper portion of the chamber wall and that is fluid communication with the ice chamber, and

wherein the chamber wall comprises a surface-treatment portion that is provided at an inner surface of the upper portion of the chamber wall and that is configured to increase attaching force between the inner surface of the chamber wall and a surface of ice in the ice chamber.

12. The refrigerator of claim 11, wherein the surface-treatment portion comprises a bump or a pattern that is

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provided at the inner surface of the chamber wall to increase a roughness of the inner surface of the chamber wall.

13. The refrigerator of claim 12, wherein the surface-treatment portion is provided by etching.

14. The refrigerator of claim 12, wherein the second tray is configured to be disposed below the first tray based on the second tray moving to the closed position,

wherein the opening is defined at an upper side of the first tray, and

wherein a roughness of an inner surface of the first tray is greater than a roughness of an inner surface of the second tray.

15. The refrigerator of claim 11, wherein the surface-treatment portion comprises a hydrophilic coating or a hydrophilic film.

16. The refrigerator of claim 11, wherein the surface-treatment portion comprises (i) a hydrophilic coating that is provided by applying a hydrophilic paint on the inner surface of the chamber wall or (ii) a hydrophilic film that is attached to the inner surface of the chamber wall.

17. The refrigerator of claim 11, wherein the second tray is configured to be disposed below the first tray based on the second tray moving to the closed position,

wherein the opening is defined at an upper side of the first tray, and

wherein the surface-treatment portion is provided only on an inner surface of the first tray and is not provided on an inner surface of the second tray.

18. The refrigerator of claim 11, wherein the first and second trays are configured to, based on the second tray moving to the closed position, define a plurality of ice chambers that are arranged in a direction away from a cold-air hole configured to provide cold air into the ice maker, and

wherein the surface-treatment portion is provided only on an inner face of a chamber wall that defines one of the plurality of ice chambers disposed closest to the cold-air hole among the plurality of ice chambers.

19. The refrigerator of claim 11, further comprising; an opening-defining wall that has a predetermined height and is disposed around the opening; and a connection rib that connects an outer surface of the opening-defining wall to an outer surface of the chamber wall.

20. The refrigerator of claim 11, wherein the chamber wall comprises (i) a first chamber wall that is disposed at the first tray and defines the first chamber and (ii) a second chamber wall that is disposed at the second tray and defines the second chamber,

wherein the second tray further comprises a side wall configured to surround at least a portion of the first chamber wall based on the second tray moving to the closed position, and

wherein the surface-treatment portion is provided only at an inner face of the first chamber wall and is not provided on an inner surface of the second chamber wall.

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